ANNEX 1

Horizon Europe (HORIZON)

Description of the action (DoA)

Part A

Part B

DESCRIPTION OF THE ACTION (PART A)

Associated with document Ref. Ares(2022)7761208 - 10/11/2022

COVER PAGE

Part A of the Description of the Action (DoA) must be completed directly on the Portal Grant Preparation screens.

PROJECT *Grant Preparation (General Information screen) — Enter the info.*	
Project number:	101091801
Project name:	A Digital value chain Integration Traceability framework for process industries for Circularity and low Emissions by waste reduction and use of secondary raw materials
Project acronym:	DigInTraCE
Call:	HORIZON-CL4-2022-RESILIENCE-01
Topic:	HORIZON-CL4-2022-RESILIENCE-01-01
Type of action:	HORIZON-RIA
Service:	HADEA/B/03
Project starting date:	fixed date: 1 January 2023
Project duration:	48 months

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PROJECT SUMMARY

Project summary

Grant Preparation (General Information screen) — Provide an overall description of your project (including context and overall objectives, planned activities and main achievements, and expected results and impacts (on target groups, change procedures, capacities, innovation etc)). This summary should give readers a clear idea of what your project is about.

Use the project summary from your proposal.

DigInTraCE aims at the development of i)transparent and interoperableDecentralised Traceability platform using novel tracking, sensing and sorting techniques ii)Dynamically updated DPPschemes supporting certification and quality validation iii)AI based and decision making mechanisms for process and lifecycle optimisation iv)up-cycling, reuse and upgrade technologies for improved secondary raw materials use v)contribute to standardization, open and easy accessible data vi)business models creating new economic opportunities and learning resources for employees, promoting new digital skills and meeting regional social needs. The solutions are demonstrated in the sectors of a) Pulp & Paper focused in 2 cases: composite wood and furniture; and wood and Pulp & Paper and b) Chemicals focused in 2 cases: plastic parts from ICT equipment and automotive market; and polymers and textiles: in 4 countries, while composite wood case combines both the 2 main sectors.

LIST OF PARTICIPANTS

PARTICIPANTS

Grant Preparation (Beneficiaries screen) — Enter the info.

Number	Role	Short name	Legal name	Country	PIC
1	COO	ICCS	INSTITUTE OF COMMUNICATION AND COMPUTER SYSTEMS	EL	999654356
2	BEN	NTUA	ETHNICON METSOVION POLYTECHNION	EL	999978142
3	BEN	UST	KEEN SOFTWARE SL	ES	887713155
4	BEN	CIRCE	FUNDACION CIRCE CENTRO DE INVESTIGACION DE RECURSOS Y CONSUMOS ENERGETICOS	ES	999516907
5	BEN	DS	FONDEN DANSK STANDARD	DK	951792131
6	BEN	IRIS	IRIS TECHNOLOGY SOLUTIONS, SOCIEDAD LIMITADA	ES	998993786
7	BEN	VTT	TEKNOLOGIAN TUTKIMUSKESKUS VTT OY	FI	932760440
8	BEN	MERIT	MERIT CONSULTING HOUSE - OLOKRIROMENES SYMVOULEFTIKES IPIRESIES EPIXEIRISEON IDIOTIKI KEFALAIOUXIKI ETAIREIA	EL	896542192
9	BEN	HTECH	HYPERTECH (CHAIPERTEK) ANONYMOS VIOMICHANIKI EMPORIKI ETAIREIA PLIROFORIKIS KAI NEON TECHNOLOGION	EL	996329681
10	BEN	CHIMAR	CHIMAR (HELLAS) AE - ANONYMI VIOMICHANIKI KAI EMPORIKI ETAIREIA CHIMIKON PROIONTON	EL	999925762

PARTICIPANTS

Grant Preparation (Beneficiaries screen) — Enter the info.

Number	Role	Short name	Legal name	Country	PIC

11	BEN	AGRST	AGROINVEST ANONYMI EMPORIKI KAI VIOMICHANIKI ETAIRIA	EL	951850331
12	BEN	MXS	MENEXES - ANONYMI ETAIREIA KATASKEVIS KAI EMPORIAS EPIPLON	EL	887256091
13	BEN	TECNL	FUNDACION TECNALIA RESEARCH & INNOVATION	ES	999604110
13.1	AE	TNLCERT	TECNALIA R&I CERTIFICACION SL	ES	925415212
14	BEN	ASTI	ASTIGARRAGA KIT LINE SL	ES	887215545
15	BEN	CTB	CENTRE SCIENTIFIQUE & TECHNIQUE DEL'INDUSTRIE TEXTILE BELGE ASBL	BE	998868171
16	BEN	SIOEN	SIOEN INDUSTRIES NV	BE	985990257
17	BEN	DGS	DGS SPA	IT	906212607
18	BEN	SIGIT	SIGIT - SOCIETA ITALIANA GOMMA INDUSTRIALE TORINO - SPA	IT	889368266
19	BEN	UVQ	UNIVERSITA DEGLI STUDI DELL'AQUILA	IT	999859511
20	BEN	EUPC	EUROPEAN PLASTICS CONVERTERS	BE	998424978
21	AP	UBRUN	BRUNEL UNIVERSITY LONDON	UK	999749610

LIST OF WORK PACKAGES

Work packages

Grant Preparation (Work Packages screen) — Enter the info.

Work Package No	Work Package name	Lead Beneficiary	Effort (PersonMonths)	Start Month	End Month	Deliverables
WP1	Project Management and Ethics	1 - ICCS		1	48	D1.1 – DigInTraCE Project management handbook, QA, ethics and legal framework and reports on innovation and data management v1 D1.2 – DigInTraCE Project management handbook, QA, ethics and legal framework and reports on innovation and data management v2 D1.3 – DigInTraCE Project management handbook, QA, ethics and legal framework and reports on innovation and data management v3 D1.4 – DigInTraCE Project management handbook, QA, ethics and legal framework and reports on innovation and data management FINAL D1.5 – Data Management Plan v1 D1.6 – Data Management Plan v2 D1.7 – Data Management Plan FINAL

WP2	Holistic Framework for digitalisation of circular value chains	21 - UBRUN		1	48	D2.1 – Methodology of Traceability v1 D2.2 – Methodology of Traceability v2 D2.3 – Methodology of Traceability FINAL D2.4 – Digital Product Passport Concept v1 D2.5 – Digital Product Passport Concept FINAL D2.6 – Real-time information and secure data exchange requirements v1

Work packages

Grant Preparation (Work Packages screen) — Enter the info.

Work Package No	Work Package name	Lead Beneficiary	Effort (PersonMonths)	Start Month	End Month	Deliverables
						D2.7 – Real-time information and secure data exchange requirements v2 D2.8 – Real-time information and secure data exchange requirements FINAL D2.9 – Specifications definition and architecture of digital tracing v1 D2.10 – Specifications definition and architecture of digital tracing FINAL

WP3	Sensing, Sorting Technologies and Process Optimization Algorithms	6 - IRIS		6	42	D3.1 – Novel Sensing technologies v1 D3.2 – Novel Sensing technologies FINAL D3.3 – Sorting mechanisms for secondary streams valorisation v1 D3.4 – Sorting mechanisms for secondary streams valorisation FINAL D3.5 – Smart Tags v1 D3.6 – Smart Tags FINAL D3.7 – CLSC tool v1 D3.8 – CLSC tool FINAL D3.9 – Online LCA computation tool based on Ex-Ante LCA v1 D3.10 – Online LCA computation tool based on Ex-Ante LCA FINAL
WP4	Digital tools and platform development	1 - ICCS		9	48	D4.1 – Dynamic Product Passport for Unit of Product through the value chain v1 D4.2 – Dynamic Product Passport for Unit of Product through the value chain FINAL D4.3 – Blockchain and Distributed ledger v1 D4.4 – Blockchain and Distributed ledger FINAL

Work packages

Grant Preparation (Work Packages screen) — Enter the info.

Work Package No	Work Package name	Lead Beneficiary	Effort (PersonMonths)	Start Month	End Month	Deliverables

						D4.5 – Financial tool v1 D4.6 – Financial tool FINAL D4.7 – MR enabler tool v1 D4.8 – MR enabler tool FINAL D4.9 – DigInTraCE platform and system integration v1 D4.10 – DigInTraCE platform and system integration FINAL
WP5	Technologies for secondary materials upgrade and process optimization	4 - CIRCE		1	42	D5.1 – Secondary materials quality characterization for recycle,reuse upgrade D5.2 – Proteins recovery for wood-resin composite production v1 D5.3 – Proteins recovery for wood-resin composite production FINAL D5.4 – Wood by-products sorting and characterization for process optimisation v1 D5.5 – Wood by-products sorting and characterization for process optimisation FINAL D5.6 – Fingerprints and markers for recycled material content v1 D5.7 – Fingerprints and markers for recycled material content FINAL D5.8 – Plastic parts and components analysis and origin for reuse optimisation v1 D5.9 – Plastic parts and components analysis and origin for reuse optimisation FINAL
WP6	Piloting and Demonstration	2 - NTUA		6	48	D6.1 – Waste streams mapping D6.2 – Systems design and description v1

Work packages

Grant Preparation (Work Packages screen) — Enter the info.

Work Package No	Work Package name	Lead Beneficiary	Effort (PersonMonths)	Start Month	End Month	Deliverables
						D6.3 – Systems design and description FINAL D6.4 – Evaluation and Impact assessment
WP7	Guidance, Training, Cooperation, Exploitation and communication	8 - MERIT		1	48	D7.1 – Standardization of novel technologies and certification schemes v1 D7.2 – Standardization of novel technologies and certification schemes FINAL D7.3 – Co-design training activities and lifelong learning programmes on digital skills and replication v1 D7.4 – Co-design training activities and lifelong learning programmes on digital skills and replication FINAL D7.5 – Exploitation and Business Innovation Plan and activities v1 D7.6 – Exploitation and Business Innovation Plan and activities v2 D7.7 – Exploitation and Business Innovation Plan and activities FINAL D7.8 – Communication and Dissemination plan and activities v1 D7.9 – Communication and Dissemination plan and activities v2 D7.10 – Communication and Dissemination plan and activities FINAL D7.11 – Policy briefs

Work package WP1 – Project Management and Ethics

Work Package Number	WP1	**Lead Beneficiary**	1. ICCS
Work Package Name	Project Management and Ethics		
Start Month	1	**End Month**	48
Objectives			
● Project Management according to EU Grant Agreement provisions and conformingly to the respective Project Boardorganisation ● consortium coordination of the activities of the Project, and ensure that partners work together effectively, by monitoring the overall technical coordination and progress of the Project ● managing the overall legal, contractual, financial, and administrative aspects of the consortium ● controlling and refining the objectives of the project by Quality Assurance (QA), performance management and conflictresolution to ensure scientific excellence for all outputs and internal processes ● establish Executive Board (ExB), assign managers (Technical, Risk and QA) ● coordination of the involvement of an External Expert Advisory Board in the project ● develop an efficient data management plan and open science guidelines towards maximisation of the exploitationpotential ● set out project's GDPR and gender balance requirements and effectively manage legal and ethics matters.			
Description			

T1.1 Project Coordination and Management (including quality assurance and risk management) M1-M48 (ICCS, all WP participants): Task 1.1 will provide the overall project coordination, management and administration by developing and implementing management processes for an efficient and successful project execution, including the QA and risk management. The following tasks will be performed: i) preparation of technical and financial interim reports in a 6-month basis so that deliverables and milestones of each period are closely followed and technical activities follow resource consumption; ii) preparation of the external EC reports, monitoring technical and financial activities as well as audit reports in applicable cases; iii) organisation of consortium meetings; iv) distribution of the EC financial contribution and costs coordination and controlling; v) communication with EC and submission of deliverables to the EC; vi) ensuring the communication between partners at all levels and the proper information exchange; vii) preparation and maintenance of an appropriate communication and collaborative environment for the consortium members (e.g., mailing lists, project repository, internal wikis, reporting tools); viii) support of the evolution of the work plan; ix) implementation of QA procedures with particular focus on monitoring review activities and submitting quality review reports –to undertake this task; x) ensuring that proactive measures are taken to mitigate risks which could jeopardise the planned project outcomes, as well as monitoring the potential critical implementation risks and xi) management of the external advisory board to provide advice, and validation of the vision, global impact, and outreach of the project. Outputs: D1.1, D1.2, D1.3, D1.4 T1.2 Technical Management and Scientific Connection (including data management) M1-M48 (NTUA, ICCS, CIRCE): will monitor the overall technical progress and work quality to support Project Coordinator and follow up the technical part of the review meetings. It will guarantee the smooth technical cooperation between partners and interconnections between WPs and tasks, participate in the consideration of, and responding to, the opinions and contributions of the relevant working groups, and will suggest alternative technical routes in case of stumbling blocks. Additionally, it will track the milestones and coordinate the technical meetings, and will ensure that the scientific and technological outcomes comply with the project's work plan, and the results fulfil the technical requirements for effective progress towards the achievement of the project's goals. In accordance with EC Guidelines regarding data management, this task will also set up a Research Data Management Plan defining how research data collected and/or generated throughout the project will be handled during (and after) the project duration. Following FAIR principles, T1.2 will produce an overview of how research data will be made Findable, Accessible, Interoperable and Reusable elaborating, among other, on the data types, the accessibility to datasets, the use of standards and the publication methodology. In this context, special emphasis will be put on making the research data accessible through open repositories, to maximise the visibility of the project's data results, aiming also, to assist other research activities to find and utilise the knowledge produced by DigInTraCE. The task will deliver the project's open science guidelines (as part of D1.1-D1.4) and Data Management Plan (D1.5) by M6 (MS2), capturing the envisioned practices at that early stage. Any updates at a later stage will be reported by M26 and M48. Outputs: D1.1, D1.2, D1.3, D1.4 , D1.5
T1.3 GDPR, Gender Balance, Legal and Ethical Issues Management M1-M48 (CIRCE, ICCS): will design and monitor the procedures and protocols necessary for handling ethical issues during the project. A framework of ethical requirements will be constructed to ensure DigInTraCE carrying out of responsible research practice with specific reference to the European Code of Conduct for Research Integrity (ALLEA) and the Data Protection legislation and regulation. It will also set out the ethics requirements (e.g. procedures, criteria to identify/recruit research participants, templates for informed consent, etc.) and manage the ethical issues concerning the design of the information and consent forms to be used in case of involvement of humans through the use cases demonstrations, while ensuring GDPR and Privacy Compliance Policy by producing and properly disseminate informed consent forms for all project participatory activities (demonstration activities, online surveys, evaluation forms, workshops, etc.) according to the EU GDPR regulations and national data protection legislation. It will also ensure that SSH is embedded in the DigInTraCE as an enabler of RRI. Outputs: D1.1, D1.2, D1.3, D1.4

Work package WP2 – Holistic Framework for digitalisation of circular value chains

Work Package Number	WP2	**Lead Beneficiary**	21. UBRUN
Work Package Name	Holistic Framework for digitalisation of circular value chains		
Start Month	1	**End Month**	48
Objectives			

- Develop DigInTraCE's traceability framework and methodologies
- set up the concept for the dynamic DDP
- set up the requirements for the real-time access to information and secure data exchange
- define the specifications and the architecture for the DigInTraCE Decentralized Traceability platform along the valuechains.

Description

T2.1 Methodologies for digital tracing across value chains M1-M48 (UBRUN): aims to develop a complete strategy for obtaining data from operations and products, discretisation of products, services and business operations facilitating the alignment of data acquisition with Blockchain key features. Key product attributes fully describing the state, origins, constitution, production process (e.g. quality, carbon footprint, circularity, availability/inventory) and methods to acquire such data will be determined. This definition covers both unit of raw material and finished emerging products (assembly of raw material). The former provides key information for the latter. Cost and benefit analysis will be conducted to ascertain the level of tracking & tracing, i.e. the higher level the higher investment on hardware and software solutions required. The modus operandi for data acquisition (automatic, operator instigated or simulated), innovative methods to fuse the acquired information for process and product modelling (digital twin) will be developed. Through bespoke learning algorithms pending the nature of product and corresponding processes, digital representation of unit of product is introduced to Blockchain for implementing business operations. The key outcomes are qualification and quantification of data, valorisation and where necessary anonymisation (driven by Blockchain protocols) to clearly define Unit of Product. Outputs: D2.1,D2.2, D2.3

T2.2 Conceptual development of dynamically updated Digital Product Passport M1-M24 (ICCS, UBRUN, CIRCE): aims to prepare DPPs implementation in demonstrators in data and system architecture level. Existing regulated or voluntary product information schemes will be enhanced (as they hold data not accessible to relevant stakeholders), regulations (e.g. REACH) initiatives and tools (BOMs, LCA) will be studied in the context of data incorporation and related usage scenarios will be developed for all the cases. DPPs' content will be attended to have as generic structure as possible. Captured and registered properties and movement patterns of products can be continuously compared to regulatory, environmental, quality and commercial criteria in the form of look-up tables with specific metrics, rules, features to be checked, following the analogous well established approach in identifying legitimate and non-security threat travellers abroad. Sustainability and circularity metrics will be included in DPP and updated though a live LCA environmental tool designed in T2.2 and developed in T3.5. A workshop with project's industries, bodies and relevant stakeholders for consultation will be performed. Cooperation and alignment with projects of related topics will be established through T7.2. A DPP features list and related system architecture(s) will be provided to T2.4 with adaptations for plastic and wood sectors of DigInTraCE. Output: D2.4, D2.5

T2.3 Real-time access to information and secure data exchange requirements M1-M48 (ICCS, UST): aims to analyze data sharing aspects to overcome barriers and support projects' innovations. The blockchain-based traceability will ensure secure data exchange channels and establish partners' control on the information shared, operating as a neutral data facilitator to face trust or policies issues and using AI, DA & Big Data models extracting value -statistics, conclusionsfrom (meta)data without exposing confidential data. The application of zero knowledge proofs and digital identity will be explored as means to maintain that confidentiality and self-custody of owned data (e.g. data owners use timestamped encryptions as unquestionable proof of validity without publicly disclosing the original data,). Methodologies and practices will be proposed as potential part of DigInTraCE Decentralized Traceability platform (WP4) providing access to useful data and information throughout supply chain to all the stakeholders. Existing information sources e.g. ERPs, existing passports and BOMs etc. explored in T2.2 will be integrated and used, while a safe culture will be diffused to consortium to support managerially trusted and secure information channels with reliable and FAIR data aligned with WP1 and accompanied by training and best practices adoption activities. The set of required data and related requirements will be validated with digital and technology providers and demo leaders to ensure applicability and feasibility of the online monitoring. Business, exploitation and training activities requirements from WP7 will be also considered. Output: D2.6, D2.7 , D2.8

T2.4 Specifications definition and architecture for DigInTraCE platform M8-M28 (UST, ICCS, IRIS, VTT, UBRUN, CTB): aims to harmonise the requirements produced in T2.3 with the traceability framework and the schemes developed in T2.1, T2.2 to produce the technical specifications and the architecture for the Decentralized Traceability platform of DigInTraCE and the modules to be developed in WP3. The aspects of system's energy efficiency and compatibility with standards such as GAIA-X will be also considered. Also, the users of the platform, their roles (access rights, actions and use cases) will be defined, including administrators, operators, supervisors, managers, auditors, researchers and roles related to standardization, certification, training, assessments, technical and quality assurance, exploitation and

dissemination etc. from the industries, providers, intermediates and all stakeholders throughout the supply chains. Output: D2.9 ,D2.10

Work package WP3 – Sensing, Sorting Technologies and Process Optimization Algorithms

Work Package Number	WP3	**Lead Beneficiary**	6. IRIS
Work Package Name	Sensing, Sorting Technologies and Process Optimization Algorithms		
Start Month	6	**End Month**	42

Objectives

- Develop novel sensing technologies for the required data gathering in the 4 demonstrators
- Develop sorting mechanisms to improve secondary streams valorisation
- Develop methodologies and algorithms for the supply chain optimization of the secondary materials●

Develop an LCA and environmental performance analysis tool.

Description

T3.1 Novel Sensing technologies M6-M36 (IRIS, ICCS): After a definition of requirements for real-time monitoring in the demo sites, systems design and development will be conducted, and validation of the final systems will be performed. For plastics based value chains, the sensing technologies will be implemented by IRIS, and the parameters analyzed might include, but are not limited to i) identification of a range of plastics in a mix (such as PP, PA, PET, etc.), ii) heavy metals detection, bromine and phosphate detection, etc. Hyperspectral imaging assets will be utilized to categorise the type of polymers detected and contaminated components assisted with NIR technology, whereas potential usage of LIBS will be explored, to support the detection of black plastic components. The control system will need to be designed and developed at all levels (sensors, hardware and process connection design, IT hardware, software back-end/frontend and modelling). Data mining and new chemometric models will be developed. An edge computing infrastructure for real-time assessment of monitoring signals and application of PAT models, as well as data treatment strategies, will be available for the user at demonstrator sites. The best combined monitoring solution for each process and demo will be selected considering the accuracy, complementarity and specific needs. Some iterations of tests will be performed at IRIS and at the demo site in order to adjust and validate the integration of the PAT platform embedded models and sensing devices. For wood based value chains, the sensing technologies will be implemented by ICCS, and the parameters analyzed might include, but are not limited to i) moisture content, ii) size and shape of wood chips and scraps, iii) wood species etc. Near-infrared hyperspectral imaging (NIR-HSI) will be utilized to classify the wood residues based on their quality, therefore a set of industrial cameras will be installed. In all cases, sensing techniques to be used will also support the implementation of the sorting mechanisms in T3.2. Output: D3.1, D3.2

T3.2 Sorting mechanisms for wood and plastics secondary streams valorisation M6-M36 (ICCS): this task aims to develop and test a sorting mechanism with high accuracy, capable to improve the quality of the feedstock for the upcycling processes by identifying the type of waste and separating the detected unsuitable items. The datasets generated by sensing, identification and characterization methods in T3.1 will be utilised, while robotic arms assisted by machineor deep- learning techniques, will be used for the removal of the defective items and impurities resulting from waste assessment analysis. Task activities include a) determination of the position, the shape and type of the waste using visionbased techniques (both wood and plastic value chain), b) expand the sorting capabilities of the existing NIR plastic sorting systems (only plastic value chain), and c) benefit from robotics solutions that remove the flawed compounds in real-time, to enhance waste valorization potential and retain greater value (both wood and plastic value chain). A wood sorter will be designed for and installed in each pilot involving industrial wood by-products (Greece and Spain), since the quality requirements established by receivers of secondary resources are similar but not identical. In particular for the wood value chain, sieving is required to separate sawdust from small wood pieces, whereas a robotic arm mechanism is required for sorting large wood scraps Output: D3.3, D3.4

Description

T4.1 Dynamically updated Digital Product Passport implementation M9-M42 (ICCS, UST, CIRCE, UBRUN): aims to develop the DPPs and the dynamic update mechanism, as described in T2.2, thus integrate existing product information schemes and tools, cover usage and capture and compare information to meet features and related system architecture
T3.3 Smart Tags for real time sensing M6-M36 (VTT): develops smart tags and use scenarios in the wood-based demonstrator in Greece to support waste reduction and management, use of secondary materials and create new circular business opportunities. Smart tags are able to connect items to data infrastructure, engage stakeholders, digitalize services and provide environmental capabilities. State-of-the-art identification technologies (e.g. 2D barcodes, NFC, RFID) will be combined with sensing where feasible (e.g. functional inks, IC chips, humidity loggers) to achieve item/batch life cycle monitoring, and automatic traceability. Tags may include product data to support circular business models, while the concept will be developed with decentralized, GAIA-X compliant data management and storage, methodology for integration of the smart tag to item (e.g. direct marking of wood based items) taking into account durability throughout product life cycle, connectivity and end-of-life (waste management and raw-material recovery). Efficient ways to engage stakeholders will be investigated (e.g. mock-ups), user groups will test tags use (e.g. product deposit, carbon credit, refurbishing, as-a-service and other circular business or earnings models) and opportunities related to assess wear and tear for lifetime extension and safety will be also explored. Output: D3.5, D3.6 T3.4 Supply chain optimization for secondary raw materials M6-M36 (DGS): DGS will: (i) build a recipe management logic that will be provided as a service within the IoT connectivity/middleware layer of SIGIT plant and (ii) integrate AI algorithms within a more complex end-to-end supply chain management system, based on a modular approach and on cloud technologies (subcontracting involved). The system manages planning and scheduling production receipts (including all necessary information like needed materials, quantity, machinery temperatures, working phases, etc.) and through the project will be extended in order to provide pegging management from the product to the raw materials. Methodologies and algorithms will be analysed and a Closed Loop Supply chain (CLSC) tool will be developed and will be virtually tested and validated. Output: D3.7, D3.8 T3.5 Online LCA computation tool based on Ex-Ante LCA M6-M42 (CIRCE): aims to develop the LCA and environmental performance analysis tool. This live LCA tool will be integrated to the update mechanism of T4.1 and will feed DPP's with environmental indicators. It will be initially created for the textile sector and will be extended beyond the textiles, applied to all project demonstrators in order to analyse their environmental impact. The tool will also contribute to the Product Environmental Footprint (PEF). Output: D3.9, D3.10

Work package WP4 – Digital tools and platform development

Work Package Number	WP4	**Lead Beneficiary**	1. ICCS
Work Package Name	Digital tools and platform development		
Start Month	9	**End Month**	48
Objectives			
● Develop the live update mechanism and implement the Dynamic DPP ● develop the Blockchain and Distributed ledger technologies● develop financial and price definition modules ● compile all technological and digital components into one DigInTraCE system assuring interoperability, visualizationand use of tools' full potential on facilitating decision making processes ● develop the Decentralized Traceability platform of DigInTraCE.			

concepts proposed to adapt tools to the different sectors involved. Linked to T2.1, T2.3 two live traceability methods will be used to feed DPP: i) technologies from T3.1, T3.3 along with existing plant (e.g. SCADA), Logistics and Operator instigated input (HMI) systems, to collect the necessary information about the unit of product. ii)in cases that these techniques are not possible, the deployment of bespoke event-based process simulation to generate Simulation-based Traceability will be proposed. The product information in DPP will be enriched as events affects product throughout its life cycle. The data will be managed by the DLH of T4.4 to keep items -materials, products- updated dynamically along the whole value chain and sharing online data to systems or end-users. Output: D4.1, D4.2

T4.2 Blockchain and Distributed ledger and secure systems development M9-M42 (UST, ICCS, UBRUN): aims to provide a distributed, Decentralized, transparent, immutable and secure data structure and implement both technical: scalability, transaction speed, data capacity, etc. user needs, potential future exploitation, sustainability scenarios; and governance requirements: needed benefits of blockchain such as cooperation, collaboration, immutability, automation and eliminating adversarial vector value (e.g. smart contracts can automate procedures eliminating needs for manual double checking). Implementation will initially use cloud-based nodes at UST for testing and ramp up and following the parallel developments in this rapidly evolving field. As the project matures, we have the option to extend the nodes to the participants – although running a node is not necessary for read/write to the blockchain. To eliminate risk, liability and costs and energy waste, a private blockchain or simulated a public chain (e.g. Ethereum testnet) will be used avoiding harmful proof of work. DLT security, resilience, scalability and performance will be monitored in addition to the fit for purpose aspects viz a viz the use cases. To provide track and trace capability, a layer will support tokenisation (the blockchain behind NFTs, cryptocurrencies, Decentralized finance paradigm) allowing rights and responsibilities to be attached to tokens that can be transferred between wallets and controlled with smart contracts and enabling us to implement, model and observe the financial services of T4.3, tied or separated from the raw material asset e.g. a financial derivative allowing the token holder a predefined price for a future delivery of the raw material can be coded into a token. These tokens can then trade independently and open up possibilities for scenarios where market makers fulfil cash flow, risk assumption and capital injection needs, just as they do in financial commodity markets Output: D4.3, D4.4

T4.3 Financial Services and price definition tool M9-M42 (HTECH, UST): aims to design and develop a toolkit of financial services for supporting stakeholders in a secondary raw materials market to: a) generate strategies for hedging against secondary raw material price fluctuation risks, b) factor-in currency movements when planning cross currency secondary raw material trades, c) apply technical analysis methodologies for visualizing key price trends and forecasting short term secondary raw material price moves, d) identification and visualization of correlations of secondary raw materials with other assets traded in international exchanges and also with key macro-economic indexes and indicators, e) definition and modelling of financial derivatives tied to secondary raw materials. An initial review of platforms and marketplaces facilitating buying/selling secondary materials and an assessment of available data sources for material prices and impact of digital tracing on trading will be performed. Market needs will be listed in terms of pricing decision support and indicatively services will be developed for visualizing historic prices and trends on secondary and virgin material prices and prediction, fluctuation risk assessment/protection (hedging) services. Output: D4.5, D4.6

T4.4 MR enabler for data visualization and human machine optimisation M9-M42 (ICCS): will create a Mixed Reality (MR) tool and application to make information more understandable, readable and accessible, providing interactive visualization and controlling functionalities to end-users (e.g. operators, workers, auditors, evaluators etc. as they will navigate or focus in specific points) along the value chain by revealing material and product information and statuses and facilitate their decisions and actions, thus boost their efficient intervention and engagement. The MR tool will be integrated to the DigInTraCE Decentralized Traceability platform by both retrieving information from the DPPs and provide information upon user interactions or other actions performed during the usage of the tool. It will be also used for on the job training to boost digital skills of the end-users and to engage citizens/raise awareness through gamification, along with and supporting T7.2 activities. The application (Android/iOS) will include authorization tool for content customisation while visualization will be based on realistic 3D environment mixing virtual and augmented reality though models of items and processes of the real value chain. VR headsets use will be examined while MR enabler will be piloted at all the demo sites. Output: D4.7, D4.8

T4.5 System integration and development of the DigInTraCE platform M13-M48 (ICCS, UST, IRIS, CIRCE): aims to integrate all the tools developed in WP3, WP4 into a common multifunctional and interoperable Decentralized Traceability platform for the complete monitoring, tracking and tracing of materials and products through dynamically updated DDP in the whole value chains by the end-users. Platform will include visualization, controlling features, AI-Based optimisation and decision making functionalities and will facilitate modelling, simulation and assessments designed. DigInTraCE platform will support the tasks of end-uses (operators, evaluators, researchers, policy makers etc.); required interactions, communication and coordination of stakeholders along the supply chain; exploitation of digital traceability and technologies to create CE/IS communities for resource exchange and matchmaking; documentation of results, statistics, best practices, barriers; dissemination and knowledge exchange in WP7. CIRCE will contribute to the data acquisition and integration providing SCADA and multiprotocol experience, software

modules for data aggregation and remote communication. These resources can be applied on gateways or devices with Linux-based

distributed developments, using docker modules or the encapsulation needed for a better portability of these components. Output: D4.9, D4.10

Work package WP5 – Technologies for secondary materials upgrade and process optimization

Work Package Number	WP5	**Lead Beneficiary**	4. CIRCE
Work Package Name	Technologies for secondary materials upgrade and process optimization		
Start Month	1	**End Month**	42
Objectives			
● Characterise the secondary materials of the new value chains ● define technical specification for proteins recovery for wood-resin composite and by-products sorting and characterization processes and technologies to be developed for the 2 wood based demonstrators ● define technical specification for fingerprints and markers for recycled Polyester and Plastic parts, componentsanalysis, origin processes to be developed for the 2 plastic based demonstrators.			
Description			

T5.1 Secondary materials quality characterization for recycle, reuse and upgrade (per value chain analysis) M1-M8 (CIRCE, NTUA): divided into two different subtasks. Initially, the four different sectors (polyester, wood, resins and plastics) will be analysed among EU and national regulation (Directives) to check whether secondary markets are suitable. In a second approach recycling/reuse and upcycling opportunities will be considered to provide T5.1 potential pathways and main actors. To this end, demo owners will be asked to identify main actors along value chains and quality requirements will be gathered to ensure that waste are suitable to be converted into secondary materials. Output: D5.1
T5.2 Upgrade and valorisation of secondary materials through proteins recovery for wood-resin composite production M6-M42 (NTUA, CHIMAR, ICCS): Agricultural by-products from oilseed crops (sunflower, rapeseed, cotton press cakes etc.) will be used to produce bioactive compounds (proteins), using recovery processes, based on protocols developed by NTUA. Hydrolysis, enzymatic hydrolysis, conventional extraction, microwave and ultrasonic assisted extraction, as well as Pressurized Liquid Extraction, will be tested and optimized, according to the composition of biomass, using classical and green solvents, aiming to efficient production of concentrated sustainable extracts. Commercially available microbial cultures and enzyme systems will be studied to determine a hydrolysis process which will guarantee high yield of proteins with low production costs. Extraction parameters, such as solvent(s), solvent systems ratio, temperature, time, power and wavelength will be examined and optimised for the efficient production of maximum yield. Extracts will be evaluated using analytical techniques (UV-Vis, HPLC-DAD etc.) and dried for obtaining proteins in powder form in order to be used as structural elements, replacing part of phenol in phenolic resins. Solutions will be demonstrated in WP6. Output: D5.2, D5.3
T5.3 Wood by-products sorting and characterization for process optimisation M6-M42 (TECNL, ICCS, IRIS, CHIMAR): Most relevant characterization parameters to be considered in terms of upgrading wood transformation byproducts are tree species, by-product type (bark, sawdust, sand dust, chips and scraps), moisture content, fiber size and scrap geometry. Tasks aiming at characterization and sorting by size ranges and geometry will be implemented. These activities consider that there is already some pre-classification in the wood manufacturing process: bark, sawdust, sand dust and wood scraps are nowadays stored in different silos after their generation. In consequence, each by-product will have its own sorting and characterization process. In the case of bark, smaller size ranges (<1 mm) will be separated, classified, and stored according to their size and moisture content. Sawdust and sanding dust with specific humidity ranges and small particles (below 1 mm) will be sorted, collected, and labelled for the manufacture of further highvalue products. A Computer Vision based architecture for the automatic detection via video that captures the properties of the wood scraps will be developed. Those properties will mainly be the dimensions of the pieces and its prominent colours. The aim is to create a wood scraps catalogue that will later be used by Generative Design algorithms to create furniture design by using those scraps. The task will develop a camera-based solution and the necessary computer vision algorithms to determine the properties and a database-based solution to create the catalogue of the scraps for posterior use. In a close relationship with WP3, the developed sorting and characterization system will be tuned to provide the highest quality possible of the by-products to manufacture the new products (solutions) in WP6. AI and data analysis will be harnessed to enhance the correlation between the properties and characteristics of the secondary resources with the quality of the final products, thus the creation of digital models for optimisation of recipes for particleboard panels and prediction of final products' quality will be performed. CHIMAR will be performing the secondary resources quality tests and performance evaluation of final products corresponding to different substitution levels (of either adhesives or wood chips), ICCS caries out the data analytics and development of AI-based algorithms. Solutions will be demonstrated in WP6. Output: D5.4, D5.5
T5.4 Fingerprints and markers for recycled material content M6-M42 (CTB, CIRCE): will perform lab-scale PET recycle in controlled environment (small scale filament extrusion trials) and look for 'fingerprints' that distinguish recycled material from virgin (characterized with advanced chemical methods). The goal is to identify at least one such 'fingerprint' that will be used to test recycled PET material from the project activities. Criteria to define a 'good fingerprint' are that it exists of at least one very specific degradation product or a specific combination of non-compatible additives. Two types of products will be considered: i)degradation products from recycling: the development of such degradation molecules as function of repeated extrusion cycles will be studied and heat treatments will be evaluated; ii)fingerprints based on additives: focus on coextrusion polymers, often used in packaging foils, but other typical additives as chain extenders or crystallizing agents should also be considered, but other typical additives as chain extenders or crystallizing agents should also be considered. Large volume dynamic headspace and gas chromatography coupled to mass spectrometry (DHS/GC-MS) and sensitive Pyrolysis-gas chromatography (Py-GC) will be used since these techniques are the most sensitive techniques for these analyses. Since the tests are only valid when high specificity can be achieved, fairly large amounts of virgin materials will be tested to show the lack of the fingerprints in the original materials (specificity of the marker for recycled material) and recycled materials to calculate the sensitivity of the test for identification of recycled content. The goal is to find a few markers with sensitivity of >40% and specificity of close to 100%. The combination of these markers will give a combined sensitivity to evaluate the recycled content of most of the recycled materials from project activities. Using fingerprints, commercially available offered products claiming

recycled PET content (granulates, actual textile products) will be tested. Solutions will be demonstrated in WP6 (Belgium). Output: D5.6, D5.7

Objectives

● Map and assess current situation on the 4 demonstrators ● establish proteins recovery system for wood-resin composite production and wood by-products sorting andcharacterization ● establish fingerprints and markers for recycled material content in polyester and plastic parts and components analysis and origin ● operate and track and trace the new solutions
T5.5 Plastic parts and components analysis and origin for reuse optimisation M6-M42 (DGS, SIGIT, UVQ): is focused on the detection of (i) flakes in pure polymers, not previously filtered, increasing the efficiency of the process (target is to increase recycling 8-9% of the total mass of polymers in weight, five polymers are considered, being PP, ABS, HIPS, PA, PC-ABS); (ii) flakes with reusable compounds, by identifying their main composition (target is to increase recycling 5-7% of total weight of polymers); (iii) flakes with pure polymers added with substances with high molecular weight to prepare input for chemical recycling process by pyrolysis and/or metanalysis (near future). This will result in an overall reduction of the wasted mass to be incinerated and of dangerous substances in the ashes. Initially a classification of a fair quantity of flakes, after the waste grinding and primary separation process, to detect and identify the chemical composition of each flake, will be performed. Due to the non-homogeneous age of the input waste (WEEE as main source), a chemical analysis will be performed to detect the presence of unusable substances and pigments. ML techniques will be designed to monitor the process parameters, i.e., physical, thermal and rheological data. SIGIT will provide its injection moulding know-how, requirements, input data, and support for the validation phase. UVQ will develop novel data-driven techniques for optimising processes and recipes and predictive analysis for the injection moulding control. Using information of the physical process the value functions to be optimised will be determined based on cause-effect experience, lessons learnt, historic database, machines-moulds and raw material technical data, etc. Optimal experiment design will be used to derive a relevant dataset for supervised learning techniques to be used. UVQ will exploit Control Theory and ML (e.g. Regression Trees, Random Forests, Kernel regression, Support Vector Machines, Neural Networks) and novel techniques that combine Optimal Control, System Identification and ML for comparing and improving them in accordance with related KPIs. DGS will integrate the CLSC tool developed in T3.4. Solutions will be demonstrated in WP6 (Italy) (subcontracting involved). Output: D5.8, D5.9

Work package WP6 – Piloting and Demonstration

● evaluate the performance and the impact of all processes and develop plan for upscaling.

Work Package Number	WP6	**Lead Beneficiary**	2. NTUA
Work Package Name	Piloting and Demonstration		
Start Month	6	**End Month**	48
Description			

T6.1 Initial assessment (waste streams mapping) and systems design per demo M6-M9 (CIRCE, UBRUN, ICCS, all WP participants): an initial assessment of waste streams will be performed for the foreseen piloting and demonstration activities including: i)collection of waste fractions involved in the circular value chains (characterized in T5.1) ii)drafting of maps of waste generation and collection points (fractions, content, etc.) in each involved value chain iii)analysis of the data of transportation, final treatment and alignment with the waste hierarchy iv)assessment of material efficiency and sorting degrees v)identification of gaps and best practices for the involved waste fractions. Baseline scenario will identify and map potential hotspots and key drivers/causes impacting value chains' circularity and sustainability, aiming their optimisation. T6.1 will focus on the evaluation of production quality, generated costs, waste, emissions, energy consumption and raw materials use, employees and community protection, product safety and compare initial and optimised chains. Output: D6.1

T6.2 Establishment of the new solutions in the wood based value chains M9-M48 (NTUA, CHIMAR, AgroInvest, MXS, TECNL, ASTI): Greek case (combining also plastic): Two new wood-based composites will be developed; plywood produced with bio-based adhesives and particleboards produced with the addition of wood waste. Proteins, developed in Task 5.2, will be used by CHIMAR as building blocks for replacing part of phenol in phenolic resins, using condensation polymerization. Properties of the bio-based resins, such as rheology, reactivity, free formaldehyde, etc. will be evaluated. The resins will be then mixed with various additives, such as fillers, hardeners/catalysts, etc., to produce woodresin composite materials for developing plywood panels. Recipes (dosage of adhesives, wood, additives) and various parameters (e.g. press temperature, press time) will be optimised and the properties of the produced plywood panels will be tested and evaluated, according to the prevailing European Standards (Bonding quality-EN:314, formaldehyde release-EN ISO 12460-3 etc.) and compared to the properties of panels produced with conventional binders. In addition, moisture characterization and size sorting (T3.1,3.2) will be applied for the selected wood residues for the development of particleboards. Sorted woodchips will be mixed with fresh wood chips to develop particleboards, using a conventional Urea-Formaldehyde resin, as adhesive. Evaluation tests will be performed at the lab and pilot facilities of CHIMAR and the properties of the produced particleboards will be tested, according to the prevailing European Standards (tensile strength-EN:319, thickness swelling after immersion in water-EN:317, modulus of elasticity-EN:310 etc.). The performance of the produced boards will be compared with that of reference conventional particleboards. Both the developed wood-based composites, will be evaluated by MXS in the furniture industry. The AI-based algorithms of T5.3 and the Smart tags of T3.3 will also be applied. Spanish case: Moisture characterization and size sorting (T3.1, T3.2) will be applied for the selected by-product streams (bark, sawdust, woodchips) for the development of particleboards, fiberboards or insulation foams. Wood-based boards and foams will be manufactured and characterized: particle boards with low formaldehyde emission (containing small-size bark particles as formaldehyde scavengers); high-quality MDF and/or HDF panels (manufactured with small fiber sizes); fiber-based insulation products (manufactured with small fiber sizes); self-supporting wood surfaces (manufactured with scraps). Recipes (dosage of particles/fiber, binder, additives) and a series of processing steps and parameters (gluing, mat forming or press cycles, temperature and time) will be optimised and the new products will be upscaled at TECNL facilities including intermediate steps (pressurised refiner, homogeniser, cyclone, hot-press etc) and optimized parameters for different products. Also, wood scraps will be sorted (T3.2) to produce self-supported mosaic surfaces, using the catalogue of existing scraps' characteristics (shape, size, colour, etc.) and automatically generative design and applying optimisation to minimise the production of new byproducts. The mathematical model (T5.3) will create the wooden surface layout and the gluing method will be analysed by comparing the behaviour of various glue types and physical joints. Depending on their physical properties, final surfaces will be used as tabletops, cladding or decorative surfaces. All wood-based products will be traced according to European EN Standards (density, moisture content, moisture resistance, formaldehyde emission, bending strength, internal bond, water absorption, thickness swelling, thermal conductivity). Output: D6.2 D6.3 D6.4

T6.3 Establishment of the new solutions in the plastic based value chains M9-M48 (CTB, CIRCE, SIOEN, DGS, SIGIT, UVQ, EUPC): Belgian case: a polyester textile will be developed starting from recycled materials coming from textile and packaging waste. The polyester waste will be sorted, monitored and further processed. Effects of selected pre-treatment processes will be tested as well as raw material feed in order to guarantee the polymer properties of textile extrusion. The appropriate recycling/upcycling route will be correlated to the extent of degradation of materials for recycling (from T5.4). Two main approaches and their combinations will be examined (restabilization and chain extension). For low degradation degrees, additives blends optimized for recycling will apply restabilization based on the concentration of the active stabilizers remaining in the material for recycling. A number of different additive formulations will be screened to minimize the downgrading of MW and properties. For higher degradation degrees, multifunctional compounds will be used as chain extenders to induce "re-building" in a meltbased process (reactive extrusion) varying critical parameters (amount, residence time). The properties of the produced recyclates will be characterized with conventional and SoA techniques (i.e. DSC and TGA, ATR-FTIR, GPC, tensile tests, DMA) in collaboration also with

Objectives

● Contribute to standardization and certification of secondary materials and digital technologies● co-design training activities and lifelong learning programmes on digital skills
● identify and study the best practices
● develop replication guidelines and methodology with technical and practical information to replicate a successful case● develop exploitation activities and business models

T5.4. The developed recycled PET compounds will be used for multifilament extrusion, using different filament titers (60 –600 dtex). The filament properties will be evaluated to guarantee a successful further reprocessing into woven or knitted textiles. A full evaluation of the textile properties will be performed and their compliance to for instance Oekotex regulation will be tested. Through the processing of data regarding: (a) the properties of the recycled polymer pellets (b) the production parameters and characteristics of the polyester fibers and (c) the final properties of the textile by SIOEN, the possibility of developing a digital model will be explored, in order to identify the characteristics of the recycled polyester resin raw material as well as the textile process parameters. The data from the various processes will be combined and introduced into a data management system and through application of IT tools (artificial intelligence, machine learning etc.) knowledge will be extracted related to the optimum processes and conditions. Moreover, the use of functional tags for sensing humidity and temperature will be tested by VTT on PET packaging foils. Simulations of different storages conditions will be tested and linked with the processing into textiles making it possible to collect relevant date on the history of the recycled polyester to predict their re-use in high demanding textile application and predict the needed reprocessing. Italian case: A first testing will be performed in SIGIT facilities with a 100% predetected quantity of flakes, selecting and create mixes of different materials (already uploaded on IMDS) as reference, by E&D, central Lab and facility teams and supported by ICCS and IRIS for sorting and sensing established in a recycling plant, processing high volumes per year under SIGIT supervision and providing recycled raw materials for SIGIT injection moulding. Then detection and separation of input flakes in the production line will be addressed, with a wide panel of mix of materials. SIGIT will support the detection tuning and activate a chemical analysis by its laboratory to enable a stable and reliable detection of potential contaminations reducing sensors' efficiency. A statistical analysis will be carried out afterwards, based on DGS optimization module (T3.4) to perform analytic evaluation (up to 400.000 cycles/hour) for increase and feed the AI models of sensors (T3.1). The end-to-end supply chain management platform running in SIGIT (T5.5) will be extended by DGS (subcontracting involved), integrating the UVQ AI functionalities and interfacing it to SIGIT injection press machine and other fundamental service devices of the injection process (i.e., mould, mould's thermos-conditioning systems, hot runner systems, etc.). Initially, SIGIT will simulate normal production and faulty conditions. Design and data collection, physical, thermal and rheological data will be acquired with the largest redundancy (i.e. using all measurable data and the smallest achievable sampling time). A selection of sensors and sampling times will be performed to derive a non-redundant dataset that will be used for the supervised learning algorithms. In a later phase the system will switch from passive data acquisition to active parameters correction. The validation will be performed through extended tests monitoring several hundred cycles. Output: D6.2 D6.3 D6.4 T6.4 Evaluation and impact assessment M20-M48 (CIRCE, NTUA, CIRCE, UBRUN, TECNL, DGS, ICCS): comprises the real scale operation of the demonstrators, evaluation and impact assessment of the different value chains. T6.4 will use the generated data of WPs 2,3,4 to demonstrate their importance, sufficiency and adequateness to both assess and optimise circularity of value chains. The circularity and sustainability performance will be compared with T6.1 results. Alternative scenarios will be tested for predictions of circularity performance and emissions reduction that would enable a further optimisation of the value chain in the future. Also, an LCC-based dynamic economic feasibility analysis will be performed in value chains, from waste streams to final products. It will be flexible assessing how main parameters and considerations (e.g. logistic operation, process technology, final use of the product, etc.) can influence economic feasibility balance of each stakeholder and global balance of the value chain. The results will support

Work Package Number	WP7	**Lead Beneficiary**	8. MERIT
Work Package Name	Guidance, Training, Cooperation, Exploitation and communication		
Start Month	1	**End Month**	48

replication and exploitation. A full Environmental and Circularity index for each unit of product will be produced at the end of life cycle of each end product-process. Enabled by Blockchain capabilities the necessary information will be delivered to players within the Value Chain to make best choices of sourcing raw material, identifying the best logistics and assembly processes to produce the end product. Output: D6.4

Work package WP7 – Guidance, Training, Cooperation, Exploitation and communication

- perform IPR Management
- develop DigInTraCE dissemination and communication activities.

Description

T7.1 Standardization and certification schemes M9-M48 (DS, CTB, TECNL, ICCS, NTUA, UST, SIGIT, MXS, TECNL, EUPC): aims to support Secondary Material, AI and Blockchain standardization. To fulfil the objectives of T7.1, for Secondary Materials two specific mutually linked activities will be carried out: an analysis of the applicable standardization landscape and the contribution to the ongoing and future standard developments. T7.1 will provide starting point for other WPs, ensure compatibility and interoperability, and link with what already exists in the market through standards. Relevant existing standards and standardization related activities such as the Circular Plastics Alliance, will be taken into consideration to create a synergy with EU-strategies on secondary materials. An overview of standardization efforts and a liaison to the most relevant committees in CEN and ISO will be made, e.g. CEN/ TC248 WG39 'Circular Textiles Chain' (which is chaired by CTB) and the CEN SABE workgroup mapping standards influencing Circular Economy. Collaboration with the relevant standardization Technical Committee(s) will be set up to ensure that results and knowledge is integrated into ongoing and future standardization developments. DigInTraCE will use the standardization system as a tool for dissemination of the project results and interaction with the market stakeholders. Also, AI and Blockchain standardization activities will be performed, aiming to facilitate the acceptance and utilisation of the developed solutions by the market. Standardization acts as a catalyst and builds trust in new technologies and innovations as it is a formal and transparent process. T7.1 will contribute to ongoing and future developments in the specific topics, promoting the inclusion of the findings of the project in new or revised standards that can be used by the European or international industry. Depending on the maturity of the technologies and the sector, this can involve providing information to standardization Technical Committees, mutual knowledge sharing, submitting technical proposals and even promoting the elaboration of new standard documents. DS holds the secretariats of CEN-CENELEC Joint Technical Committee 21 Artificial Intelligence and ISO/TC 307 WG 5 Blockchain and distributed ledger technologies – Governance, hereby being a frontrunner in standardization in trustworthy AI systems and Blockchain governance. As for certification schemes, it will be investigated how the digital tracing explored within the project can be useful for related activities. CTB as described in 1.2.3 will combine digital tracking with fingerprinting (T5.4) to do a certification based on measurements of end product recycled content of the end product, potentially extending the existing PolyCertEurope scheme for PET and other polymers and results serving as input to standardization. In parallel, TNLCERT will define certification scheme for subproducts types and intended uses. The applicable specifications will be determined for each case, incorporating environmental and functional requirements, be shared and agreed with consortium and interested parties. The applicable scheme will be determined and evaluation activities as well as technical competence requirements applicable to their personnel will be defined. A pilot application of the scheme will be developed. Outputs: D7.1, D7.2

T7.2 Co-design training activities and lifelong learning programmes on digital skills, replication and cooperation M9M48 (UBRUN, ICCS): will summarise the main results, the lesson learnt and the best practices along the project for knowledge documentation and sharing. Aligned with EU digital compass this task will contribute to the development of digitally skilled population and workforce. For this purpose, upskilled pathways for the employees and the citizens will be developed focusing on the identification of digital skills required due to current status and thus development and provision of tailored learning offers. Training will include non-formal and informal learning by utilizing internal and external experts. The progress will be tracked and the acquired skills will be validated frequently to improve guidance, training and mentoring services and based on the special regional needs of the value chains. Solutions demonstrated in the project along with business models of T7.3 will be used for the creation replication packages (RPs) specialized into specific sectoral/regional/national. For clustering and cooperation they will be investigated synergies with projects and initiatives, funds, research centres, experts along the innovation chain. Task is linked to T7.4 contributing to dissemination while performing a best practice exchange exercise among projects and relevant EU/national initiative.
Outputs: D7.3, D7.4

T7.3 Exploitation, Business models and IPR Management M1-M48 (MERIT, UBRUN): the consortium under the guidance of MERIT will conduct a series of activities towards the definition and refinement of a plan for the collective as well as partner-specific exploitation of the project's tangible and intangible results. It includes: i)early definition of the project's exploitation strategy focusing towards on maximising societal, commercial and scientific impact and with specific focus on cross-sectoral collaboration and industrialization of the solutions ii) identification of the project's tangible and intangible results and the relevant consortium partners (or clusters of them) responsible for their exploitation; iii) the definition of target markets and the relevant actors and stakeholders for the marketization of project outputs; iv) definition of monetization routes for project results; and iv) preparation of collective and individual exploitation roadmaps defining roles, responsibilities, milestones and IPR ownership. In support of these activities, the consortium will actively pursue and benefit from all the relevant tools and services provided by European Commission and its partners such as the European IPR helpdesk and Horizon Results Booster Platform to address non-technological

exploitation issues and support SME partners towards commercialization of their results. The foreseen high-level educational activities and financial services towards circular economy stakeholders will be closely linked to the project exploitation activities and will act as yet another source of valuable feedback regarding market and customer needs that could be addressed by the novel DigInTraCE offerings. Finally, a preliminary business innovation plan consolidates the value proposition of the project results while being supported by relevant customer side cost-benefit analysis and financial plans addressing pricing strategies and cost revenue models. Outputs: D7.5, D7.6, D7.7 T7.4 Dissemination and communication activities, policy recommendations M1-M48 (ICCS, All partners): A Communication and Dissemination Strategy will be developed in order to present the project's concept and outcomes to the general public and to address a wide audience, while the appropriate communication and dissemination channels will be determined. DigInTraCE will develop 2 policy briefs for sectors of focus: pulp and paper, chemicals and plastics (Output D.7.5) to present policy recommendations. An official DigInTraCE website (M3) to serve as the main gateway for all interested parties and social media accounts will be developed (M1). Social media campaigns will be held on important milestones. A stakeholders' hub will be established (M6) to involve different actors of the supply chains. By project's end a final conference will be organised. Outputs: D7.8, D7.9, D7.10 , D7.11

LIST OF DELIVERABLES

Deliverables

Grant Preparation (Deliverables screen) — Enter the info.

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D1.1	DigInTraCE Project management handbook, QA, ethics and legal framework and reports on innovation and data management v1	WP1	1 - ICCS	R — Document, report	SEN - Sensitive	6
D1.2	DigInTraCE Project management handbook, QA, ethics and legal framework and reports on innovation and data management v2	WP1	1 - ICCS	R — Document, report	SEN - Sensitive	18
D1.3	DigInTraCE Project management handbook, QA, ethics and legal framework and reports on innovation and data management v3	WP1	1 - ICCS	R — Document, report	SEN - Sensitive	36
D1.4	DigInTraCE Project management handbook, QA, ethics and legal framework and reports on innovation and data management FINAL	WP1	1 - ICCS	R — Document, report	SEN - Sensitive	48
D1.5	Data Management Plan v1	WP1	1 - ICCS	DMP — Data Management Plan	SEN - Sensitive	6

D1.6	Data Management Plan v2	WP1	1 - ICCS	DMP — Data Management Plan	SEN - Sensitive	26
D1.7	Data Management Plan FINAL	WP1	1 - ICCS	DMP — Data Management Plan	SEN - Sensitive	48
D2.1	Methodology of Traceability v1	WP2	21 - UBRUN	R — Document, report	PU - Public	8

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D2.2	Methodology of Traceability v2	WP2	21 - UBRUN	R — Document, report	PU - Public	24
D2.3	Methodology of Traceability FINAL	WP2	21 - UBRUN	R — Document, report	PU - Public	48
D2.4	Digital Product Passport Concept v1	WP2	1 - ICCS	R — Document, report	PU - Public	8
D2.5	Digital Product Passport Concept FINAL	WP2	1 - ICCS	R — Document, report	PU - Public	24
D2.6	Real-time information and secure data exchange requirements v1	WP2	1 - ICCS	R — Document, report	SEN - Sensitive	8
D2.7	Real-time information and secure data exchange requirements v2	WP2	1 - ICCS	R — Document, report	PU - Public	24

D2.8	Real-time information and secure data exchange requirements FINAL	WP2	1 - ICCS	R — Document, report	SEN - Sensitive	48
D2.9	Specifications definition and architecture of digital tracing v1	WP2	3 - UST	R — Document, report	PU - Public	18
D2.10	Specifications definition and architecture of digital tracing FINAL	WP2	3 - UST	R — Document, report	PU - Public	28
D3.1	Novel Sensing technologies v1	WP3	6 - IRIS	R — Document, report	PU - Public	18
D3.2	Novel Sensing technologies FINAL	WP3	6 - IRIS	R — Document, report	PU - Public	36
D3.3	Sorting mechanisms for secondary streams valorisation v1	WP3	1 - ICCS	R — Document, report	PU - Public	18

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D3.4	Sorting mechanisms for secondary streams valorisation FINAL	WP3	1 - ICCS	R — Document, report	PU - Public	36
D3.5	Smart Tags v1	WP3	7 - VTT	R — Document, report	PU - Public	18
D3.6	Smart Tags FINAL	WP3	7 - VTT	R — Document, report	PU - Public	36

D3.7	CLSC tool v1	WP3	17 - DGS	OTHER	SEN - Sensitive	18
D3.8	CLSC tool FINAL	WP3	17 - DGS	OTHER	SEN - Sensitive	36
D3.9	Online LCA computation tool based on Ex-Ante LCA v1	WP3	4 - CIRCE	OTHER	SEN - Sensitive	18
D3.10	Online LCA computation tool based on Ex-Ante LCA FINAL	WP3	4 - CIRCE	OTHER	SEN - Sensitive	42
D4.1	Dynamic Product Passport for Unit of Product through the value chain v1	WP4	1 - ICCS	OTHER	SEN - Sensitive	18
D4.2	Dynamic Product Passport for Unit of Product through the value chain FINAL	WP4	1 - ICCS	OTHER	SEN - Sensitive	42
D4.3	Blockchain and Distributed ledger v1	WP4	3 - UST	OTHER	SEN - Sensitive	18
D4.4	Blockchain and Distributed ledger FINAL	WP4	3 - UST	OTHER	SEN - Sensitive	42
D4.5	Financial tool v1	WP4	9 - HTECH	OTHER	PU - Public	18
D4.6	Financial tool FINAL	WP4	9 - HTECH	OTHER	PU - Public	42

Deliverables

Grant Preparation (Deliverables screen) — Enter the info.

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)

D4.7	MR enabler tool v1	WP4	1 - ICCS	OTHER	PU - Public	18
D4.8	MR enabler tool FINAL	WP4	1 - ICCS	OTHER	SEN - Sensitive	42
D4.9	DigInTraCE platform and system integration v1	WP4	1 - ICCS	OTHER	SEN - Sensitive	18
D4.10	DigInTraCE platform and system integration FINAL	WP4	1 - ICCS	OTHER	SEN - Sensitive	48
D5.1	Secondary materials quality characterization for recycle,reuse upgrade	WP5	4 - CIRCE	R — Document, report	PU - Public	8
D5.2	Proteins recovery for wood-resin composite production v1	WP5	2 - NTUA	R — Document, report	PU - Public	18
D5.3	Proteins recovery for wood-resin composite production FINAL	WP5	2 - NTUA	R — Document, report	PU - Public	42
D5.4	Wood by-products sorting and characterization for process optimisation v1	WP5	13 - TECNL	R — Document, report	PU - Public	18
D5.5	Wood by-products sorting and characterization for process optimisation FINAL	WP5	13 - TECNL	R — Document, report	PU - Public	42
D5.6	Fingerprints and markers for recycled material content v1	WP5	15 - CTB	R — Document, report	PU - Public	18

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D5.7	Fingerprints and markers for recycled material content FINAL	WP5	15 - CTB	R — Document, report	PU - Public	42
D5.8	Plastic parts and components analysis and origin for reuse optimisation v1	WP5	17 - DGS	R — Document, report	PU - Public	18
D5.9	Plastic parts and components analysis and origin for reuse optimisation FINAL	WP5	17 - DGS	R — Document, report	PU - Public	42
D6.1	Waste streams mapping	WP6	4 - CIRCE	R — Document, report	PU - Public	9
D6.2	Systems design and description v1	WP6	2 - NTUA	R — Document, report	PU - Public	24
D6.3	Systems design and description FINAL	WP6	2 - NTUA	R — Document, report	PU - Public	36
D6.4	Evaluation and Impact assessment	WP6	1 - ICCS	R — Document, report	PU - Public	48
D7.1	Standardization of novel technologies and certification schemes v1	WP7	5 - DS	R — Document, report	PU - Public	24
D7.2	Standardization of novel technologies and certification schemes FINAL	WP7	5 - DS	R — Document, report	PU - Public	48

D7.3	Co-design training activities and lifelong learning programmes on digital skills and replication v1	WP7	21 - UBRUN	R — Document, report	PU - Public	24
D7.4	Co-design training activities and lifelong learning programmes on digital skills and replication FINAL	WP7	21 - UBRUN	R — Document, report	PU - Public	48

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EU classified —RESTREINT-UE/EU-RESTRICTED, CONFIDENTIEL-UE/EU-CONFIDENTIAL, SECRET-UE/EU-SECRET under Decision 2015/444

Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D7.5	Exploitation and Business Innovation Plan and activities v1	WP7	8 - MERIT	R — Document, report	PU - Public	6
D7.6	Exploitation and Business Innovation Plan and activities v2	WP7	8 - MERIT	R — Document, report	PU - Public	24
D7.7	Exploitation and Business Innovation Plan and activities FINAL	WP7	8 - MERIT	R — Document, report	PU - Public	48
D7.8	Communication and Dissemination plan and activities v1	WP7	1 - ICCS	R — Document, report	PU - Public	6
D7.9	Communication and Dissemination plan and activities v2	WP7	1 - ICCS	R — Document, report	PU - Public	24

D7.10	Communication and Dissemination plan and activities FINAL	WP7	1 - ICCS	R — Document, report	PU - Public	48
D7.11	Policy briefs	WP7	1 - ICCS	R — Document, report	PU - Public	45

Deliverable D1.1 – DigInTraCE Project management handbook, QA, ethics and legal framework and reports on innovation and data management v1

Deliverable Number	D1.1	**Lead Beneficiary**	1. ICCS
Deliverable Name	DigInTraCE Project management handbook, QA, ethics and legal framework and reports on innovation and data management v1		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	6	**Work Package No**	WP1
Description			
An internal document setting the basis for the project management, the communication channels and methods, the periodicity of reporting, gender and ethics issues to serve as a reference point during the duration of the project. Related Task(s): T1.1, T1.2, T1.3			

Deliverable D1.2 – DigInTraCE Project management handbook, QA, ethics and legal framework and reports on innovation and data management v2

Deliverable Number	D1.2	**Lead Beneficiary**	1. ICCS
Deliverable Name	DigInTraCE Project management handbook, QA, ethics and legal framework and reports on innovation and data management v2		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	18	**Work Package No**	WP1
Description			
An internal document setting the basis for the project management, the communication channels and methods, the periodicity of reporting, gender and ethics issues to serve as a reference point during the duration of the project. Related Task(s): T1.1, T1.2, T1.3			

Deliverable D1.3 – DigInTraCE Project management handbook, QA, ethics and legal framework and reports on innovation and data management v3

Deliverable Number	D1.3	**Lead Beneficiary**	1. ICCS
Deliverable Name	DigInTraCE Project management handbook, QA, ethics and legal framework and reports on innovation and data management v3		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP1
Description			
An internal document setting the basis for the project management, the communication channels and methods, the periodicity of reporting, gender, and ethics issues to serve as a reference point during the duration of the project. Related Task(s): T1.1, T1.2, T1.3			

Deliverable D1.4 – DigInTraCE Project management handbook, QA, ethics and legal framework and reports on innovation and data management FINAL

Deliverable Number	D1.4	**Lead Beneficiary**	1. ICCS

Deliverable Name	DigInTraCE Project management handbook, QA, ethics and legal framework and reports on innovation and data management FINAL		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	48	**Work Package No**	WP1
Description			
An internal document setting the basis for the project management, the communication channels and methods, the periodicity of reporting, gender and ethics issues to serve as a reference point during the duration of the project. Related Task(s): T1.1, T1.2, T1.3			

Deliverable D1.5 – Data Management Plan v1

Deliverable Number	D1.5	**Lead Beneficiary**	1. ICCS
Deliverable Name	Data Management Plan v1		
Type	DMP — Data Management Plan	**Dissemination Level**	SEN - Sensitive
Due Date (month)	6	**Work Package No**	WP1
Description			
Document including: Data sets related to secondary materials identification and characterization and their descriptions; Standards and metadata; Data sharing, maintenance and storage procedures. Related Task(s):T1.2			

Deliverable D1.6 – Data Management Plan v2

Deliverable Number	D1.6	**Lead Beneficiary**	1. ICCS
Deliverable Name	Data Management Plan v2		
Type	DMP — Data Management Plan	**Dissemination Level**	SEN - Sensitive
Due Date (month)	26	**Work Package No**	WP1
Description			
Document including: Data sets related to secondary materials identification and characterization and their descriptions; Standards and metadata; Data sharing, maintenance and storage procedures. Related Task(s): T1.2			

Deliverable D1.7 – Data Management Plan FINAL

Deliverable Number	D1.7	**Lead Beneficiary**	1. ICCS
Deliverable Name	Data Management Plan FINAL		
Type	DMP — Data Management Plan	**Dissemination Level**	SEN - Sensitive
Due Date (month)	48	**Work Package No**	WP1
Description			

Document including: Data sets related to secondary materials identification and characterization and their descriptions; Standards and metadata; Data sharing, maintenance and storage procedures. Related Task(s): T1.2

Deliverable D2.1 – Methodology of Traceability v1

Deliverable Number	D2.1	**Lead Beneficiary**	21. UBRUN
Deliverable Name	Methodology of Traceability v1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	8	**Work Package No**	WP2
Description			
Document defines a complete strategy for discriminating the level of tracking & tracing materials and methods to acquire associated data, with special attention to digital representation of unit of product in compliance with blockchain technology. Related Task(s):T2.1			

Deliverable D2.2 – Methodology of Traceability v2

Deliverable Number	D2.2	**Lead Beneficiary**	21. UBRUN
Deliverable Name	Methodology of Traceability v2		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	24	**Work Package No**	WP2
Description			
Document defines a complete strategy for discriminating the level of tracking & tracing materials and methods to acquire associated data, with special attention to digital representation of unit of product in compliance with blockchain technology. Related Task(s): T2.1			

Deliverable D2.3 – Methodology of Traceability FINAL

Deliverable Number	D2.3	**Lead Beneficiary**	21. UBRUN
Deliverable Name	Methodology of Traceability FINAL		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP2
Description			
Document defines a complete strategy for discriminating the level of tracking & tracing materials and methods to acquire associated data, with special attention to digital representation of unit of product in compliance with blockchain technology. Related Task(s): T2.1			

Deliverable D2.4 – Digital Product Passport Concept v1

Deliverable Number	D2.4	**Lead Beneficiary**	1. ICCS
Deliverable Name	Digital Product Passport Concept v1		
Type	R — Document, report	**Dissemination Level**	PU - Public

Due Date (month)	8	**Work Package No**	WP2
Description			
Document describes the required data captured and incorporated in DPPs in accordance with existing product information schemes and regulations, as well as a list of DPP features for the corresponding system's architecture. Related Task(s): T2.2			

Deliverable D2.5 – Digital Product Passport Concept FINAL

Deliverable Number	D2.5	**Lead Beneficiary**	1. ICCS
Deliverable Name	Digital Product Passport Concept FINAL		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	24	**Work Package No**	WP2
Description			
Document describes the required data captured and incorporated in DPPs in accordance with existing product information schemes and regulations, as well as a list of DPP features for the corresponding system's architecture. Related Task(s): T2.2			

Deliverable D2.6 – Real-time information and secure data exchange requirements v1

Deliverable Number	D2.6	**Lead Beneficiary**	1. ICCS
Deliverable Name	Real-time information and secure data exchange requirements v1		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	8	**Work Package No**	WP2
Description			
Deliverable analyses data sharing techniques as a set of required data and related requirements, in accordance to FAIR data management for project's partners and external stakeholders. Related Task(s): T2.3			

Deliverable D2.7 – Real-time information and secure data exchange requirements v2

Deliverable Number	D2.7	**Lead Beneficiary**	1. ICCS
Deliverable Name	Real-time information and secure data exchange requirements v2		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	24	**Work Package No**	WP2
Description			
Deliverable analyses data sharing techniques as a set of required data and related requirements, in accordance to FAIR data management for project's partners and external stakeholders. Related Task(s): T2.3			

Deliverable D2.8 – Real-time information and secure data exchange requirements FINAL

Deliverable Number	D2.8	**Lead Beneficiary**	1. ICCS
Deliverable Name	Real-time information and secure data exchange requirements FINAL		

Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	48	**Work Package No**	WP2
Description			
Deliverable analyses data sharing techniques as a set of required data and related requirements, in accordance to FAIR data management for project's partners and external stakeholders. Related Task(s): T2.3			

Deliverable D2.9 – Specifications definition and architecture of digital tracing v1

Deliverable Number	D2.9	**Lead Beneficiary**	3. UST
Deliverable Name	Specifications definition and architecture of digital tracing v1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP2
Description			
Presents produced technical specifications of the Decentralized Traceability platform architecture and the modules to be developed, including access rights, confidentiality issues and user roles. Related Task(s): T2.4			

Deliverable D2.10 – Specifications definition and architecture of digital tracing FINAL

Deliverable Number	D2.10	**Lead Beneficiary**	3. UST
Deliverable Name	Specifications definition and architecture of digital tracing FINAL		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	28	**Work Package No**	WP2
Description			
Presents produced technical specifications of the Decentralized Traceability platform architecture and the modules to be developed, including access rights, confidentiality issues and user roles. Related Task(s): T2.4			

Deliverable D3.1 – Novel Sensing technologies v1

Deliverable Number	D3.1	**Lead Beneficiary**	6. IRIS
Deliverable Name	Novel Sensing technologies v1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP3
Description			
Defines real-time monitoring requirements for each demonstrator and describes each system's final design, development, and validation procedure. Related Task(s): T3.1			

Deliverable D3.2 – Novel Sensing technologies FINAL

Deliverable Number	D3.2	**Lead Beneficiary**	6. IRIS
Deliverable Name	Novel Sensing technologies FINAL		

Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP3
Description			
Defines real-time monitoring requirements for each demonstrator and describes each system's final design, development, and validation procedure. Related Task(s): T3.1			

Deliverable D3.3 – Sorting mechanisms for secondary streams valorisation v1

Deliverable Number	D3.3	**Lead Beneficiary**	1. ICCS
Deliverable Name	Sorting mechanisms for secondary streams valorisation v1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP3
Description			
Document presents a set of requirements and associated technical specifications for sorting and describes the final sorting system's design and development for each value chain and demonstrator. Related Task(s): T3.2			

Deliverable D3.4 – Sorting mechanisms for secondary streams valorisation FINAL

Deliverable Number	D3.4	**Lead Beneficiary**	1. ICCS
Deliverable Name	Sorting mechanisms for secondary streams valorisation FINAL		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP3
Description			
Document presents a set of requirements and associated technical specifications for sorting and describes the final sorting system's design and development for each value chain and demonstrator. Related Task(s): T3.2			

Deliverable D3.5 – Smart Tags v1

Deliverable Number	D3.5	**Lead Beneficiary**	7. VTT
Deliverable Name	Smart Tags v1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP3
Description			
Deliverable exhibits Smart Tags use scenarios, presents tested real-time sensing and identification technologies and related results, approaches to support data integrity and integration, as well as stakeholders' engagement strategies. Related Task(s): T3.3			

Deliverable D3.6 – Smart Tags FINAL

Deliverable Number	D3.6	**Lead Beneficiary**	7. VTT

Deliverable Name	Smart Tags FINAL		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP3
Description			
Deliverable exhibits Smart Tags use scenarios, presents tested real-time sensing and identification technologies and related results, approaches to support data integrity and integration, as well as stakeholders' engagement strategies. Related Task(s): T3.3			

Deliverable D3.7 – CLSC tool v1

Deliverable Number	D3.7	**Lead Beneficiary**	17. DGS
Deliverable Name	CLSC tool v1		
Type	OTHER	**Dissemination Level**	SEN - Sensitive
Due Date (month)	18	**Work Package No**	WP3
Description			
Document includes methodologies and algorithms for the expansion of a planning and scheduling system in order to integrate recipe management logic and AI-based supply chain management from the product to the raw materials. Related Task(s): T3.4			

Deliverable D3.8 – CLSC tool FINAL

Deliverable Number	D3.8	**Lead Beneficiary**	17. DGS
Deliverable Name	CLSC tool FINAL		
Type	OTHER	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP3
Description			
Document includes methodologies and algorithms for the expansion of a planning and scheduling system in order to integrate recipe management logic and AI-based supply chain management from the product to the raw materials. Related Task(s): T3.4			

Deliverable D3.9 – Online LCA computation tool based on Ex-Ante LCA v1

Deliverable Number	D3.9	**Lead Beneficiary**	4. CIRCE
Deliverable Name	Online LCA computation tool based on Ex-Ante LCA v1		
Type	OTHER	**Dissemination Level**	SEN - Sensitive
Due Date (month)	18	**Work Package No**	WP3
Description			

Deliverable describes the module's requirements and the data models created for environmental performance analysis to analyse the impact and contribute to the Product Environmental Footprint (PEF). Related Task(s): T3.5

Deliverable D3.10 – Online LCA computation tool based on Ex-Ante LCA FINAL

Deliverable Number	D3.10	**Lead Beneficiary**	4. CIRCE
Deliverable Name	Online LCA computation tool based on Ex-Ante LCA FINAL		
Type	OTHER	**Dissemination Level**	SEN - Sensitive
Due Date (month)	42	**Work Package No**	WP3
Description			
Deliverable describes the module's requirements and the data models created for environmental performance analysis to analyse the impact and contribute to the Product Environmental Footprint (PEF). Related Task(s): T3.5			

Deliverable D4.1 – Dynamic Product Passport for Unit of Product through the value chain v1

Deliverable Number	D4.1	**Lead Beneficiary**	1. ICCS
Deliverable Name	Dynamic Product Passport for Unit of Product through the value chain v1		
Type	OTHER	**Dissemination Level**	SEN - Sensitive
Due Date (month)	18	**Work Package No**	WP4
Description			
The dynamic update mechanism and overall design of DPP are presented, including a set of features and online data sharing routines in compliance with the whole proposed system's architecture. Related Task(s):T4.1			

Deliverable D4.2 – Dynamic Product Passport for Unit of Product through the value chain FINAL

Deliverable Number	D4.2	**Lead Beneficiary**	1. ICCS
Deliverable Name	Dynamic Product Passport for Unit of Product through the value chain FINAL		
Type	OTHER	**Dissemination Level**	SEN - Sensitive
Due Date (month)	42	**Work Package No**	WP4
Description			
The dynamic update mechanism and overall design of DPP are presented, including a set of features and online data sharing routines in compliance with the whole proposed system's architecture. Related Task(s):T4.1			

Deliverable D4.3 – Blockchain and Distributed ledger v1

Deliverable Number	D4.3	**Lead Beneficiary**	3. UST
Deliverable Name	Blockchain and Distributed ledger v1		
Type	OTHER	**Dissemination Level**	SEN - Sensitive
Due Date (month)	18	**Work Package No**	WP4

Description
Data structure implemented is defined, procedures of in-house and on-site testing of the utilised cloud-based nodes, as well as tokenization techniques for tracing, tracking and financial services are described. Related Task(s):T4.2

Deliverable D4.4 – Blockchain and Distributed ledger FINAL

Deliverable Number	D4.4	**Lead Beneficiary**	3. UST
Deliverable Name	Blockchain and Distributed ledger FINAL		
Type	OTHER	**Dissemination Level**	SEN - Sensitive
Due Date (month)	42	**Work Package No**	WP4
Description			
Data structure implemented is defined, procedures of in-house and on-site testing of the utilised cloud-based nodes, as well as tokenization techniques for tracing, tracking and financial services are described. Related Task(s): T4.2			

Deliverable D4.5 – Financial tool v1

Deliverable Number	D4.5	**Lead Beneficiary**	9. HTECH
Deliverable Name	Financial tool v1		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP4
Description			
Financial services' toolkit designed after an initial review of available relevant marketplaces is described and related features in terms of market needs and pricing decision support on secondary and virgin materials are analyzed. Related Task(s): T4.3			

Deliverable D4.6 – Financial tool FINAL

Deliverable Number	D4.6	**Lead Beneficiary**	9. HTECH
Deliverable Name	Financial tool FINAL		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	42	**Work Package No**	WP4
Description			
Financial services' toolkit designed after an initial review of available relevant marketplaces is described and related features in terms of market needs and pricing decision support on secondary and virgin materials are analyzed. Related Task(s): T4.3			

Deliverable D4.7 – MR enabler tool v1

Deliverable Number	D4.7	**Lead Beneficiary**	1. ICCS
Deliverable Name	MR enabler tool v1		

Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP4
Description			
Functionalities and characteristics of the MR enabled tool, e.g. type of data displayed, are presented, integration to the overall platform is described and demonstrated activities in the pilots are analyzed. Related Task(s): T4.4			

Deliverable D4.8 – MR enabler tool FINAL

Deliverable Number	D4.8	**Lead Beneficiary**	1. ICCS
Deliverable Name	MR enabler tool FINAL		
Type	OTHER	**Dissemination Level**	SEN - Sensitive
Due Date (month)	42	**Work Package No**	WP4
Description			
Functionalities and characteristics of the MR enabled tool, e.g. type of data displayed, are presented, integration to the overall platform is described and demonstrated activities in the pilots are analyzed. Related Task(s): T4.4			

Deliverable D4.9 – DigInTraCE platform and system integration v1

Deliverable Number	D4.9	**Lead Beneficiary**	1. ICCS
Deliverable Name	DigInTraCE platform and system integration v1		
Type	OTHER	**Dissemination Level**	SEN - Sensitive
Due Date (month)	18	**Work Package No**	WP4
Description			
Document illustrates the integration of all the designed tools and modules along with their associated requirements into the Decentralized Traceability platform, as well as describing common data acquisition, storage and update routes. Related Task(s): T4.5			

Deliverable D4.10 – DigInTraCE platform and system integration FINAL

Deliverable Number	D4.10	**Lead Beneficiary**	1. ICCS
Deliverable Name	DigInTraCE platform and system integration FINAL		
Type	OTHER	**Dissemination Level**	SEN - Sensitive
Due Date (month)	48	**Work Package No**	WP4
Description			
Document illustrates the integration of all the designed tools and modules along with their associated requirements into the Decentralized Traceability platform, as well as describing common data acquisition, storage and update routes. Related Task(s): T4.5			

Deliverable D5.1 – Secondary materials quality characterization for recycle,reuse upgrade

Deliverable Number	D5.1	**Lead Beneficiary**	4. CIRCE
Deliverable Name	Secondary materials quality characterization for recycle,reuse upgrade		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	8	**Work Package No**	WP5
Description			
Document analyses secondary materials markets for the four different value chains and pinpoints the recycling/reuse and upcycling quality requirements of each potential valorization pathway. Related Task(s): T5.1			

Deliverable D5.2 – Proteins recovery for wood-resin composite production v1

Deliverable Number	D5.2	**Lead Beneficiary**	2. NTUA
Deliverable Name	Proteins recovery for wood-resin composite production v1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP5
Description			
Extraction techniques, along with associated examined and optimised parameters for achieving maximum production yield will be presented, as well as analytical techniques for evaluation and drying method for obtaining final proteins described. Related Task(s): T5.2			

Deliverable D5.3 – Proteins recovery for wood-resin composite production FINAL

Deliverable Number	D5.3	**Lead Beneficiary**	2. NTUA
Deliverable Name	Proteins recovery for wood-resin composite production FINAL		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	42	**Work Package No**	WP5
Description			
Extraction techniques, along with associated examined and optimised parameters for achieving maximum production yield will be presented, as well as analytical techniques for evaluation and drying method for obtaining final proteins described. Related Task(s): T5.2			

Deliverable D5.4 – Wood by-products sorting and characterization for process optimisation v1

Deliverable Number	D5.4	**Lead Beneficiary**	13. TECNL
Deliverable Name	Wood by-products sorting and characterization for process optimisation v1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP5
Description			

Deliverable involves characterization and classification requirements, in terms of by-products being directed towards specific valorization, and data analysis correlating secondary resource' properties with the final products' quality. Related Task(s): T5.3

Deliverable D5.5 – Wood by-products sorting and characterization for process optimisation FINAL

Deliverable Number	D5.5	**Lead Beneficiary**	13. TECNL
Deliverable Name	Wood by-products sorting and characterization for process optimisation FINAL		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	42	**Work Package No**	WP5
Description			
Deliverable involves characterization and classification requirements, in terms of by-products being directed towards specific valorization, and data analysis correlating secondary resource' properties with the final products' quality. Related Task(s): T5.3			

Deliverable D5.6 – Fingerprints and markers for recycled material content v1

Deliverable Number	D5.6	**Lead Beneficiary**	15. CTB
Deliverable Name	Fingerprints and markers for recycled material content v1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP5
Description			
Fingerprints distinguishing recycled from virgin material, identified for either degraded molecules or additives, along with testing conditions is described to provide a commercially available recycled PET content methodology. Related Task(s): T5.4			

Deliverable D5.7 – Fingerprints and markers for recycled material content FINAL

Deliverable Number	D5.7	**Lead Beneficiary**	15. CTB
Deliverable Name	Fingerprints and markers for recycled material content FINAL		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	42	**Work Package No**	WP5
Description			
Fingerprints distinguishing recycled from virgin material, identified for either degraded molecules or additives, along with testing conditions is described to provide a commercially available recycled PET content methodology. Related Task(s):T5.4			

Deliverable D5.8 – Plastic parts and components analysis and origin for reuse optimisation v1

Deliverable Number	D5.8	**Lead Beneficiary**	17. DGS
Deliverable Name	Plastic parts and components analysis and origin for reuse optimisation v1		

Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP5
Description			
Identification of the chemical composition and classification of polymeric flakes is analysed, machine learning techniques controlling and optimizing processes and recipes are presented and integration of CLSC tool is evaluated. Related Task(s): T5.5			

Deliverable D5.9 – Plastic parts and components analysis and origin for reuse optimisation FINAL

Deliverable Number	D5.9	**Lead Beneficiary**	17. DGS
Deliverable Name	Plastic parts and components analysis and origin for reuse optimisation FINAL		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	42	**Work Package No**	WP5
Description			
Identification of the chemical composition and classification of polymeric flakes is analysed, machine learning techniques controlling and optimizing processes and recipes are presented and integration of CLSC tool is evaluated. Related Task(s): T5.5			

Deliverable D6.1 – Waste streams mapping

Deliverable Number	D6.1	**Lead Beneficiary**	4. CIRCE
Deliverable Name	Waste streams mapping		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	9	**Work Package No**	WP6
Description			
Initial assessment in terms of waste streams mapping per demonstrator is described, as an establishment of a baseline			
scenario focusing on potential hotspots related to production waste quality, cost, emissions, energy consumption and raw materials use. Related Task(s): T6.1			

Deliverable D6.2 – Systems design and description v1

Deliverable Number	D6.2	**Lead Beneficiary**	2. NTUA
Deliverable Name	Systems design and description v1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	24	**Work Package No**	WP6
Description			

Establishment of the real scale operation of the new solutions in all four demonstrators (plastic- and wood-based value chains) is analysed, evaluation tests performed are described and each system's design is finalized. Related Task(s): T6.2 , T6.3

Deliverable D6.3 – Systems design and description FINAL

Deliverable Number	D6.3	**Lead Beneficiary**	2. NTUA
Deliverable Name	Systems design and description FINAL		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP6
Description			
Establishment of the real scale operation of the new solutions in all four demonstrators (plastic- and wood-based value chains) is analysed, evaluation tests performed are described and each system's design is finalized. Related Task(s): T6.2 , T6.3			

Deliverable D6.4 – Evaluation and Impact assessment

Deliverable Number	D6.4	**Lead Beneficiary**	1. ICCS
Deliverable Name	Evaluation and Impact assessment		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP6
Description			
Document includes the circularity and sustainability performance and impact of all demonstrators, potential alternative scenarios tested, LCC-based dynamic economic feasibility analysis results and an environmental and circularity index for each unit of product. Related Task(s): T6.2, T6.3 , T6.4			

Deliverable D7.1 – Standardization of novel technologies and certification schemes v1

Deliverable Number	D7.1	**Lead Beneficiary**	5. DS
Deliverable Name	Standardization of novel technologies and certification schemes v1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	24	**Work Package No**	WP7
Description			
Possible contributions in secondary materials or/and AI and Blockchain standardization and certification are involved in this deliverable. All relevant analyses and evaluation activities performed are described and all the results are analyzed in details. Related Task(s): T7.1			

Deliverable D7.2 – Standardization of novel technologies and certification schemes FINAL

Deliverable Number	D7.2	**Lead Beneficiary**	5. DS
Deliverable Name	Standardization of novel technologies and certification schemes FINAL		
Type	R — Document, report	**Dissemination Level**	PU - Public

Due Date (month)	48	**Work Package No**	WP7
Description			
Possible contributions in secondary materials or/and AI and Blockchain standardization and certification are involved in this deliverable. All relevant analyses and evaluation activities performed are described and all the results are analyzed in details. Related Task(s): T7.1			

Deliverable D7.3 – Co-design training activities and lifelong learning programmes on digital skills and replication v1

Deliverable Number	D7.3	**Lead Beneficiary**	21. UBRUN
Deliverable Name	Co-design training activities and lifelong learning programmes on digital skills and replication v1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	24	**Work Package No**	WP7
Description			
Documents all the produced knowledge by summarising the main results, lessons learnt and best practices along the project. Training activities and replication packages are described, and potential clustering and cooperation are assessed. Related Task(s):T7.2			

Deliverable D7.4 – Co-design training activities and lifelong learning programmes on digital skills and replication FINAL

Deliverable Number	D7.4	**Lead Beneficiary**	21. UBRUN
Deliverable Name	Co-design training activities and lifelong learning programmes on digital skills and replication FINAL		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP7
Description			
Documents all the produced knowledge by summarising the main results, lessons learnt and best practices along the project. Training activities and replication packages are described, and potential clustering and cooperation are assessed. Related Task(s): T7.2			

Deliverable D7.5 – Exploitation and Business Innovation Plan and activities v1

Deliverable Number	D7.5	**Lead Beneficiary**	8. MERIT
Deliverable Name	Exploitation and Business Innovation Plan and activities v1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	6	**Work Package No**	WP7
Description			
Initial (v1) version of the exploitation and business innovation plan and undertaken activities. Related Task(s):T7.3			

Deliverable D7.6 – Exploitation and Business Innovation Plan and activities v2

Deliverable Number	D7.6	**Lead Beneficiary**	8. MERIT
Deliverable Name	Exploitation and Business Innovation Plan and activities v2		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	24	**Work Package No**	WP7
Description			
Updated (v2) version of the exploitation and business innovation plan and undertaken activities. Related Task(s):T7.3			

Deliverable D7.7 – Exploitation and Business Innovation Plan and activities FINAL

Deliverable Number	D7.7	**Lead Beneficiary**	8. MERIT
Deliverable Name	Exploitation and Business Innovation Plan and activities FINAL		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP7
Description			
Final versions of the exploitation and business innovation plan and undertaken activities. Related Task(s): T7.3			

Deliverable D7.8 – Communication and Dissemination plan and activities v1

Deliverable Number	D7.8	**Lead Beneficiary**	1. ICCS
Deliverable Name	Communication and Dissemination plan and activities v1		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	6	**Work Package No**	WP7
Description			
Initial (v1), of the dissemination and communication plan and undertaken activities. Related Task(s): T7.5			

Deliverable D7.9 – Communication and Dissemination plan and activities v2

Deliverable Number	D7.9	**Lead Beneficiary**	1. ICCS
Deliverable Name	Communication and Dissemination plan and activities v2		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	24	**Work Package No**	WP7
Description			
updated (v2) version of the dissemination and communication plan and undertaken activities. Related Task(s): T7.5			

Deliverable D7.10 – Communication and Dissemination plan and activities FINAL

Deliverable Number	D7.10	**Lead Beneficiary**	1. ICCS
Deliverable Name	Communication and Dissemination plan and activities FINAL		

Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	48	**Work Package No**	WP7
Description			
Final version of the dissemination and communication plan and undertaken activities. Related Task(s): T7.5			

Deliverable D7.11 – Policy briefs

Deliverable Number	D7.11	**Lead Beneficiary**	1. ICCS
Deliverable Name	Policy briefs		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	45	**Work Package No**	WP7
Description			
2 policy briefs (1 per sector of focus: pulp and paper, chemicals and plastics) targeting EU policy-makers, agencies. Related Task(s): T7.5			

LIST OF MILESTONES

Milestones

Grant Preparation (Milestones screen) — Enter the info.

Milestone No	Milestone Name	Work Package No	Lead Beneficiary	Means of Verification	Due Date (month)
1	Establishment of Internal Collaborative Tool, visual id, virtual presence, Risk Procedure and Quality Management Plan	WP1	1-ICCS	Internal Collaborative tool operates with access to all partners. Documents are uploaded. Official website and social media accounts are developed.	4
2	Establishment of Data management, Exploitation, Communication and Dissemination plans	WP7	1-ICCS	First versions of D1.5, D7.5, D7.8 are submitted	6
3	The definition phase is completed for traceability framework and value chains	WP5, WP2, WP6	1-ICCS	D2.1, D2.3, D2.5, D5.1 and D6.1 are submitted.	8
4	An external Advisor Board is set up	WP1	1-ICCS	Adv. Board participates in a plenary session and feedback is documented	8
5	The development Phase is completed for all the Pilots	WP5, WP6	2-NTUA	All prototypes are up and running, relevant research outputs (e.g. first version of the 4 study cases) are published.	24
6	The Implementation Phase is completed for all the pilots	WP5, WP6	2-NTUA	All the technologies are up and running for all the value chains. Research outputs updated accordingly.	36
7	Traceability systems	WP4, WP3	1-ICCS	Sensing and sorting technologies, digital tools and services supporting real implementation of DDP and traceability are up and running.	36
8	Results of successful real demonstration	WP7	1-ICCS	Standardization contribution, training, learning programs on digital skills, replication packages, exploitation, business innovation plans are available.	48
9	Project ends successfully – Final event	WP1	1-ICCS	A conference is organized by coordinator.	48

Milestones

Grant Preparation (Milestones screen) — Enter the info.

Milestone No	Milestone Name	Work Package No	Lead Beneficiary	Means of Verification	Due Date (month)
				Submission of D1.4 and all final deliverable versions are submitted.	

LIST OF CRITICAL RISKS

Critical risks & risk management strategy

Grant Preparation (Critical Risks screen) — Enter the info.

Risk number	Description	Work Package No(s)	Proposed Mitigation Measures
1	Project milestones or deliverables are delayed, schedule is partially mismatched P:M I:M	WP5, WP7, WP2, WP4, WP6, WP1, WP3	In the scope of project management continuously monitoring activities, detailed analysis will be done on both global project and lower (WP/Task) project implementation levels. Thus, it will be ensured that cases that could delay any milestones or deliverables are recognised in early stages, ensuring timely and effective corrective actions in the work plan. In crucial cases, PC can work on the plan adaptation in close collaboration with the Project Officer (PO). M M
2	Underestimated partner resource or Overspending P:L I:M	WP5, WP7, WP2, WP6, WP4, WP1, WP3	Situations will be detected by T1.1. ExB will analyse the following possibilities to ensure that planned work can be completed: i) re-arranging resources among partners ii) committing further internal resources of partners; iii) re-planning activities iv) mobilise more skilled and efficient human resources. Ad-hoc recovery plans could be put in place. L M
3	One or more partners leaving the consortium P:L I:M	WP5, WP7, WP2, WP6, WP4, WP1, WP3	Given the good reputation of all partners, most of them with proven success records in H2020 Projects and related experience, this possibility is considered to be very unlikely. In the unlikely case of that happening, immediate actions will be taken to reassign the responsibilities either within the consortium or outside if needed. L M
4	One of the partners (SME or start-up) goes bankrupt P:M I:M	WP5, WP7, WP2,	The economic uncertainty driven by the Covid-19 pandemic has pushed struggling companies over the edge into bankruptcy proceedings. To overcome a potential partner

Critical risks & risk management strategy

Grant Preparation (Critical Risks screen) — Enter the info.

Risk number	Description	Work Package No(s)	Proposed Mitigation Measures
		WP6, WP4, WP1, WP3	bankrupt, advanced allocation of funding will be carefully organised so as to minimise this risk, and the ExB will identify existing or new partners to fill the gap. M M
5	Delays due to the processes complexity P:L I:H	WP5, WP6	Pilots' research and technology partners have already tested and used this state of the art technologies and equipment for other applications, while its adaptation and its establishment for piloting and demonstration is performed by highly experienced and skilled staff of both research and industrial pilots' partners. L H
6	Missing/delayed equipment P:M I:H	WP5, WP4, WP3	Potential delays (due to chipset/equipment unavailability or transport restrictions) may occur in the delivery/availability of hardware equipment, especially if a new pandemic wave or other unexpected event occurs. Partners involved in the technical development and pilot specifications will closely monitor the supply chain of the identified components, to mitigate delays, and if needed, provide alternative solutions and will be expected to regularly report the progress. Special attention will be paid by all partners to place the orders well in advance. M H
7	Reluctance from partners to provide data due to security issues. Insufficient data from value chains P:M I:M	WP2, WP4, WP6, WP3	The CA will include and define the terms regarding access to data and existing knowledge, with emphasis on the pilots. Secure mechanisms will be deployed for data sovereignty and sharing for data which is characterised as "sensitive" and "confidential". T1.2 will undertake the data management, which will run thought the project duration, also the enhanced security offered by blockchain and project's solutions will be explained in detail to the partners. M M

8	Deviations from technical expectations, invalid design, specifications or requirements P:M I:M	WP5, WP2, WP4, WP6, WP3	Iterative development periodically verified in all project meetings will guarantee an early detection of deviations and invalid specifications. Specific effort is assigned to the technical part of the project in the definition phase, potential technical difficulties are expected to be detected at early stage. Special attention was given to the amount of time assigned for the development stage, assuring that there is enough time for solving potential problems. The integration phase of mature technologies should further minimise these risks. TM will monitor the technical design and development processes (T1.2) with WP leaders and technical partners, to avoid alleviate misalignment of technical visions, to ensure detailed specifications extraction of the software/hardware components and architecture alignment and to quickly identify ineffective designs and take proactive measures. The technological tasks are undertaken by experienced research and technological partners further mitigating possible risk of unforeseen obstacles disturbing the project performance. M M

Critical risks & risk management strategy

Grant Preparation (Critical Risks screen) — Enter the info.

Risk number	Description	Work Package No(s)	Proposed Mitigation Measures
9	Not enough residues available or/and impurities in the secondary material flows for the implementation of the circular loops or/and contamination. P:L I:H	WP5, WP6	The secondary materials/waste providers involved in the consortium are industrial partners (CHIMAR, MXS, AGRIST, ASTI, SIGIT) that have the capacity to provide amounts of materials for the needs of the project. In case of inappropriateness of material or batch of material the alternative sources of materials and wastes for the pilot cases will be investigated within their network of contacts or other industries. L H
10	Interoperability problems between heterogeneous components and difficulties to connect the process data P:L I:M	WP5, WP2, WP4, WP6, WP3	Best practices will be used to prevent issues while assessing the design specifications of components, aligned with open and interoperability standards (protocols to enable easy connection, APIs, manually data input as workarounds etc.) Consortium partners possess deep knowledge of interoperability with the technical partners to agree on any technical decision that may impact interoperability, with the Technical Coordinator taking the final decisions. L M
11	Disputes over ownership of IPR amongst partners P:M I:L	WP5, WP7, WP2, WP4, WP6, WP3	Standard IPR and access rights clauses will be included in the CA which will be signed before the project starts. Further support to be provided by the legal experts involved in the project for IP conflict resolution. If necessary, ownership will be decided according to CA. M L

12	Competition and Low End-users Acceptance P:L I:L	WP5, WP7, WP2, WP4, WP6, WP3	Market intelligence activities upon exploitation planning, will ensure continuous monitoring of competition landscape. MERIT will ensure the thoroughness and quality of the resulting reports. Well defined user requirements and cost-benefit validation of the solution will be implemented. Engagement activities will ensure co-creation with end-users to enhance acceptance. L L
13	Blockchain / DLT P:L I:M	WP3	Blockchain is a fast-moving field on the technological, legal and regulatory axes raising the risk that new capabilities, vulnerabilities or restrictions emerge during the project lifecycle. These are out of the project control but will be mitigated by monitoring and adaption of the workplan. L M
14	New superior relevant Methodologies and Practices arise P:M I:M	WP5, WP2, WP4, WP3	Trends within research areas will be constantly monitored, and strong liaisons with relevant projects and initiatives will be established to join forces. The technical and scientific choices of the project will be re-oriented, considering the latest advances in order to avoid project's approach to be obsolete. M M

Critical risks & risk management strategy

Grant Preparation (Critical Risks screen) — Enter the info.

Risk number	Description	Work Package No(s)	Proposed Mitigation Measures
15	Coordination with relevant projects, initiatives and associations P:L I:M	WP7	T7.1 and 7.2 will support cooperation with relevant activities and projects, ensure proper links with different Working Groups and the rest projects of same topic (TT-01-01), to build on consensus future systems with different perspectives applied in various environments and beyond. L M

PROJECT REVIEWS

Project Reviews

Grant Preparation (Reviews screen) — Enter the info.

Review No	Timing (month)	Location	Comments

RV1	18	tbd	
RV2	36	tbd	
RV3	48	tbd	

History of Changes

Version Number	Date	Part A,B or Annex I; section and page	Reason/Justification	Specific Change (Short Elaboration)
V1.0	29/08/2022	Annex 1, Part A (List of Deliverables; List of Work Packages), Part B (sections 3.1.1 & 3.1.2)	Correction following **bullet point** "In the description of each deliverable, please verify that you indicated to which task(s) and/or WP(s) it is linked"	List of deliverables updated to include one deliverable per date and the related tasks per deliverable
V1.0	29/08/2022	Annex 1, Part B section 1.1.3(Pages 7,8)	Correction following shortcoming, **Criterion 1 – Excellence about** elaboration of sensing/sorting techniques	Sensing Techniques beyond SoA part was elaborated with more details; Sorting techniques beyond SoA part was elaborated with more details;
V1.0	29/08/2022	Annex 1, Part B section 1.1.3 (Pages 10,11)	Correction following shortcoming, **Criterion 1 – Excellence** about the upcycling of plastics and starting TRL for protein recovery	Section for the Upcycling of plastic (state of the art and beyond) was added to describe the details; Protein recovery chemical processes beyond SoA part was updated to discuss the starting TRL;
V1.0	29/08/2022	Annex 1, Part B section 2.2.3 (Pages 30-32)	Correction following shortcoming **Criterion 2 – Impact** about EU market and four demonstrations relation in exploitation	Exploitation strategy section was updated accordingly to consider EU market context and to present the relation among the four demonstrations via joint venture of exploitation:
V1.0	29/08/2022	Annex 1, Part A, List of Critical Risks	Correction following minor shortcoming 1, **Criterion 3 – Quality and efficiency of the implementation** related to technological risks and mitigation measures	The table with DigInTraCE potential critical risks for implementation and contingency plan were enhanced and updated. Namely the risks #5, #9 were added to address additional technological risks. The mitigation measures of risks, #6 , #8 , #10 were updated .
V1.0	29/08/2022	Annex 1, Part A (List of Work Packages), Part B (sections 3.1.2)	Correction following **bullet point** "For subcontracting activities, pls indicate in the costs table the WP(s) and/or task(s) related; indicate that also in the WP and/or task description"	Subcontracting table was updated with their relevant WP/task information, the related WPs descriptions is updated; accordingly,
V1.0	29/08/2022	Annex 1, Part B, section 3.1.3	Correction following **bullet point** "For purchase costs please also indicate WP and or task."	Purchase costs table was updated with relevant WP/task information;
V1.0	29/08/2022	Annex 1, Part B (Section 3.1 section 3.1.3)	Correction due to minor mistake identified	Purchase costs for NTUA, title minor corrections on title and amounts justified;
V1.0	27/9/2022	Annex 1, Part A	Correction of tasks starting date	The starting date of the T7.3 is updated from M9 to M1. Task 7.4 related deliverables were updated to include D7.8, D7.9, D7.10, D7.11.
V1.0	27/9/2022	Annex 1, Part A (List of Deliverables)	Correction of deliverable dates for more efficient monitoring of the progress	The first version of deliverables in WPs3,4,5 is moved from M24 to M18. More specifically it covers the deliverables: D3.1, D3.3, D3.5, D3.7, D3.9 ,D4.1 , D4.3 ,D4.5 , D4.7 ,D4.9 ,D5.2,D5.4 D5.6 , D5.8

				The tasks T6.2 and T6.3 will also contribute to D6.4
V1.0	27/9/2022	Annex 1, Parts A and B and in Annex 2	Status of UK partner Correction of the order of the partners and of the numbering	The UK partner will participate as an associated partner. The status of the partner and the budget are updated accordingly it was done in parts A and B of Annex 1 and in Annex 2 The order of the partners and the numbering has been updated
V1.0	27/9/2022	Annex 1, Parts A and B and in Annex 2	Subcontract need identified for beneficiary EUPC	The subcontract amount is amended in the part B of Annex 1 and in Annex 2. The total allocated budget of the beneficiary remains the same. The subcontracting budget of the beneficiary is gathered by reducing 1PM in WPs2, 6,7.
V1.0	27/9/2022	Annex 1, Part B, Section 3.1	Gantt Chart Updated	th document Ref. Ares(2022)7761208 - antt chart updated to include the new dates of deliverables and the updated outcome of tasks
V1.0	27/9/2022	Annex 1, Part B, Section 4	Section 4 added	Section 4 Ethics self assessment has been added

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1. Excellence

Associated with document Ref. Ares(2022)7761208 - 10/11/20

1.1. *Objectives and Ambition*

1.1.1. Motivation and Vision

The Sustainable Products Initiative (SPI) was developed by the EC, widening Ecodesign Directive[1] to reach Green Deal (GD) objectives[2], leading to sustainable products and making European market and industry more resource-efficient, competitive, resilient and innovative. SPI considers a wide range of products, prioritized by CEAP[3] (e.g., electronics, ICT, textiles, furniture, steel, cement, chemicals) to improve their durability, reusability, upgradability and reparability, address hazardous chemicals presence, increase energy and resource efficiency and establish sustainability and circularity. Lack of transparency and traceability of products, their components, substances and environmental impacts, represent a barrier for circular economy business models implementation[4]. Supply chains involving diverse actors (e.g., materials suppliers, producers, recyclers, consumers) complicated information management and combination of all the different needs and technical capabilities of companies across the value chain[4]. Recently 2030 Digital Compass highlighted the importance of digital solutions for the achievement of GD objectives. The uptake of digital technologies and schemes like Digital Product Passport (DPP) enable sharing of product-related data across the product lifecycle and along value chains, by tagging, tracing and localising down to item-level identification[5]. Several digital technologies (e.g. blockchain, big-data, artificial intelligence, digital twins) are proposed for traceability of component and products along value chains and they are considered key enablers promoting circularity, increasing inherent recyclability of materials and upgrading secondary resources, in terms of manifesting their composition as potential secondary raw materials in the same or different process. **Enhanced tracing via**

[1] 2009/125/EC
[2] COM (2019) 640
[3] COM/2020/98 final
[4] Rizos, V., Bryhn, J., Alessi, M. et al., Barriers and enablers for implementing Circular Economy Business Models Evidence from the electrical and electronic equipment and agri-food value chains, CEPS Research Report (2021) [5] COM/2021/118 final

***DigInTraCE** aims at the development of i)transparent and interoperable **Decentralized Traceability platform** using **novel tracking, sensing and sorting techniques** ii)**Dynamically updated DPP** schemes supporting certification and quality validation iii)AI based and decision making mechanisms for process and lifecycle optimisation iv)**up-cycling, reuse and upgrade technologies** for improved secondary raw materials use v)contribute to **standardization**, open and easy accessible data vi)**business models** creating new economic opportunities and learning resources for employees, promoting **new digital skills** and meeting regional social needs. The solutions are developed at TRL6 by the end of the project and demonstrated in the sectors of a) **Pulp & Paper** focused in 2 cases: composite wood and furniture; and wood and Pulp & Paper and b) **Chemicals** focused in 2 cases: plastic parts for ICT equipment and automotive market; and polymers & textiles.*

digital tools combined with **novel sensing and sorting techniques**, is considered an **imperative success factor** for upgrading of secondary resources, to enable cleaner production and recycling, ensure toxic-free material cycles limiting substances of concern and produce informed consumers regarding the origin and the way materials are processed. Industrial sectors with high digital traceability needs as highlighted by the P4P roadmap include chemicals, pulp and paper (along with the whole forest-based industry and wood value chain), ceramics and cement[5]. DPP, as an emerging concept for digitally recording product and process information to cover the entire product life cycle by focusing mainly on item level identification, although being a quite challenging endeavour, it is required for enabling circular economy adoption, scaling and practical implementation, as it provides an easily accessible, centralized registry of diverse data [6]. A physical link (tagging such as QR code) between the product and the data is utilized, allowing value chain stakeholders, end-users and authorities to directly access a unique digital profile with quantitative/qualitative, static/dynamic, standardized and machine-readable data related to the product's origin, composition, repair and dismantling possibilities, end of life handling etc.[7] [9]. The implementation of DPP requires multiple data sources supported by existing tools (such as EPREL database, EU eco-labels, existing BOMs etc.) via a step-by-step and flexible approach, based on information already available and regulated while DPP used as a link to decentralized information databases[8]. Project's methodology and results are aligned with Processes4Planet (P4P) roadmap, GD, CEAP, Digital Single Market and 2030 Digital Compass objectives and the guidelines of Orgalim[8], EIT RawMaterials, CEN & CENELEC[9] and SPI.

1.1.2. Objectives and Related KPIs

DigInTraCE will pursue its realisation through 6 SMART objectives described below:

O.1: To design, develop and implement solutions fostering and optimizing use of secondary raw materials and reducing waste as part of circular value chains.

DigInTraCE will assure circularity of processes of sectors belonging to the process industry, by enhancing digitalisation Associated with document Ref. Ares(2022)7761008 - 10/11/2022 (WP3, WP4), upgrading by-products and optimising processes, with special focus on secondary materials, but in the context of whole value chain (WP2, WP5). DigInTraCE is aiming to establish Decentralized Traceability Platform (based on blockchain) applied to holistically optimized circular value chains, assuring availability and access to reliable information enhancing cross-sectoral collaboration (WP2, WP3, WP4). DigInTraCE will produce an advanced transparent information system with central point being the DPP, supported by live update mechanism (T2.2, T2.3, T4.1), enhancing the use of secondary raw materials of the own industry and improving cross-sectoral collaboration. Overall, DigInTraCE will comprehensively approach the optimization of secondary materials use through application of horizontal tracing systems combined with demo/case specific **chemical** (e.g. protein recovery process innovation, bio-based adhesive development- T5.2) and **digital** (data-driven algorithms-T5.3,5.5 together with sensing-T3.1 and sorting-T3.2) **processes innovation** leading to waste reduction and lower emissions.
Relation to the call: *"Propose new solutions for improved use of secondary raw materials along the value chain of the own industry or in other industries."*

[5] Processes4Planet 2050 Roadmap. (2021, January)
[6] Walden, J., Steinbrecher, A. and Marinkovic, M., Digital Product Passports as Enabler of the Circular Economy. Chemie Ingenieur Technik, 93: 1717-1727 (2021). https://doi.org/10.1002/cite.202100121
[7] Orgalim Policy Exchange: The Sustainable Products Initiative and the EU Digital Product Passport, (2021, June 29th) Virtual Event [9] CM\1231965EN.docx
[8] DIGITALEUROPE's supporting paper on the public consultation on the Sustainable Products Initiative. (2021, June)
[9] European Committee for Standardization & European Committee for Electrotechnical Standardization
[10] Hyperspectral imaging & Laser-induced breakdown spectroscopy

KPI O1.1: At least 6 examples (products or process related) of up-cycling, reuse and upgrade technologies of secondary raw materials implemented. **KPI O1.2:** At least 4 AI based models developed for secondary raw material optimization processes
O.2: To develop and demonstrate new concepts for material tracing while offering a digital decentralized platform allowing tracing and certification of secondary raw materials
DigInTraCE will develop information systems and digital tools for material tracing (WP4) integrated **into multifunctional and interoperable decentralized traceability platform** (T4.6) with **automatic and real-time update** (T4.1) of information on material composition, quality, history, and other crucial data updated along the value chain, which will be defined through integration of existing regulations and specific identified needs (T2.2). Those new concepts will be supported by **new standardisation and certification schemes** (T7.1) integrated into digital solutions (T4.5). Standardization deliverables for materials and technologies as well as certification for subproducts types and intended use will contribute to better acceptance of the solutions and efficient launching to the market through support of the transparent and digitalized information system. Through incorporation of the project findings into new or revised standardization deliverables, that will be available for the industry, the outcomes of the project will be long lasting. To assure applicability of the concepts in real-life settings, solutions will be applied and demonstrated in 4 pilots representing crucial sectors of i) pulp & paper and ii) chemicals and plastics.
Relation to the call: *"Propose methodologies for digital tracing and certification of secondary raw materials; Proposals should contribute to standardisation, wherever possible"*
KPI O2.1: Additional Services developed in WP3, WP4 integrated in the Decentralized Traceability platform by M46. **KPI O2.2:** Decentralized traceability platform being developed at TRL6 by M48, having at least 20 active users. **KPI O2.3:** At least 3 certification proposals produced for secondary raw materials based on the analyzed value chains.
O.3: To design and deliver novel near real time sensing and sorting mechanisms enhancing data exchange through a dynamic Digital Product Passport
DigInTraCE is based on a **dynamic DPP scheme** indicating properties and history of the products and materials on real closed-loop value chains optimised in terms of waste, by-products and secondary raw materials with technologies using and supporting computer vision and HIS & LIBS[12], AI and Machine Learning, robotic sorting, recovery process based on novel extraction methods (microwave and/or ultrasonic assisted). **Novel real time sensing and sorting mechanism** will contribute to better identification, characterization, quantification and separation of by-products of the value chains (T3.1, T3.2, T5.3, T5.5). The collected data dynamically included in the DPP will allow to trace all produced materials (T4.1). The integration of existing PolyCertEurope certification scheme will be examined **and novel certification schemes** will be defined for subproduct types (T7.1).
Relation to the call: *"This should include real-time access to information on material compositions and material quality along the whole value chain; Propose digital tools for integration of product passport and/or certification schemes;"*
KPI O3.1: 2 sensing technologies and at least 2 sorting mechanisms deployed at least 3 demonstrator sites. **KPI O3.2:** At least 5 material parameters are added in DPP and continuously updated along the value chain.
O.4: To facilitate access to composition and origin of material data through smart tags, smart contracts, open software and the usage of immersive technologies
DigInTraCE will provide innovative solutions to facilitate access to information, that is crucial for efficient cross-sectoral symbiotic collaboration. Those activities will be proceeded by waste streams mapping (T6.1) to assure that maximal potential of the collaborations is achieved and accompanied by circularity and sustainability performance of the demonstrators (T6.4). Innovative solutions like **functional SmartTags** (T3.3) enabling item/batch life cycle monitoring and dynamically transmitting data to the decentralized traceability platform will be tested and implemented in relevant demonstration settings. Blockchain technology will assist **smart contracts** providing automated interoperable financial services greatly facilitating financial procedures within circular economy setting (T4.2, T4.3), which can be challenging

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for less experienced stakeholders. Circular collaboration will be facilitated through **secure and safe data channels** with corresponding accessibility rights enabling intra- and inter-firm communication via **open data/software/hardware** (T4.5).

Relation to the call: *"Propose open-source software, open hardware design, and easy access to data, in order to facilitate access to information for the own and for other industries; Develop means and tools to indicate the composition and origin of recycled materials (bar code could be an option) indicating the composition and origin"*

KPI O4.1: SmartTags implemented in at least 1 demonstrator.
KPI O4.2 Smart contracts developed and integrated in the decentralized traceability platform by M46 with additional financial services
KPI O4.3: Safe and secure data channels with access control functional in the decentralized traceability platform by M40.

O.5: To validate the DigInTraCE technologies in 4 different value chains.

DigInTraCE deals with two sectors: i) Pulp & Paper (focused in 2 cases: composite wood and furniture; and wood and Pulp & Paper) and ii) Chemicals & Plastics (focused in 2 cases: plastic parts from ICT equipment and automotive market; and polymers and textiles) with the **application of digital and data driven innovations along with mechanical and chemical solutions, for the upgrade and optimization of value chains in terms of circularity, low emissions and increased waste reduction** (WP6). End products of the new value chains include novel wood composites (panels, boards/foams), mosaic surfaces, polyester textiles etc. The project will develop **business models** (T7.3) and will support the primary effort for standardization and certification (standardization deliverables) of secondary materials and technologies allowing interaction with market stakeholders and contributing to dissemination of the project result facilitating acceptance of the developed solutions (T7.1).

Relation to the call: *"Proposals submitted under this topic should include a business case and exploitation strategy, as outlined in the introduction to this Destination. Proposals should contribute to standardisation, wherever possible"*

KPI O5.1: To demonstrate the application of digital technologies at TRL 6 in 4 demonstrator sites in Europe for waste reduction and processes optimisation.
KPI O5.2: At least 5 cross sectorial business models developed.
KPI O5.3: At least 5 technologies/products are exploited and have a clear plan for uptake after the end of the project.

O.6: To empower local and regional bodies by involving them in designing educational resources for both workplaces and educational institutions. Stimulate adoption of the DigInTraCE solutions by the wider community and enhance knowledge transfer to maximise the impact.

To increase adoption of key digital and enabling technologies in line with EU digital compass, DigInTraCE will engage in a number of educational activities. Through **co-designing of learning resources** and linking them to replication and exploitation activities DigInTraCE aims at contributing to increased adoption of novel technologies and enhancement of the digital skills. DigInTraCE will produce replication packages based on developed solutions along with business models (T7.2), which will be specific sectoral/regional/national levels. To maximize the impact DigInTraCE will provide **exploitation strategy** for results and will produce an implementable plan, including finance viability validation on value chains, to attract stakeholders to access and utilise its results. **Achievements and knowledge acquired will be disseminated and communicated** towards the target groups and interested stakeholders. DigInTraCE will join its forces with other projects of the topic and operating initiatives for collaboration and capacity building (T7.2-7.4).

Relation to the call: *"Consider the co-design of learning resources together with local and regional educational organisations for current and future generations of employees, integrating them in existing curricula and modules for undergraduate level and lifelong learning programme. new skills and should propose innovative learning-teaching methods that meet regional social needs and have a high potential for replication.; Proposals submitted under this topic should include a business case and exploitation strategy."*

KPI O6.1: At least 2 co-design activities performed.
KPI O6.2: Dissemination, Communication and Exploitation KPIs fulfilled as presented in Section 2.2.
KPI O6.3: 4 value chains are validated for financial viability and are maintained after the project implementation.

1.1.3. Ambition and R&I maturity

DigInTraCE aims to provide a cross sectoral, from sensing to sorting, from materials to products, digitalisation solution and propose a dynamic digital product passport scheme as landmark and in parallel develop and combine these digital with process and material innovations, to increase waste reduction, improve use of secondary raw materials, thus lead to circular and low emission value chains

I. Decentralized Traceability platform

State of the Art & Current Challenges: Supply chain traceability in complex networks is a key emerging benefit of blockchain[11]. Decentralization, immutability and transparency lead to accurate, reliable data that is trusted despite the absence of trust relationships among participants. Multisource data can be collected (RFID, barcodes, IoT, GPS, etc.) and added to the append-only blockchain ledger to update the status of each specific item in the supply chain journey. Combined with the fact that the tokenisation aspect of blockchains is also used for trading, market transparency, auctions (e.g. wholesale energy

or carbon credits) and even financial instruments (bonds, derivatives), as well as hosting currency (crypto, stablecoins) there

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is enormous potential to reduce the frictions inherent in trading and distributing secondary raw materials. Furthermore, the combination of unquestionable provenance data via traceability provides the opportunity to develop economically viable new products that incorporate secondary materials, meet the green initiatives, and maximize circularity. Accurate and updated secondary materials information can improve product deletion analysis and decision-making. Blockchain can provide more accurate reusability and recyclability of materials and products information[12].

Progress beyond SoA: Blockchain is already being applied to supply chains and to commodity (financial) derivatives. In DigInTraCE, we will investigate how a) traceability and smart contracts can be optimized with respect to the other technology specific to secondary raw materials, and b) break new ground by combining tokenized financial products associated with the blockchain-traced items in the same platform. Public blockchain based on proof-of-work (PoW) such as bitcoin is notorious for its egregious energy consumption. This is not a factor in DigInTraCE because we will not be using a public, PoW chain but rather a private chain with no greater consumption than a standard cloud-hosted virtual machine.

Tool / Technology	Key Advantage & Progress – Start -> Foreseen End TRL
Decentralized Traceability platform	A traceability platform based on blockchain technology allowing secure data exchange secondary materials while also offering the possibility of additional financial services, **TRL5❼TRL6**

II. Dynamically updated Digital Product Passport

State of the Art & Current Challenges: SPI is expected to set up the ground for DPPs that collect **dynamically updated** information about the product as its moves across the value chain. Related first efforts are focusing on the electric vehicles' batteries and in construction sector[13], with APIs reach cloud-based databases and connect materials passport to the Building Information Modelling (BIM)[14], while advanced DPP requires dynamic update and access control. As the European Blockchain Partnership supports the development of services across Europe, this technology is considered potentially suitable tool for DPPs[15] [16] with some successful cases available[17].

Progress beyond SoA: In DigInTraCE novel real time sensing tools will be developed, coupled with smart tags and more established tracing solutions (e.g. barcodes). All the captured data will be made available through a decentralized traceability platform and propagated towards a dynamically updated DPP. The newly developed concept is to be validated in 2 sectors and 4 value chains, **pulp and paper** and **chemicals** (plastics).

[11] Westerkamp, M., Victor, F., Küpper, A. Tracing manufacturing processes using blockchain-based token compositions, Dig. Com. & Net. (2020). https://doi.org/10.1016/j.dcan.2019.01.007

[12] Kouhizadeh, M., Sarkis, J., Zhu, Q. At the Nexus of Blockchain Technology, the Circular Economy, and Product Deletion. Appl. Sc. 9, 8:1712 (2019). https://doi.org/10.3390/app9081712

[13] MATERIALS PASSPORTS, BAMB, EU Horizon 2020 project, GA No 642384.

[14] Atta, I., Bakhoum, E., Marzouk, M. Digitizing material passport for sustainable construction projects using BIM, Jrnl Build. Engnr.,43 (2021). https://doi.org/10.1016/j.jobe.2021.103233

[15] Tezel, A., Papadonikolaki, E., Yitmen, I., et al. Preparing construction supply chains for blockchain technology: An investigation of its potential and future directions. Front. Eng. Manag.7(4): 547–563 (2020). https://doi.org/10.1007/s42524-020-0110-8

[16] Alaranta, J., Topi Turunen, T. How to Reach a Safe Circular Economy? - Perspectives on Reconciling the Waste, Product and Chemicals Regulation, Jrnl Env. Law,33(1) 113–136 (2021). https://doi.org/10.1093/jel/eqaa016

[17] Bacchetta, A.v.B., Krümpel, V., Cullen, E. Transparency with Blockchain and Physical Tracking Technologies: Enabling Traceability in Raw Material Supply Chains. Mater. Proc. 5, 1 (2021). https://doi.org/10.3390/materproc2021005001

Tool / Technology /Activity	Key Advantage & Progress – Start -> Foreseen End TRL
Dynamically updated DPP	A new concept and implementation of a dynamically updated digital product passport, with a focus on secondary raw materials, **TRL4❼TRL6**

III. Smart Tags for real time sensing

State of the Art & Current Challenges: Smart Tags are functional, information transmitting elements combining 2D barcodes and RFID technologies for environmental sensing into a cost-effective tag[18,19] that can contain data to identify individual items. Item-level identification is used in some industries (e.g., healthcare, car parts) but it is not widespread having challenges on durability and tag integration to the item throughout lifecycle. Further work could also improve engagement of value-chain stakeholders and end-users for waste reduction, recycling procedures, enable new business models (e.g., asa-service, deposit-base) aligned with EU initiatives reshaping digital infrastructures. Future-proof data management and sharing need to be compatible with EU principles to guarantee cross-sectoral interoperability, data privacy and security.

Progress beyond SoA: In DigInTraCE the full smart tag concept from the data management and storage (especially for traceability of raw materials), durable physical tagging methodology (including the sensing when feasible) as well as connectivity features supporting stakeholder engagement for waste reduction and raw material recycling will be developed. The developed tags will allow the tracing of material parameters across the value chain, considering at least temperature and humidity, while continue being cost effective.

Tool / Technology /Activity	Key Advantage & Progress – Start -> Foreseen End TRL
Smart tags for real time sensing	Functional Smart Tags, dynamically transmitting information to the Decentralized platform, with a focus on humidity and temperature, **TRL4❼TRL6**

IV. Novel Sensing Techniques

State of the Art & Current Challenges: Process monitoring and quality control systems are necessary for safe and optimal

operation in industrial processes with standardized feedstock as wood Associated with document Ref. Ares(2022)7761208 - 10/11/2020and plastic recycling sectors. Rapid analytical

techniques are required for continuous product and process control. Spectroscopic methods are frequently used, being rapid, reliable and often non-destructive analytical techniques. Process Analytical Technologies (PAT) used for real-time monitoring of critical quality parameters, are the next step for the competitive operation of many industries. Numerous automated identification systems have been developed [22], such as near infrared (NIR) mainly used to recognize and sort the plastic polymer type by analyzing the reflected spectrum from material's surface[21], whereas identification of presence of elements such as bromine, lead, mercury and chromium is performed via XRF, LIBS or FTIR methods [22]. Challenges exist in recycling of plastic composites as fibre-reinforced plastics and multi-layered structures, in characterization of black components[23] and in identifying recycled content, as the only accurate method uses very expensive auditing systems estimating values of certain parameters. Some analytical testing methods with moderate sensitivity and high specificity, such as DHS/GC-MS and Py-GC, are promising for the identification of degradation indicators differentiating recycled from virgin polymers [24] [25]. For wood by-products, to detect the quality and classify the suitable materials for valorization, spectroscopic methods are employed, as near-infrared

[18] Gligoric, N., Krco, S., Hakola, L. et al. SmartTags: IoT Product Passport for Circular Economy Based on Printed Sensors and Unique Item-Level Identifiers. Sensors, 19, 586 (2019). https://doi.org/10.3390/s19030586

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hyperspectral imaging (NIR-HSI) for the detection of moisture content[26], the classification of wood according to species [27], and to differentiate composites (preservative-treated) and untreated waste wood[28]. In case of wood chips, laser scanning, machine vision and image processing are applied to calculate and categorize according to their size, discover and determine percentage of rot in residues [29][30].

Progress beyond SoA: In DigInTraCE spectroscopic techniques and sensors using multiple bands will be developed. These will be tailored-made for monitoring many key parameters according to the value chain, either wood or plastic residues (i.e., the ratio of polymers and heavy metals in a mix of plastic flakes, size, shape and moisture content in a mix of wood chips etc.). For the Chemicals process industry (polymers, textiles) demonstrator, the aim is to focus on the PET separation. High purities are needed in order to produce high tenacity textile fibers, so impurities such as cotton, elastane, PA, residues of spin oils, presence of additives (flame retardants, water repellents, softeners, etc.) can influence the reprocessing of the PET and should be monitored. In this sense, the sensing technologies that will be used to identify the presence of impurities are: Hyperspectral Imaging (HSI) for polymer identification combined with Laser Induced Breakdown Spectroscopy (LIBS) for detection of anatomic impurities such as flame retardants. Also, a monitoring system based on the combination of HSI technology (for plastic identification) and LIBS technology (for heavy metals detection, bromine and phosphate detection, among other compounds) will be implemented in the recycling process of process industry producing plastic components for the automotive market. Customized algorithms will allow process control for the optimization of processes and final product development. For the estimation of recycled content in polyester waste, 'fingerprints' will be examined of either degradation molecules or combination of additives, determining virgin by recycled polyester, as indicators and detection methods[32].

Tool / Technology /Activity	Key Advantage & Progress – Start -> Foreseen End TRL
Novel spectroscopy techniques and algorithms for plastic waste and wood by-products	Sensors with novel functionalities and applications, which could be customized in function of the key parameters that should be monitored within the processes, **TRL4❼TRL6**
Recycled content identification and certification scheme	Identification of patterns within recycled materials used for integrity validation and certification for them, **TRL4❼TRL6**

V. Novel Sorting Techniques

State of the Art & Current Challenges: To physically separate waste or by-products properly, indirect sorting techniques are applied using sensing to detect quality (as described above) and location of unsuitable materials, so that automated machines or robots can be harnessed to sort the detected recyclables[22]. In automated sorting, size reduced infeed materials are provided on conveyer belts and scanned via photonic spectrometers in order to govern air nozzles to blow out specific

particles. As recently reported, the particle size of such processes is decisive for the sorting outcome[31]. The sorting efficiency

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of indirect technologies is limited by the executing agencies, such as compressed air jets, suction pads and robot manipulators, however indirect sorting technologies present higher recognition efficiency but lower processing capability. Although the existing sorting techniques offer accurate results, optical sorters and/or AI-equipped sorters can be combined to improve product yield or sorting efficiency, as AI-driven identification and robotic manipulation are currently at the forefront. Machine- or deep-learning techniques should be utilized to improve data processing, to sort a wide range of materials, offer reliability and enable real-time recognition response, capable of operating on

[26] Ma, T., Morita, G., Inagaki, T. et al. Moisture transport dynamics in wood during drying studied by long-wave near-infrared hyperspectral imaging. Cellulose 29, 133–145, (2022). https://doi.org/10.1007/s10570-021-04290-y

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[31] Lubongo, C., Alexandridis, P. Assessment of Performance and Challenges in Use of Commercial Automated Sorting Technology for Plastic Waste. Recycling 7, 11 (2022). https://doi.org/10.3390/recycling7020011

robotic-based mechanical sorters[33]. One main challenge for robotic sorting is dealing with light particles[32] such as wood chips, sawdust and plastic flakes, where it is inefficient and mechanical separation techniques need to be utilized such as sieving, disc screening and air classification[33] [34].

Progress beyond SoA: In DigInTraCE, a multi-spectrum analysis will be performed allowing a robotic-based sorting mechanism to accurate separate residues in real-time, assisted by machine- or deep-learning techniques. The classification of waste will be relied on the developed sensing techniques and additional industrial camera system that identifies all the dimensions of objects. The separation will be based on multi robots across the same line, upgrading the existing recycling processes, providing better classification results and identifying value chains with the lowest environmental impact. Specifically, robotic arms assisted with sieving and real-time by vision-based, machine- or deep- learning techniques, will be used for the removal of the defective items and further separation of the plastic flakes (having the weight of approx. 80400g) in plastic-based pilot (Italy) and for wood residual side streams of different often irregular shapes and sizes ranging from several mm to larger pieces e.g., small wooden rectangles (20x30 cm), with larger pieces containing composites in respective wood-based pilots.

Tool / Technology /Activity	Key Advantage & Progress – Start -> Foreseen End TRL
Automated Plastic Sorting system	An automated sorter for plastic flakes, **TRL4❼TRL6**
Automated Wood Sorting system	An automated wood sorting system serving the needs of the pulp & paper sector, **TRL4❼TRL6**

VI. Data-driven process optimization for increased uptake of secondary raw materials

Optimal control and AI package for recycled plastic

State of the Art & Current Challenges: Optimizing the parameters of injection moulding process has a substantial impact on the quality of the derived products and is industrial and scientific challenge: complex process depending on multiple crossrelated variables, optimal injection parameters selection can prevent defects and improve quality. Many optimization approaches and techniques are proposed e.g. Taguchi method, numerical simulation based on Computer Aided Engineering (CAE), genetic algorithms and artificial neural networks[35]. Taguchi method is simple, practical and effective in obtaining optimised process parameters by means of orthogonal test; however, it can suffer nonlinear behaviors. The optimisation of CAE is based on the simulation software, but its accuracy is strongly dependent on calibration procedures. Genetic algorithms are very efficient but strongly depend on inherent randomized approach while neural networks allow considering of system nonlinearities, are robust and quickly adaptable to different case studies and experiment characteristics. In general, the use of AI for injection moulding parameters is still in its infancy and its comprehensive impact analysis is an open and challenging problem requiring interdisciplinary scientific approach (e.g. optimal control and system identification, ML etc.) as well as a real industrial environment for appropriate case studies and related validation metrics and KPIs.

Progress beyond SoA: DigInTraCE will fill the gap exploiting a combination of academic and industrial competences in Control Theory and Machine Learning (ML) for Automation (UVQ) and ICT data collection systems (DGS, IRIS), as well as a relevant use-case. More precisely, we will exploit and apply to the Italian pilot ML techniques (e.g. Regression Trees, Random Forests[36], Kernel regression, Support Vector Machines and Neural Networks[37] [38]) and novel techniques recently developed at UVQ that combine Optimal Control, System Identification and ML[39] [40] [41], with the final aim of comparing and

[32] Ku, Y., Yang, J., Fang, H. et al. Deep learning of grasping detection for a robot used in sorting construction and demolition waste. J Mater Cycles Waste Manag 23, 84–95 (2021). https://doi.org/10.1007/s10163-020-01098-z

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improving such techniques in accordance with the KPIs of the experimental setup.

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Tool / Technology /Activity	Key Advantage & Progress – Start -> Foreseen End TRL
Optimal control and AI package for recycled plastic	Optimization package increasing the usage of recycled plastic during the process of moulding with a control on the process itself, **TRL4❼TRL6**

Optimisation and prediction of recipes for particleboard panels

State of the Art & Current Challenges (A): Post-industrial wood waste is considered a vital source of feed in manufacture of particleboard (or chipboard), as it exhibits adequate performance for this application while also saves on the use of fresh forest wood. The substitution levels of fresh wood with recycled wood in the production of particleboard (post-industrial and post-consumer) are claimed to range from 20% to even 80%, while the processing methods and parameters and mostly the features of the secondary wood materials used differ widely depending on location, factory and other conditions. The regained wood chips may be bonded with conventional adhesives, both organic or inorganic, to form new particleboard panels dedicated for use e.g. in furniture, insulation. Wood scraps resulting after cutting of planks are currently underutilized in the wood-processing industry, as they are used for production of wood chips. One approach to retain value of those pieces is to utilize them for manufacturing of mosaic, self-supporting surfaces for furniture and cladding applications. In this way minimization of generated waste and value addition will be achieved. One main challenge for assembling the final selfsupporting surface is the heterogeneity and variety of sizes, shapes and colors of the obtained wooden scraps.

Progress beyond SoA (A): In DigInTraCE, data analysis and AI tools will allow to form a correlation between the properties and characteristics of the recycled wood chips feed with the performance of the obtained panels, which will be evaluated following the prevailing European standards. The substitution levels of fresh wood chips with recycled ones will be tested, where conventional resins will be employed for bonding the wood chips and forming the panels used (urea-formaldehyde). The optimisation of use of wood by-products will be conducted via AI-based algorithms and ultimately prediction of final products' quality will be performed. Development of digital optimisation tool, able to automatically identify the pieces, calculate and design of original arrangements between the remnants to create a mosaic-type surface will be performed. By using the catalogue of existing remnants properties (shape, size, color, etc.), the tool will utilize mathematical models (AI algorithms) to optimise the usage of wood scraps, thus.

State of the Art & Current Challenges (B): Several different wood by-products with different characteristics are produced in-site along the value chain (e.g. sawdust, woodchips and wood scraps), currently considered underutilized. Efforts are being deployed towards more efficient recycling and cascading use of industrial wood-based resources according to the quality of the residues and their potential to be valorised[44]. Attributes which make wood by-products good candidates for certain applications vary, i.e. wood element used (particles or fibers), moisture content, particle size and scrap shape. The potential applications involve particleboards, fiberboards, insulation boards, MDF and HDF panels, products with high interest in the building and construction sector. Adhesives used to bond the wood elements are usually petroleum-derived components (formaldehyde-based thermosetting adhesives), but focus has been made to substitution of those adhesives' formulations by biobased, non-toxic sources such as tannins, starch or lignin, to boost the sustainability of the panels production [45,46].

Progress beyond SoA (B): This innovation concerns wood in-site by-products being transformed and upgraded towards higher added-value bio-based construction products (panels, boards and insulation foams). Within the project, higher quality applications are targeted by means of the identification and classification of each material flow, in terms of key parameters such as moisture content, particle size and scrap shape. These three parameters would make a huge difference in the quality and performance of the by-products, as raw materials for the production of higher value wood-based products such as low formaldehyde emission particle boards, wood-plastic composites (WPC), high performance fiberboards (MDF, HDF, etc.), acoustic and thermal insulation panels and solid wooden surfaces.

Tool / Technology /Activity	Key Advantage & Progress – Start -> Foreseen End TRL
AI based wood by products characterization algorithms	Digitalisation to enhance the correlation between the properties and characteristics of the secondary resources with the quality of the final products, **TRL4❼TRL6**
Machine vision-based wood re-design toolkit	Fast data acquisition of parameters needed for further steps Composition of the mosaic-like surface with small wood scraps, **TRL4❼TRL6**

Wood by products automated valorization flow selection	Higher value products validation Development and characterization of new high value products with sorted/classified by-products as secondary raw materials causing its value retention, **TRL4❼TRL6**

Closed-loop supply chain solutions based on low emission value chains

State of the Art & Current Challenges: Due to the increasing worldwide waste production, continuously growing costs and environmental impacts dictate a circular model of production and consumption[47]. From business perspective, this trend

44 Vis, M., Mantau, U., Allen B. Study on the optimised cascading use of wood (Eds.) No 394/PP/ENT/RCH/14/7689. Final report. Brussels. (2016).

45 Antov, P., Krišťák, L., Réh, R. et al. Eco-Friendly Fiberboard Panels from Recycled Fibers Bonded with Calcium Lignosulfonate. Polymers, 13, 639 (2021). "https://doi.org/10.3390/polym13040639"

46 Kariuki, S.W., Wachira, J., Kawira, M., et al. Formaldehyde Use and Alternative Biobased Binders for Particleboard Formulation: A Review. Journal of Chemistry, (2019). https://doi.org/10.1155/2019/5256897

47 Jauhari, W.A. et al. A closed-loop supply chain model with rework, waste disposal, and carbon emissions, Operations Research Perspectives, vol. 7 (2020). https://doi.org/10.1016/j.orp.2020.100155

is leading to closed-loop supply chain (CLSC) solutions, integrating both forward flowsAssociated with document Ref. Ares(2022)7761208 - 10/11/2022 (raw materials manufactured into products) and reverse flows (used products collected and remanufactured). The forward flow goes from the purchase of raw materials from suppliers, the design and manufacturing of the single components, the assembly of the final product up to the product distribution and sale. The reverse flow includes product and/or sub-product acquisition, component analysis, disposal or refurbish, and redistribution[48]. In conventional supply chains, product demand can be foreseen with conventional forecasting techniques. In CLSC, the high level of unpredictability and uncertainty of returned products and components makes the forecast more challenging. Different methods and approaches, ranging from mathematical to operations research techniques, are proposed in literature for different parts of the management process. Game Theory (GT) approaches can be applied to model CLSCs and analyze the behavior of rational players, stakeholder choices, and the development of voluntary extended producer responsibility policies. Decision making can be focused on the pricing decisions, channel operational management, remanufacturing strategy and coordination mechanisms[49].

Progress beyond SoA: In this project, decision making mechanisms and AI approaches will be applied in order to optimise the CLSC management in the real use cases. The objectives are: (i)improve the production process introducing analytics mechanism for the evaluation of the data collected by the digital factory framework with respect to the recipe and the production machine behavior while there are using recycled plastic; (ii)define AI algorithms able to support the Supply Chain mechanism with specific reference to the usage of recycled plastic.

Tool / Technology /Activity	Key Advantage & Progress – Start -> Foreseen End TRL
CLSC management toolkit optimization plastic recycling	AI based optimization toolkit tackling the Supply chain management and the production processing, **TRL4❼TRL6**

Online LCA computation tool

State of the Art & Current Challenges: LCA methodology and circularity evaluation of circular value chains have been developed in several EU H2020 projects, e.g., CIRC-PACK, POLYNSPIRE, from a static point of view. The inclusion of dynamic process parameters (real-time data) in the environmental module leads a multi-modular approach able to calculate the main environmental impacts caused by the inclusion of new technologies under different operating conditions. Therefore, the dynamic monitored data (operation variables), used to feed several KPIs, allows to evaluate the environmental impacts of the whole value chain under different operation scenarios.

Progress beyond SoA: LCA algorithms will be integrated for a dynamic multi-criteria assessment that will support the optimization of the developed solutions considering the full value chain impacts. In addition, the tool will be used to provide automatic information for the Product Environmental Footprint under the value chain approach (decision making tool).

Tool / Technology /Activity	Key Advantage & Progress – Start -> Foreseen EndTRL
Online LCA computation tool	Dynamic multicriteria assessment with integrated PEF, **TRL4❼TRL6**

VII. Up-cycling, reuse and upgrade technologies of secondary raw materials

Protein recovery chemical processes

State of the Art & Current Challenges: Conventional processes for recovery of proteins from seeds biomass, include hydrolysis, enzymatic hydrolysis and solvent extraction. These technologies face challenges, such as low yielding, high cost, use of harsh solvents with great environmental impact. It is important to find cost-effective sustainable technologies to maximize the recovery of high-value compounds from the secondary raw materials and

to overcome the difficult conditions for extraction, resulting from the complex matrix in which the proteins are embedded [50].

Progress beyond SoA: In the framework of DigInTraCE project, green and sustainable recovery techniques for secondary raw materials valorization will be used and developed, in order to overcome these challenges. Pressurized liquid extraction (PLE), Ultrasound (UAE) and Microwave assisted (MAE), in combination with green solvents, will be used for protein isolation, offering high yielding, selectivity and reproducibility (in min instead of h), leading to a simplified manipulation and work-up, as well as a reduced solvent and energy consumption. NTUA owns laboratory equipment for implementing the pre-mentioned protein recovery processes and NTUA team has already developed protocols for protein recovery from various natural sources and agricultural by-products of similar composition [51][52][53]. These protocols are going to be tested and optimized using the by-products from oilseed crops (sunflower, rapeseed, cotton press cakes etc.). The project aims to validate the concept up to technical scale (TRL 5).

Tool / Technology /Activity	Key Advantage & Progress – Start -> Foreseen End SRL/TRL
Protein recovery chemical processes	Recovery technologies for protein isolation **TRL4❼TRL5**

48 Shekarian, E., A review of factors affecting closed-loop supply chain models, Journal of Cleaner Production, vol. 253 (2020). https://doi.org/10.1016/j.jclepro.2019.119823

49 Zhang, X., Li, Q., Qi, G. Decision-Making of a Dual-Channel Closed-Loop Supply Chain in the Context Government Policy: A Dynamic Game Theory. Discrete Dynamics in Nature and Society. (2020). https://doi.org/10.1155/2020/2313698

50 Cravotto, G. et al. Improved extraction of vegetable oils under high-intensity ultrasound and/or microwaves. Ultrasonics sonochemistry, 15(5), pp.898 (2008). https://doi.org/10.1016/j.ultsonch.2007.10.009

51 S. Papadaki, V. Oikonomopoulou, D. Mavropanos, M. Krokida, Isolation of proteins and prebiotics from chickpeas using environmentally friendly methods, 35th EFFoST International Conference 2021, 1-4 November 2021, Lausanne, Switzerland

52 C. Tsaroucha, V. Oikonomopoulou, K. Kritsotakis, E. Topakas, M. Krokida, Isolation of proteins from by-products and residues of the brewing industry, CORFU 2022 9th International Conference on Sustainable Solid Waste Management, 15-18 June 2022, Corfu, Greece

53 Liu, Y., et al., Ultrasonic-assisted extraction and functional properties of wampee seed protein. Food Science and Technology, 2018. 39.

Bio-based adhesives development Associated with document Ref. Ares(2022)7761208 - 10/11/2022 **State of the Art & Current Challenges:** Formaldehyde-based thermosetting adhesives (urea-formaldehyde (UF), melamine-formaldehyde (MF), phenol-formaldehyde (PF), and their combinations) are the main binders used in the production of composite wood panels. Although they offer advantages like constant product quality, their building blocks have petrochemical origin (non-renewable fossil resources like oil and natural gas). The environmental problems associated with petrochemicals, potential health risks from formaldehyde use, and reliance on importation of oil have recently renewed industry interest in bio-based adhesives[42]. In wood-composites the most common types of bio-based adhesives are derived from protein, tannin, or lignin. The press cakes produced during the manufacture of oil from oilseed crops (sunflower, rapeseed and even cotton) are characterized by high content of protein and amino acids[43]. Combinations with synthetic binders are currently the preferable option, and the degree of industrial application and market acceptance for bio-based adhesives is still low due to availability, technical performance, and renewable raw material costs, requiring further development and demonstration of successful formulations.

Progress beyond SoA: In this project, CHIMAR which has successfully formulated protein-based adhesives at lab scale using various proteins derived from organic waste and by-products, will use protein samples from press cakes to replace part of the petrochemical phenol that is needed in the adhesive formulation at pilot scale and at a relevant environment. The highest phenol substitution level achieved so far is 40% when using proteins from rapeseed.

Tool / Technology /Activity	Key Advantage & Progress – Start -> Foreseen End SRL/TRL
Bio-based adhesives process engineering	Increase the phenol substitution levels in adhesives by 40% **TRL4❼TRL6**

Upcycling of plastic

State of the Art & Current Challenges: In order to insure a good product quality the quality of the recycled plastics need to be controlled. Current approaches can either explore a vast chemical space by high-resolution and trace-level off-line analytics from small spot samples or rely on averaged low information inputs like NIR classification from automated sorting. Despite exhaustive publication on plastics recycling procedures, including automated sorting for industrial and waste management sorting is often done manually involving critical health and safety aspects, and to

42 Lin et al. Study on preparation and properties of phenol-formaldehyde-chinese fir liquefaction copolymer resin, Ciencia y tecnología 16(2): 159-174 (2014). DOI:10.4067/S0718-221X2014005000013

43 Yang, I., Kuo, M., Myers, D.J. et al. Comparison of protein-based adhesive resins for wood composites. J Wood Sci 52, 503–508 (2006). https://doi.org/10.1007/s10086-006-0804-5

identify critical issues affecting material separation e.g. material mixing incompatibility impairing recycling performances or dismantling costs.

Another aspect that is insufficiently monitored is the thermal and mechanical stability of each recyclate. During service lifetime, polymers may suffer from degradation caused by thermal oxidation and photoxidation. During re-processing of plastic waste, chain scission and/or crosslinking reactions occur for a range of polymers, leading to polymeric structure alterations with consequent variations in crystallinity, opacity and deterioration of mechanical properties directing the recovered polymers to low-requirement applications. However, pure polymer waste can be efficiently recycled into new added-value recyclates through adapted reprocessing - upcycling - consisting of maintenance and/or increase of the polymer waste quality (e.g. molecular weight, macromolecular structure, functionality through additivation).

Progress beyond SoA: In the framework of DigInTraCE project, a high demanding textile case is foreseen whereby different polyester waste streams will be monitored over its life time so that critical processing steps can be determined. Models for lifetime prediction using available data from real-aged polymer products will be developed and validated by simulated labbased accelerated aging and prediction models, promoting the safe and sustainable use of recycles in added-value products, targeting for example high demanding textile sectors. The degradation degree of EoL plastics (input products) will be assessed as both input for the lifetime models, but also as input for the EoL processing and treatment (recycling/upcycling etc.).By using smart sensor techniques and where needed additivation the quality of the materials will be improved leading to good quality recycled polyester making it possible to reprocess it in high tenacity textile fibers al TRL 6.

Tool / Technology /Activity	Key Advantage & Progress – Start -> Foreseen End SRL/TRL
Upcycling of plastic	Upcycling technologies for plastic **TRL3❼TRL6**

1.2. *Methodology*

1.2.1. Overall Concept and approach

DigInTraCE contributes to circular and low emissions value chains of process through digitalization. Digital tools and information systems will be integrated forming a multifunctional and interoperable decentralized traceability platform based on blockchain technology. End-to-end digitalization of the value chains will be implemented, as online monitoring, tracking and tracing of streams will be established via novel sensing (e.g. HIS) and tracking techniques (e.g. QR, smart tags) for automatically obtaining real-time information. Enhancement of DPPs through dynamic update of product information, is one of the project's main innovations. The specific innovations beyond SoA described in 1.1.3 will be clustered into the below 4 main modules and additional services that are combined shaping the overall system.

Decentralized Traceability platform – Dynamically Updated Digital Product Passport *(contributing to all Objectives)* Based on ***Decentralized Traceability Platform*** and integrating the ***Dynamically updated DPP*** DigInTraCE expands currently applied traceability schemes, paying special focus to secondary raw materials and products and frequently neglected

needs in existing practices. Thus, emphasis will be given on capturing information regarding secondary materials' properties,

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composition and origin, not only to enable quality assurance but also to ensure more efficient and sustainable transformation into products. Extracting all the secondary materials transformation-associated data throughout the processing line and process them with the use data analysis and AI offers an opportunity to facilitate optimization of process efficiency, productivity and end-products' recipes, and assist interested stakeholders' decision-making related to circularity, environmental impact and economic viability. The platform will assure secure and safe data exchange through intra- and inter-firm communication channels, by application of corresponding accessibility rights via open data/software/hardware and will operate as neutral data holder, thus supporting features that create value from data without exposing sensitive information. All the modules described below (sensing, sorting and data driven activities, certification and standardization schemes, additional financial and educational services) will be integrated into the platform, facilitating the optimization of secondary raw materials and forming the overall system. This optimization has been approached in an extensive way, as illustrated in Figure 1, by deploying the concentrated digital technologies to enhance the chemical upgrade of secondary raw materials. At product and process level, optimization to efficiently valorise the materials and consume less energy, by applying novel digital technologies (e.g. data-driven optimization, novel sensing and sorting mechanisms) and chemical innovations for upcycling (e.g. protein recovery from rapeseed cake for bio-based resins development) to be employed.

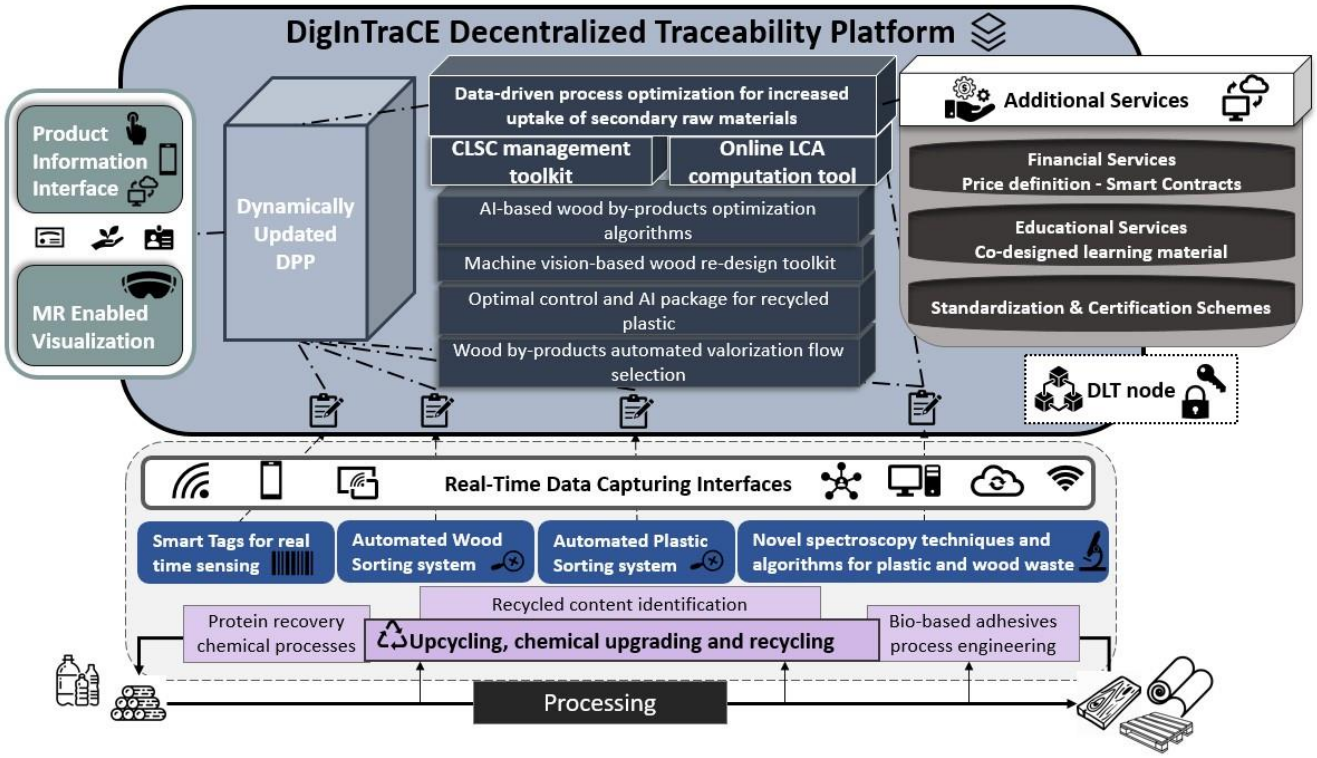

Figure 1: DigInTraCE Overall Approach (modules' colors corresponding to sections in the text)

Novel sensing, Smart Tags and Sorting Mechanisms *(contributing to Obj 1,3,4,5,6)*

DigInTraCE offers digital tools that facilitate automated and high accuracy identification of secondary resources quality properties, characteristics and environmental conditions, that will be integrated to the platform. ***Novel spectroscopy techniques & algorithms for plastic waste & wood by-products:*** Online sensors that rely on spectroscopic techniques, involving novel functionalities customised in compliance with the key parameters that should be monitored within each process will be applied. Hyperspectral imaging assets will be utilized to detect and categorise polymers and contaminated components assisted with NIR technology. Some highlighted properties, based on which sorting will be performed, include (but are not limited to) size and shape of wood chips and scraps, moisture content, previous chemical treatment, etc. Realtime identification and classification according to all the requested characteristics in terms of quality, will be performed by industrial hyperspectral cameras utilising vision-based techniques. Sorting mechanisms will be designed based on the above relevant sensing techniques, to improve the quality of the feedstock for the upcycling processes. ***Automated Plastic Sorting system:*** robotic arms assisted by machine- or deep- learning techniques, will be used for the removal of the defective items and impurities resulting from waste assessment analysis, implemented in plastic based pilots (Belgium and Italy)**.** ***Automated Wood Sorting system*****:** categorisation according to size and the removal of defective pieces using robotics arms followed by sieving, manipulated in real-time using hyperspectral imaging assets assisted by machine- or deep- learning techniques and NIR technology. This sorting mechanism will enable reduction of waste and increased identification accuracy and efficiency in recovery of high-quality secondary resources, designed for and installed in wood by-products cases (Greece and Spain). ***Smart tags for real time sensing*** for real time sensing and delivering information to the decentralised platform will be utilized.

Data-driven process optimization for increased uptake of secondary raw materials *(contributing to Obj 1,5,6)*

Tools and AI based features will be included in the Decentralised Traceability Platform for process and product optimizations

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and facilitate decision-making using real-time measurements and related data. ***Optimal control and AI package for recycled plastic***: during injection moulding designed to also control the process, in terms of auto-adapting to improve the quality of second life materials. ***AI based wood by products optimization algorithms*** will be developed, connecting materials characteristics' and production process parameters, with final products' properties. A wood by-products automated valorisation flow selection based on data analysis will be developed, in order to minimize value decay. High speed data acquisition of parameters related to composition of small wood scraps will be exploited by the ***Machine vision-based wood re-design toolkit*** which will enable the arrangement of pieces to create a mosaic-like surface. Stakeholders' decision making will be assisted in terms of supply chain management, planning and scheduling via the ***CLSC management toolkit optimization plastic recycling*** whereas environmental impact and

sustainability of selected pathways for optimisation of secondary resources will be evaluated dynamically via the ***Online LCA computation tool based on Ex-Ante LCA***.

Up-cycling, reuse, and upgrade technologies of secondary raw materials *(contributing to Obj 1,5,6)*

Project's digital framework, transversal tools and data driven optimization described will be applied in 4 different value chains in relevant industrial fields combined with technologies for the upcycling, reuse and upgrade of secondary materials and replication potential. The value chains mainly involve wood-based industry and associated pulp & paper sector, as well as chemicals sector, in terms of plastic recycling and bio-based resin development. ***Bio-based adhesives process engineering:*** Regarding first wood-related chain (Greece) wood composites are manufactured for future furniture applications, by using underutilized, high-quality wood by-products and by utilizing agricultural by-products from oilseed crops, for development of bio-based binders via ***Protein recovery chemical processes.*** In the other wood-related case (Spain) wood by-products are characterized and classified in compliance with quality parameters by means of digital technologies and transformed accordingly towards added-value products (***AI based wood by products optimization algorithms*** referred already***).*** Regarding the plastic-relevant value chains, in first (Belgium) polyester recycling for textiles application will be introduced, highlighting the determination of 'recycled content' by detecting fingerprint molecules (***Recycled content identification, certification scheme***) while in the second (Italy) the target is to increase the polymers detection and recycling efficiency and reduce the amount of materials which are incinerated, by optimizing the process of injection moulding of recycled plastic (via ***Optimal control and AI package for recycled plastic*** and ***CLSC management toolkit optimization plastic recycling***).

Additional Services *(contributing to all Objectives)*

Material Characterization, Information Access and Security, Standardization and certification Schemes

Combining several results already referred, Decentralized traceability platform captures real-time data via interfaces to detect both object properties and changes through time and reactions to different environmental conditions. Additional data from the industrial stakeholders' legacy systems will be also integrated. Data standards will be aligned to GS1 and other special standardization schemes. Blockchain and DLH will be used to provide decentralised, transparent, immutable and secure data structure for all the information gathered. End-to-end cybersecurity solutions for proper user and consent management, to ensure security, confidentiality and integrity of data will also be fostered. The automatic update of DPPs via a live update mechanism is contingent on digital models involving rules to create or retrieve information, either relying upon user actions or items' state changes and reactions detected by human or digital sensors. This mechanism will support the real-time tracking and monitoring of materials and components throughout the value chain, while also provides interactive visualization and controlling functionalities of DDPs using Mixed reality (MR). Imperative information regarding the composition, quality characterization and categorization as well as safety data sheets and other standards will be recorded and displayed in the decentralized traceability platform, especially when addressing characterization and sorting technologies for the optimisation of the materials recycle. Certification schemes will be also integrated (***Recycled content identification, certification scheme).*** AIbased tools will be integrated for optimizing value chains, identifying hidden patterns and exploring new potential opportunities in terms of resources exchange, reuse and recycling of wastes and by-products between the same or multiple industries/factories. A proper ontology and data models shall be adopted to support the complexity of the recycling and circular value chains, concerning the second (and so on) lives of materials.

Financial services, smart contracting and business models

In the circular economy actors are not trying to optimise their own cost, quality and waste parameters in isolation, but rather their contribution to the overall ecosystem's parameters. This may create misalignment between the risks and rewards when procuring materials and these may be distributed over a time scale. i.e. the circular optimum may require more capital expenditure by the primary actor in order to reduce the waste of a secondary actor five years later. Financial derivatives such as options, swaps and futures are widely used in commodity markets to move expense, risk and reward through time and between the actors, with financial traders providing liquidity, price discovery and capital flows. In the circular economy those same instruments can dissociate the capital and risk, for example the primary actor selling a future or option on the secondary raw material with an expiration date based on the primary object's lifespan. Given that DigInTraCE intends to track and trace through a blockchain / DLT data structure, and that the same blockchain infrastructure can manage tokenised financial derivatives, a testbed is provided to examine the potential of tying traceable secondary raw materials to rights enshrined in legal contracts (financial derivatives) represented by tokens. The financial traders purchasing of these rights would use this instrument in their financial strategy before selling the option on to an industrial actor requiring the by-product, close to expiry. The value of financial derivatives is highly understood and describable by mathematical equations. Such

derivatives are common today, but are used with standardized commodities such as wheat and wood where traceability is not required Associated with document Ref. Ares(2022)7761208 - 10/11/2022
(because they are widely available commodities). The missing step today is the ability to enforce the contract on materials that are not so abundant and standardized as to be a commodity, or even to non-fungible items, i.e. a specific object. (This is essentially a liquidity issue). However, by combining the financial instrument in the same blockchain as the track and trace of the underlying object, and even more so, through smart contacts tied to wallets containing monies in the form of cryptocurrency, potentially one can indeed enforce the validity of the contract. This achievement would solve imbalances in risk, reward and capex between the actors in a circular economy, supported by market makers. DigInTraCE will study and develop financial services to support stakeholders in the secondary raw materials market. This will include analysis of the viability of the above scenario from a financial perspective and enhanced by studying the capability of tokenisation and smart contracts to support such a solution by enshrining and enforcing the required instruments.

Educational services, co-design material

Educational services based on co-created training materials will be included in the DigInTraCE platform, contributing to the development of digitally skilled population and workforce. The main focus will be to identify status quo of staff members' digital competence and discover future requirements of the market, in order to provide tailor-made enlightenment, guidance, training and mentoring services for employees and citizens by internal and external experts, specialized into specific sectoral, regional or national social needs. Validation of the acquired skills and progress will be performed regularly through formal and informal channels. Workshops for validating and co-creating educational materials will be organized.

1.2.2. Scientific Methodology

DigInTraCE concept relies on the implementation of a circular approach for the whole lifecycle of secondary resources. Circularity principles require integration of the entire value chain, implementation of detailed monitoring and traceability systems and assurance of increased use of secondary materials, as well as the integration of secondary resources in the processes. These requirements imply the execution of a comprehensive methodology that follows an iterative multistep development approach, in which user requirements and corresponding solutions evolve through multiple iterative collaboration rounds involving all the stakeholders. The core of the methodology is agile approach, adaptive development and continuous improvement based on iterative rounds including the following 4 reciprocal phases given that knowledge increases over time through interactions between different phases and stakeholders.

Phase 1: Data gathering, characterization and definition phase (M1-M8) The first phase of the project sets the baseline for its execution. A preliminary analysis of the secondary materials quality characterization for recycle, reuse and upgrade per value chain. The exploration of digital traceability methodologies for secondary materials across value chains will be initiated and all the related requirements regarding digital tracing, DPP, real-time access to information and supplementary services for the development of a holistic Decentralized traceability platform will be gathered. Finally, an assessment of current situation of involved value chains in the demonstrators in terms of waste generated and uptake of secondary resource will be performed.

This phase is performed the first 8 months of the project (M1-M8) and covers WP1 (all), WP2 (all), WP3 (all), WP3 (T4.14.4), WP5 (all), WP6 (T6.1) and WP7 (T7.3-7.4).

Phase 2: Design, development and testing phase (M9-M24) Demo cases assessment is finalized and 3 parallel stages are established: **P2.1:** initial design and development of the new **technological solutions in the Pulp & Paper and Chemical value chains**; testing in laboratory level of new processes and products; and the establishment of the holistic digitalisation of circular value chains framework in the industrial ecosystems in the 4 demo cases. **P2.2:** system specifications of the monitoring tools for **digital tracing** along the value chains defined; their deployment in the field along with the development of the live update mechanism for DPP, algorithms, smart tags, sensing and sorting technologies and the rest digital tools/services (T3.4, T3.5, T4.3, T4.4). **P2.3:** Based on the above progress, activities related to research for secondary materials' **certification** and **standardization**; knowledge exchange, **learning programmes** and training; replication and transferability across sectors; cooperation and clustering with other activities; business plans and exploitation activities will be planned, initiated and performed.

This phase is performed during 16 months of the project (M9-M24) and covers WP1 (all), WP2 (all), WP3 (all), WP4 (all), WP5 (T5.2-5.5), WP6 (T6.1-T6.3) and WP7 (all).

Phase 3: Implementation Phase (M25-M36) The demonstration of DigInTraCE solutions, technological and digital innovations will take place in this phase in 3 parallel stages: **P3.1:** The real environment implementation and **operation of the technologies for secondary materials** according to the particularities of each demo-site, industry and value chain will be demonstrated. **P3.2:** digital demonstration activities will begin, making sure that the most relevant information is monitored and tracked according to the baseline captured in the first phase

for each demo-site. The methodologies for digital tracing analysed will be utilized for the establishment and the real **operation of the blockchain based decentralized traceability platform**, the **dynamically updated DPPs** for all the involved value chains, along with the sensing techniques, smart tags, AI algorithms and sorting systems. Also, the integration and testing of all the rest digital tools/services (CLSC, LCA, financial, MR) will be performed while system specifications and architecture will be enhanced and finalized. **P3.2:** The **certification** schemes will be applied and **standardization**, **educational co-created** and other transversal activities will be fed with real data and information.

This phase is performed during 12 months of the project (M25-M36) and covers WP1 (all), WP2 (T2.1,T2.3,T2.4), WP3 (all), WP4 (all), WP5 (T5.2-5.5), WP6 (T6.2-6.4) and WP7 (all).

Phase 4: Integration, Solution Refinement and Knowledge Exchange (M37-M48) The demonstrators are expected to operate at TRL6 allowing the evaluation and validation of the results, in terms of performance and impact assessment for all the designed systems. The final version of all digital tools designed and tested will also be included within the final version of decentralized traceability platform. Final conclusions from standardization and certification activities will be drawn. Moreover, once the final results have been gathered together with feedback from main stakeholders, corresponding policy recommendations final business models, enhanced learning programmes and exploitation strategy will be defined and a final exploitation meeting will be held with the consortium to set up the path for replication of the project results after the project's end.

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This phase is performed during the last 12 months of the project (M37-M48) and covers WP1 (all),WP2 (T2.1, T2.3), WP3 (T3.5), WP4 (all), WP5 (T5.2-5.5), WP6 (T6.2-6.4) and WP7 (all).

1.2.3. Demonstrators

The value chains that the project implements and demonstrates the overall concept, are selected explicitly for belonging to industrial sectors highlighted by P4P roadmap, while also allowing the validation of closed loop supply chain solutions employing concepts as recycling, reuse and upgrade of materials by deploying optimized chemical processes.

Sectors	Location	Partners involved and their role
Chemicals industry (binders, biopolymers), Pulp & Paper and Wood industry (composite wood-based panels)	Greece	CHIMAR – Demo owner; NTUA – Demo Coordinator and Research partner; AGRST – supplier of secondary resources; MXS – supplier of secondary resources and receiver of end product; VTT - Research partner, smart tags and data management; ICCS – Technical contribution
Demonstrator overview: The main value chains to be examined involve agricultural by-products from oilseed crops, supplied by AGRST (industry engaged in agribusiness), as well as wood by-products from MXS (furniture manufacturer). NTUA (research partner) performs extraction of proteins from the oilseed crops that are supplied to CHIMAR (industrial partner with competitive R&D services in the field of chemicals and adhesives), to produce bio-based adhesives. Underutilized wood by-products of MXS are also supplied to CHIMAR. Various wood-based composites are manufactured, ultimately to be utilized by MXS for the production of furniture and other applications. VTT (research partner) develops methodologies for smart tags and their GAIA-X compatible data management and storage as well as strategy supporting end-of-life management. ICCS contributes to the demonstration site with the design of a wood sorter.		
Overall Concept of the activities to be implemented		

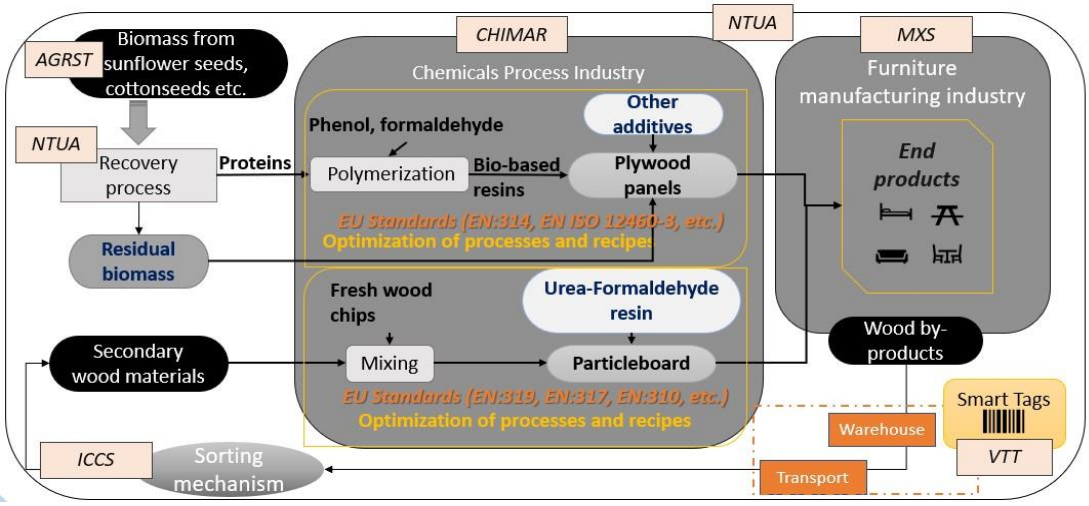

Figure 2: Greek Demonstrator

Demo concept description: Phenol-formaldehyde resins (phenolic resins) are synthetic polymers that are widely used as adhesives for bonding of plywood and other engineered wood products. These adhesives are of petrochemical origin, increasing environmental pollution and therefore, the need to replace them with natural bio-based products as substitutes for conventional wood resin adhesives is crucial. Proteins are a group of natural and renewable materials, suitable for use in wood adhesives. In DigInTraCE, **proteins and amino acids will be recovered from oilseed crops** (sunflower, rapeseed and cotton press cakes) **and utilized as building blocks in phenolic bio-based resins** by CHIMAR, using condensation polymerization, replacing part of the petroleum-based phenol material. The developed resins will be characterized in terms of rheological properties, reactivity, free formaldehyde, and other properties needed to assess the resins behaviour and performance.
The resins will be then mixed with a variety of additives, such as fillers, hardeners/catalysts, etc., **to produce wood composite materials for construction and furniture applications, and specifically plywood panels**. Furthermore, **secondary raw materials from the wood industry** (wood residues provided by MXS), that are now mainly used for heat generation in wood-based panels factories, will be mixed with fresh wood chips **to develop a wood composite material, called particleboard**, by CHIMAR, using a conventional Urea-Formaldehyde resin. The properties of both the produced plywood and particleboards will be evaluated and compared with that of reference panels produced from conventional materials. DigInTraCE will, finally, lead to the production of two new wood-based composites, i) **plywood produced with bio-based adhesives** and ii) **particleboards produced with the addition of wood waste** that will be evaluated by MXS in the furniture industry. The developed composites will be characterized by more sustainable bio-sources of raw materials, improved health and safety during manufacture and end use, enhanced performance characteristics and improved environmental profile.

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Digital technologies to increase waste reduction: In this demonstrator, wood residues of MXS will be sorted with an automated wood sorting system,co,, designed by ICCS, and selected compounds will be sent to CHIMAR to be processed and used to develop wood composite materials (particleboards). Additionally, through the processing of the data regarding: (a) the properties of the proteins extracted by NTUA, (b) the production parameters and characteristics of the phenolformaldehyde resins synthesized therewith and (c) the final properties of the plywood panels produced by CHIMAR, the possibility of developing a digital model will be explored, in order to identify the characteristics of the protein raw materials as well as the resin and panel production process parameters, that could optimise the recipes and predict the final products' properties and performance (for both the produced resins and panels). Item level tagging will be supported through durable smart tagging solutions (i.e. printed, unique 2D barcodes, potentially RFID) developed by VTT. Tagging solutions will include the means for identifying the data items as well tailoring of functional tagging for sensing and authentication as feasible, as an example humidity monitoring tagging solutions can be targeted. The data from the various processes will be combined and introduced into a data management system and through application of IT tools (artificial intelligence, machine learning etc.) knowledge will be extracted related to the optimum processes and conditions, which can then be promoted for market application. GaiaX compatible data structure/architecture to support data collection will be designed (VTT). Simulations and mock-ups apps to engage stakeholders to more efficient use/ re-use/ re-cycle/sorting (dynamic

updating of item information) as well as piloting customer/end-user data collecting and connecting the data in the platform will be carried out. The project aims to validate the concept up to TRL 5 or 6.
Valorization of use of secondary raw materials in the value chain: In this demonstrator, conventional hydrolysis, enzymatic hydrolysis, as well as green extraction processes (UAE, MAE, PLE) using friendly solvents, will be used and optimized by NTUA, concerning various process parameters (time, power, solid to solvent ratio, temperature, etc.), for efficient isolation of proteins. Proteins will be recovered based on protocols that have already been developed in TRL4 in laboratory environment and will be applied in prototype systems, reaching a TRL5.
Other important activities: A standardisation system will be carried out based on the results obtained and the technical data process developed (considering the specific requirements needed for plywood and particleboards). CHIMAR will characterize the features of the final wood-based composites, based on European Standards (Bonding quality-EN:314, formaldehyde release-EN ISO 12460-3, tensile strength-EN:319, thickness swelling after immersion in water-EN:317, modulus of elasticity-EN:310). Standardisation aspects are going to be addressed aiming to improve the development of the composites and to reduce relevant costs, considering different recipes. Co-design and replication activities will be included as part of the whole project.

KPIs	
Increase of secondary raw materials use:	>80 %
Waste reduction:	>50 %
GHG emissions reduction:	20 %
Number of up-cycling, reuse and upgrade technologies of secondary raw materials implemented:	at least 3
By-products value increase	30 %

Sectors	Location	Partners involved and their role
Pulp & Paper and Wood industry (composite wood-based panels)	Spain, Azpeitia	ASTI – Demo owner; TECNL – Demo leader; ICCS – Technical partner
Demonstrator overview: ASTI is a Spanish company that comprises the whole value chain of the wood sector, as it includes the initial processing of tree trunks originating from the forest and 1st transformation of logs (in the sawmill), and up to their final product, i.e., solid wood furniture with the claim of hard-wearing and decorative products.		
Overall Concept of the activities to be implemented		

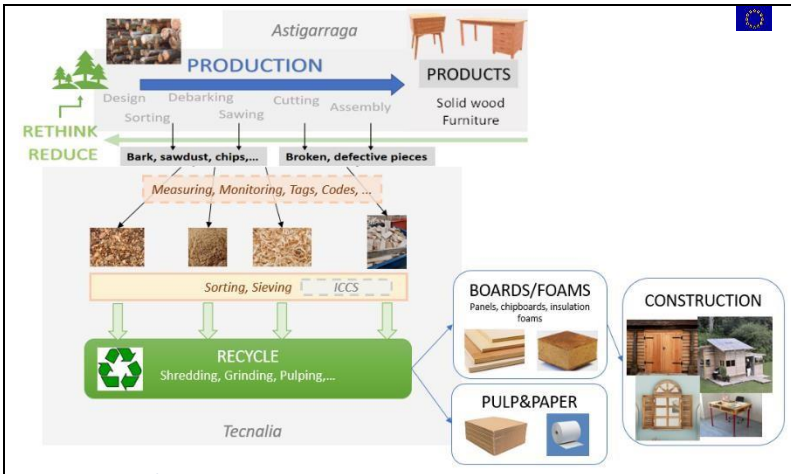

Figure 3 Spanish Demonstrator

Associated with document Ref. Ares(2022)7761208 10/11/2022**Demo concept description:** The overall objective of this demo is value retention and upgrading of woo based by-products by the use of digital technologies. At the permises of **ASTI,** along the whole wood processi including furniture production line, different types of by-products might be produced. Those residual side strea include bark, sawdust, wood chips, wood scraps, sanding dust and broken and defective pieces. The aim of the de is to identify, characterise, quantify and separate the classified by-products by means of digital technologies develop by ICCS. This classification and collected data will allow to trace all produced materials towards the next slab in t furniture value chain and to move into a different sector (such as construction, pulp&paper, packaging). Pro classification of the by-products will allow for new, added-value products to be manufactured by TECNL, such high-value wood-based boards, insulation foams, and new wooden surfaces, as illustrated in the Figure 3. The succ of the demo will allow the value retention of the by-products and, thus, the market value per m^3 (or Ton) of the b products will remain comparable with the current end product.

Digital technologies to increase waste reduction: Digital technologies introduced in the wood value chain byproducts classification and with a well-defined and traceable quality. In this demonstrator, industrial wood by-products of ASTI will be sorted with a wood sorter, designed by ICCS, and selected side streams and pieces will be valorized accordingly by TECNL, in order to facilitate increased waste reduction, upgrade of the materials and enhanced retention of value throughout the chain. The critical parameters to be examined, according to which the residual sidestreams are categorized and sorted, include the size, shape, moisture content etc. DigInTraCE platform will enable traceability of the by-products and the final products, as data from all stages will be collected regarding quality attributes (e.g. properties, composition, origin etc.). Besides, Artificial Intelligence and Computer Vision for optimization of the wood scraps by-products will be used. By using those digital technologies, the scraps (that normally are crushed into smaller pieces, losing value) will be used to create mosaic-like self-supporting functional surfaces for furniture and cladding. Result will be its value retention.

Optimisation of use of secondary raw materials in the value chain: will be realized after the appropriate separation of wood by-products. As a result of sorting by size and by quality requirements (moisture content), application into the demonstrator processes, the energetic valorization (through pellets manufacture) will be reduced and eventually eliminated. Small wood particles (<1mm) are normally mixed with particles of higher sizes to manufacture pellets or boards. However, the particles of that size is very valuable for other applications such as wood plastic composites. Novel wood-based products manufactured include **a)** particle boards containing small-sized bark chips (with low formaldehyde emission); **b)** highquality MDF and HDF panels (manufactured with small fiber sizes); **c)** wood insulation products (manufactured with small fiber sizes) and **d)** wooden self-supporting surfaces (made out of scraps for furniture and cladding).

KPIs	
Increase of secondary raw materials use:	30 %
Waste reduction:	30 %
GHG emissions reduction:	20 %
Number of up-cycling, reuse and upgrade technologies of secondary raw materials implemented:	4

By-products value increase	200%

Sectors	Location	Partners involved and their role
Chemicals process industry (polymers, textiles)	Belgium	CTB – lab-scale, coordinator; SIOEN – PET textile producer; CIRCE – value chain mapping; IRIS – Development of monitoring systems, VTT - Research partner, smart tags and data management

Demonstrator overview: Polyester is the most widely used material for textiles, both for clothing and technical applications. Although polyester recycling is already advanced for some aplications like bottles, the use of recycled polyester for high demanding textile applications is still limited. In this demo, polyester decontamination and upcycling will be tested for end-of-life thermoplastics polyester. Processing of polyester will be performed on labscale by CTB. The produced recycled polyester will be provided to SIOEN, where PET textile will be produced. SIOEN is a vertically integrated company and a large convertor of polyester (20kTon / year), of the standard quality (e.g. for non wovens) and of the high tenacity (HT) polyester (e.g. for techncial applications). CIRCE will support this pilot in the form of value chain mapping and CTB as demo coordinator will be responsible for the lab-scale PET

recycling and characterisation of the recycled materials. Supply of secondary raw materials will be provided by

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EREMA, an entity that has ensured its support via a letter of support, although not joining the cosnortium.

Overall Concept of the activities to be implemented

Demo concept description: The main target of this demo is to focus on the qualities of PET resulting from polyester recycling, in order to fit textile applications. The potential waste streams to be used are end-of-life plastics and textiles. Two types of recycled polyester qualities for textiles applications are identified, normal PET (standard quality) and high tenacity (HT) PET. Ultimately, the goal is to utilise recycled normal PET for fibre-to-fibre PET recycling, however, one of the main challenges that hampers its wide uptake is that decontamination of suitable streams and thermoplastic recycling of the material is required. In this demonstrator, two relevant waste streams are considered: (i) polyester textile waste, where monitoring of the fingerprints of degradation molecules will be performed in order to monitor the impact of its use life on the properties of the recycled polyester and (ii) polyester packaging materials having a relatively short lifetime (e.g. bottles, trays,etc.), where ratios of polymers will be monitored. The bottlenecks are situated at several levels, e.g. sorting (where the digital tracking/DPP can help) but also dealing with the contaminations present in the sorted out 'monostreams' and being able to recycle and upcycle the PET material in a cost efficient way, by processing it an a innovative recycling line combined with polymer upgrading using additives. Overall the concept of the demonstrator is illustrated in Figure 4. The recycled polyester pellets will be characterized in terms of rheological properties, molecular weight, crystallisation behaviour, and other properties needed to assess the polymer behaviour and performance. The upcycled pellets will be processed on a textile extrusion line by CTB and upscaled by SIOEN. The properties of both the produced fibers and the final textile products will be evaluated and compared with that of reference conventional materials. DigInTraCE will, finally, lead to the production of a new polyester textile with maximum recycled content. Life Cycle Assessment (LCA) will continuously be applied for the evaluation of the environmental performance of the developed textiles.

Digital technologies to increase waste reduction: In this demonstrator, textile and plastic materials will be sorted and further processed. A smart detection of ratio of polymers, moisture content (IRIS) and fingerprint of degradation molecules (CTB) will be installed. Through the processing of data regarding: (a) the properties of the recycled polymer pellets (b) the production parameters and characteristics of the polyester fibers and (c) the final properties of the textile by SIOEN, the possibility of developing a digital model will be explored, in order to identify the characteristics of the recycled polyester resin raw material as well as the textile process parameters.

The data from the various processes will be combined and introduced into a data management system and through application of IT tools (AI, ML etc.) knowledge will be extracted related to the optimum processes and conditions. Moreover, the use of functional tags for sensing humidity and temperature will be tested by VTT on packaging materials. Simulations of different storages conditions will be tested and linked with the processing into textiles making it possible to collect relevant date on the history of the recycled polyester to predict their re-use in high demanding textile application and predict the needed reprocessing. The project aims to validate the concept up to technical scale (TRL 6).

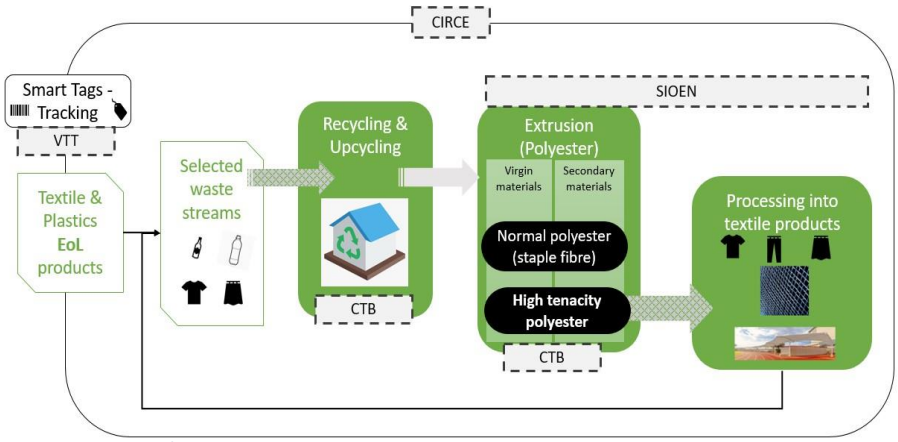

Figure 4 Belgian Demonstrator

Optimisation of use of secondary raw materials in the value chain: will be realised after the appropriate sorting and screening of both textile wasts streams and packaging waste streams. As a result of sorting by quality requirements, application into the demonstrator processes, the re-use of textile and plastic waste will be reduced and, eventually, eliminated. Within the project, CTB will use a innovative recycling line combined with additional upgrading steps to guarantee the qualitty of the raw materials to enable the processing of PET waste and process it back into a secondary raw material suitable for fibre extrusion. This will require adaptation of the equipment to include a cleaning step and to enable upcycling the material so that it has again an Intrinsic Viscosity (IV) fitted for filament extrusion. Further, this might be helped by using additives like chain extenders. Lab-scale tests for the recycling and for the subsequent filament extrusion will be performed at the lines of CTB, results can be upscaled at SIOEN (for extrusion).

Standardization and Certification Determining 'recycled content' is important for various types of materials in the textile sector. In the project we will focus on polyester, the most used synthetic polymer for textile products. The recycled content will be evaluated by fingerprint molecules. These can be either degradation molecules or combinations of additives that are not used in these combinations in virgin polymers. A good fingerprint exists of at least one very specific degradation product or a specific combination of non-compatible additives. The novelty in this project exists in determining those markers which have a high specificity, since this is needed for its use as detection method of recycled content. An overview of standardization efforts and a liaison to the most relevant committees in CEN a ISO will be made, e.g., CEN/TC248 WG39 'Circular Textiles Chain' (which is chaired by CTB) and the CEN SA workgroup mapping standards influencing Circular Economy. We will contribute to the ongoing and future standa developments in specific topics, promoting the inclusion of the findings of the project in future new or revised standa that can be used by the European or international industry. Depending on the maturity of the standardization sect this can involve providing information, submitting technical proposals and even promote the elaboration of n standard documents/deliverables. We will investigate how the digital tracing explored within the project can be use for certification activities. CTB is involved in PolyCertEurope, currently mainly based on a mass-balance system. T digital tracking combined with the fingerprinting should allow to also do a certification based on actual measureme of the recycled content of the end product. We will investigate how this can be integrated to extend the existi PolyCertEurope certification scheme. Within the project we will focus on PET for the actual tests but the conc developed will also be applicable for other polymers.

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KPIs	
Increase of secondary raw materials use:	>50 %
Waste reduction:	30 %
GHG emissions reduction:	20 %
Number of up-cycling, reuse and upgrade technologies of secondary raw materials implemented:	at least 2
By-products value increase	50 %

Sectors	Location	Partners involved and their role
Process industry producing plastic components for the automotive market	Italy	SIGIT – Demo owner; DGS – Demo Coordinator; UVQ – Demo research partner; IRIS – Development of monitoring systems; ICCS – Development of sorting systems

Demonstrator overview: The Italian demonstrator aims to improve the efficiency of a circular economy flow of recycled polymeric materials for injection moulding. The flow consists of heavy and contaminated plastics with a capacity of processing up to 2000kg/h of flakes. The aim of this demonstrator is to increase its polymers detection efficiency, enable polymeric compounds detection and recycling and reduce the amount of materials which are incinerated.

Overall Concept of the activities to be implemented

Demo concept description: The pilot covers all the steps of the plastics value chain: purchase of raw materials, transformation, moulding and finished product release, supported by sales planning and scheduling, with the aim of tracking and tracing circular practices to increase the use of recycled materials. On the raw materials side, by tracing the recipes it is possible to maximize the use of recycled materials. A more effective management of BOMs in plastic moulding cycles will increase the quality of produced materials and enable the reduction of raw material waste, increasing the reuse of recycled materials. Moreover, tracking of production waste associated with the BOMs of raw materials allows to maximize the recycling of unsold products. The n-level BOM including the list of all components, subassemblies, semi-finished products and raw materials necessary to make a product, provides information about origin of materials (first choice, second choice, recycling): these data allow to track and trace all required materials and components information like origin, reusing, first or second choices and so on. The circularity aspects will be realized managing the reuse of waste materials depending on different nature of the recycled material or component: upcycling (an unused component is reused in another product without any transformation) or downcycling (unused components are transformed and reused in production).

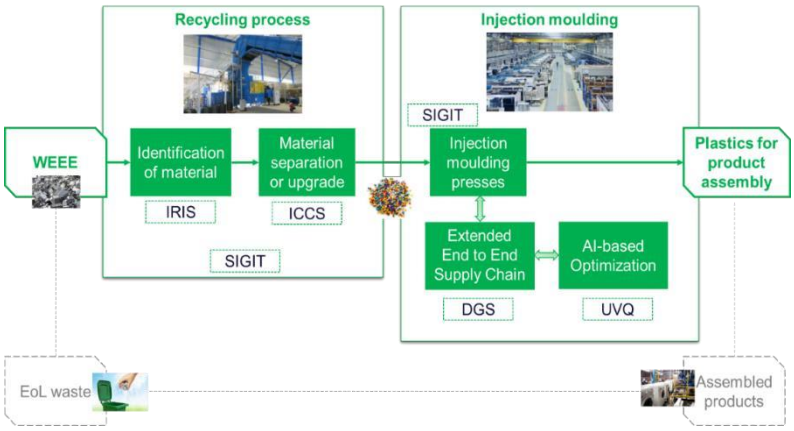

Figure 5: Italian Demonstrator

Digital technologies to increase waste reduction and Optimisation of use of secondary raw materials in the value chain: Sensing technologies will be used for identification of heavy and contaminated plastics and target polymers identification. IRIS will develop a monitoring system based on the combination of HSI technology (for plastic identification) and LIBS technology (for heavy metals detection, bromine and phosphate detection, among other compounds), which will be implemented in the recycling process. The goal is to increase polymers detection efficiency by enabling polymeric compounds detection and afterwards to implement separation of defective compounds, in order to boost recycling procedures and reduce materials to incinerator. The sorting mechanism will be designed and developed by ICCS, utilizing the real-time identification and classification provided by IRIS system. The categorization according to plastic flake size and the removal of defective pieces will be succeed using robotics arms, manipulated in real-time utilizing vision-based techniques. This sorting mechanism will enable reduction of waste, increased identification accuracy and efficiency in recovery of high-quality secondary resources. Data driven AI algorithm will be implemented in the existing infrastructure to enhance the existing supply chain software. That software, in the manufacturing execution system (MES) layer implements a recipe management system. The implementation of the AI algorithm will allow the plastic components producer to get understanding on the mistakes that originate from recycled materials or from process related mistakes

(pressure, temperature, etc.). Implementation of the AI algorithm, which will be deployed in the existing software used by Associated with document Ref. Ares(2022)7761208 - 10/11/2022 SIGIT, will allow for optimisation of the process of plastic components production from secondary raw materials. Through anticipation of mistakes during production process and more efficient use of the secondary raw materials significant increased waste reduction will be obtained and improve the quality of the goods produced with recycled materials. Moreover, quantification of recycled content and level of plastic degradation, as well as optimisation of the material composition will be allowed. Additionally, sensing technologies (near-infrared hyperspectral imaging) will be used to predict the particle size[44] and preliminary support the algorithm.	
Standardization and Certification: EUPC will support the certification and standardization activities related to the plastics produced in the demonstrator. The plastics converting industry is committed to incorporate sustainability as a key aspect of the production process, and will further contribute in this direction with a focus on recycled materials and the processing needed in order to uptake their usage. Certification proposals for recycled plastics will be produced, as part of the activities of the demonstrator.	
KPIs	
Increase of secondary raw materials use:	>15 %
Waste reduction:	>8 %
GHG emissions reduction:	20 %
Number of up-cycling, reuse and upgrade technologies of secondary raw materials implemented:	5 polymers
By-products value increase	>15 %

1.2.4. Synergies with other National or International Projects

The following list describes the projects that DigInTraCE is aiming to be based upon and upscale, ensuring its scalability and its impact. Added knowledge is complemented by relevant research projects.

Table 1 Links to other national or international projects.

Relevant Objectives	Added Value in DigInTraCE
Commercial blockchain traceability solution, UST	
Design, build and delivery of a commercial farm-to-fork blockchain traceability solution for Walmart, providing track & trace capability across multiple product lines. Rollout of Santander's OnePay Fx.	UST is a recognized market leader in blockchain services (Nelson Hall, 2021) with ample experience in commercial traceability and also blockchain payments and financial products (Santander Bank). UST builds on and combines this experience in the fields of circularity.
TRINEFLEX, (Horizon Europe) ICCS, DGS	
Provide an end-to-end service to manage the plant digital lifecycle and transition to flexible and sustainable operation, by advanced and green data acquisition, big data, process analysis, model development and digital twins with integrated multi-agent decision support systems.	DGS is responsible for energy consumption data collection and process optimisation for the glass demo case. DGS will use this expertise, know-how and methodologies for digitalization and in process optimization in supply chain management (e.g. for the CLSC tool).
ACCELWATER (H2020) ICCS, NTUA	
Offers a holistic food and beverage production framework in industries (to contribute to the reduction of the use of freshwater in the food and beverage industry resulting in a reduced and more cost-effective use of resources (water, raw materials, energy), lower waste and environmental impacts.	NTUA is involved in protein recovery from waste and byproducts, and application development, also supports LCA of process developments as well as techno-economic feasibility analysis. It will use knowledge for circular resources valorization, protein recovery while ICCS will bring knowledge on novel digital monitoring systems for IS facilitation.
PRO-ENRICH (H2020) CHIMAR	

[44] Pieszczek, Ł., Daszykowski, M. Near-infrared hyperspectral imaging for polymer particle size estimation. Measurement, Vol. 186, art. no. 110201, s. 1-10. (2021). https://doi.org/10.1016/j.measurement.2021.110201

Development of novel functional proteins and bioactive ingredients from rapeseed and other fruit side streams. CHIMAR developed bio-based adhesives for composite wood panels using the non-edible protein and phenolic fractions derived therein.	Knowledge for circular resources valorisation and development of bio-based adhesives for composite wood panels using recovered proteins
CORALIS (H2020) CIRCE, ICCS	
Pathways creation for resource decarbonization in energy intensive sector through implementation of viable industrial symbiosis approaches.	CIRCE, ICCS will use knowledge for circular resources valorization, monitoring systems and IS data sharing and management.
TagItSmart (H2020) VTT	
A Smart Tags driven service platform for enabling monitoring technologies for use connected objects, e.g. food industry, will offer expertise for development solutions for sensing and item level	ecosystems of VTT's experience in developing VTT's role was to develop Smart Tags and cases from digital identity concepts. of functional tagging authentication (Smart Tags).
BASAJAUN (H2020) TECNL	
Building A SustainAble Joint between rural and Urban Areas Through Circular and Innovative Wood Construction Value Chains, focusing on optimal utilisation of wood forest resources for construction value chain	Knowledge of different wood-based by-products generated a analysis of possible uses with value retention afterwards.
POLYNSPIRE (H2020) CIRCE, Dansk Standard	
Demonstrate innovative, cost-effective and sustainable technologies for energy and resource efficiency of postconsumer/industrial plastic recycling, targeting 100% waste streams containing at least 80% of plastic materials.	Knowledge, know-how and expertise for circular resources valorization, innovations and standardization in plastic based value chains and related resources.
C.R.AB.-Composites Researc Abruzzo (H2020) SIGIT, UVQ	
R&D of automotive market applications with thermoplastic and recyclable composites/ compounds and zero-waste hybrid processes. Focus on innovative products with new materials to reduce overall ecological footprint, using LCA/LCC, recycled polymers and fillers.	Enhancing the use of recyclable composites materials in the supply chain towards zero-waste processes.

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1.2.5. Interdisciplinary approach

In order to promote overall sustainability, new circular solutions for secondary materials and waste management, as well as value chains digitalization and traceability, have to consider many different aspects thus require the joint involvement of different scientific domains. DigInTraCE works in a strong inter-disciplinary field, as it involves expertise and methods from various disciplines including natural and applied disciplines such as chemistry and chemical engineering, physics, ICT, data science, electrical and mechanical engineering together with regional and social sciences (psychology, sociology, economics, political science, ethics, law, business, communications, knowledge exchange, training and learning, governance, management). These are interwoven within the DigInTraCE activities and will be brought together and integrated in pursuit of the project objectives forming inter- and trans-disciplinary conditions, used to develop innovations in different fields such as novel traceability, sensing and sorting techniques; data-driven optimisation and AI based algorithms for secondary raw materials; decision making mechanisms and LCA analysis; technologies including chemical processes for the up-cycling, reuse and upgrade of secondary raw materials; as well as support other activities related to e.g. standardization, business cases, exploitation and learning programs promoting new digital skills. DigInTraCE also uses the knowledge of to end user of the secondary materials thru his industrial partners, to consider the complex requirement of the specific markets in term standardization, norms constraints, performances. The project recognizes the difficulties of a trans-disciplinary collaboration (e.g. different language, terminology among different disciplines and methods of work, different levels

of scientific understanding and cognitive structures etc.)[45] and adopts a proactive approach towards them by planning to follow the fundamentals of the Methodology in Interdisciplinary Research (MIR) framework[46]. Also, at WP levels, the consortium will employ a quality assurance process (supported in WP1) to ensure that all trans-disciplinary aspects are integrated in the developed solutions, thus creating a meta-knowledge about all these different disciplines, methods and epistemologies, and purposefully and reflectively integrate and synthesize different perspectives, to better understanding and contribute to the proper monitoring and modelling of the environmental processes.

1.2.6. Integration of Social Sciences and Humanities

In the framework of its interdisciplinary approach, DigInTraCE will combine knowledge from SSH to explore how to better contribute to policy making, to enhance public understanding on cumulative impacts of climate change and the contribution of circular economy in dealing with them, how European governments and research institutes deal with diverse truth claims about the adverse effects of new digital enablers and how to offer perspectives for innovation forms of action. There are many different opportunities on how to use technologies depending on human behavior, needs and objectives and governance structures. DigInTraCE will study the special sectoral, social and regional needs and urgencies in the demo cases, the value chains and across different stakeholders upon creating the new solutions and the related training activities and lifelong learning programmes on digital skills. In order to boost innovations' impact the training activities will be co-created and codesigned based on the fact that people can develop new ideas and innovations based on the new technologies. The process of co-creation would be based on the Responsible Research and Innovation (RRI) principles: i)anticipation increasing resilience with holistic approaches of expected social risks, gaps and potentials of recycle technologies and traceability ii)reflexivity and continuous re-examination of positions, activities and assumptions iii)inclusion giving space to currently unheard voices to mitigate concerns and iv)responsiveness building capacities to adapt to changing technological and societal conditions.

1.2.7. Sex and Gender Analysis

DigInTraCE will overcome sex and gender bias by disaggregating socioeconomic and biological differences between man, woman or other gender identities, contributing to a more inclusive research and innovation by integrating related activities inspired from gendered innovations proposed by EC[59]. Project management will be engaged to gender issues including them in the reporting period's reports. To overcome gaps (considering both occupational and social status of genders) a draft study

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will be performed resulting in DigInTraCE Gender Equality Analysis with potential actions, as an integrated part of project activities. Study will be based on Gender Situation Analysis and Needs Assessment (GSANA, T1.3) to ensure gendersensitive project objectives, results, activities, partners assessment, management, indicators for monitoring and resource allocation. Project's motivation to perform these activities are both equality and the release of special creativity and abilities of all genders and individuals, creating added value for research, innovation and society. **Proportionality and balance in research:** equal participation of women, men and other genders at all project levels will be ensured, promoted already from the early stages of the project conceptualization. All partners are encouraged to increase women's and other genders' participation in activities including management, research, conceptualization, analysis, networking etc. to assure equal opportunities, recruitment and working arrangements. In addition, there is gender balance in the core project management activities and in the principal investigator per partner. All partners provide equal opportunities on employment and participation in research multinational projects, with special care to contributing lessening the disparities being a result of COVID-19 pandemic [47] (e.g. offering flexible working schedules to parents of all genders according to their special needs). **Scientific and Linguistic bias:** Addressing in texts and material (e.g., personas) written in English will follow gender-neutral language[48], ensuring data, algorithms, digital tools, technologies and visuals are free of gender bias and stereotypes. In all studies and surveys, gender dimension will be considered. Focus groups dedicated to diversity and gender issues will be included in co-creation activities. Collection of gender statistics employing AI and use of related metadata is also encouraged to identify challenges and include in policy-making, internal monitoring and to better understand gender-specific needs, issues and gaps. Annotating source data for sex and gender will be ensured. Special biological and other factors limiting or exposing to special dangers and gender-specific risk (e.g. specific professions and activities) will considered. Gender

[45] MacLeod, M. What makes interdisciplinarity difficult? (2018). https://doi.org/10.1007/s11229-016-1236-4
[46] Tobi, H., Kampen, J.K. Research design: the methodology for interdisciplinary research framework. (2018). https://doi.org/10.1007/s11135-017-0513-8
[59]European Commission, Directorate-General for Research and Innovation, (2013) Gendered innovations: how gender analysis contributes to research: report of the expert group 'Innovation through gender'. Publications Office. DOI:10.2777/11868
[47] European Parliament News, Understanding Covid-19's impact on women (infographics), (2021)
[48] Papadimoulis D., GENDER-NEUTRAL LANGUAGE in the European Parliament, (2018)

participation and real influence will be studied upon the operation and the decision-making in the governance structures of DigInTraCE as well as in the variety of entities and sectors involved. **Equal projects' results:** DigInTraCE ensures that new solutions usage (technologies, business models, processes, digital tools etc.), access to resources and services and the related trainings, foster the involvement of women and other gender identities scientists and engineers in the future landscape. **Equal dissemination:** Dissemination (quotations, visibility in publications and patents), communication in general public though releases and press, industry and business stakeholders, policy recommendations and institutional regulations, business and project's results should be addressed to all genders equally and adjusted to their needs. **Connection with established gender activities and expertise:** DigInTraCE will seek connection with EU organizations and institutes, academia studying gender, national gender equality bodies etc. for knowledge sharing, participation in activities and provision of data and tracking in EU gender statistic database[49], policy-making or studies. Existing tools will be considered to ensure the integration of gender analysis into the project such as gender specific questionnaires, checklists (i.e. Yellow Window, IGAR tool, Eige) guidelines and research methodologies. **Gender equality promoting:** project is rooted in industry environment, so it focuses on gender equality to be oriented on the participating partners making difference within their workspaces. In line with this and to assure full integration of the gender dimension, a dedicated session will be shared in each demo site, where general topic knowledge, all necessary information and guidelines (incl. gender-neutral language guidance) related to project will be held for the partners and possibly other participants at the beginning of the project. Partners will be encouraged and supported to implement their own gender equality plans, based on the knowledge gained during the project.

1.2.8. Open science actions

DigInTraCE will establish open science principles as an integral part of its methodology and its management, fully aligned with the ambition of the European Open Science Cloud as a trusted environment for hosting and processing research data to support EU science in its global leading role; aligned to Open Innovation 2.0 paradigm and supporting the principle 'as open as possible' in the goals of the Open Science Policy under Horizon Europe. Appropriate open science practices will be implemented supported by WP1 in terms of coordination, monitoring and management and WP7 included in dissemination, communication replication, learning activities etc. Specifically, the contributions are expected to be:

Research output management and impact to the industry: DigInTraCE research outputs will be made available to dedicated target groups aiming to allow the community to leverage from the project results and ensure replicability.

Research Outputs	Target Group	Means of management and Impact to the industry
Technological and Digital Innovations, Traceability	Industry and businesses, SMEs, Scientific and research community, industrial, sectoral, recycle, product and material associations (e.g. Orgalim, EIT RawMaterials, Circular Plastics Alliance, EUPC and members)	The communication and dissemination plan, (D1.2) will be made available to the public. A public version is to be produced even for deliverables that contain sensitive content.
Standardization and certification results, Policy Recommendations	Industry, businesses, European and international associations and organizations (e.g., SPIRE, P4P, DGs, SPI, Orgalim, EIT RawMaterials, Circular Plastics Alliance, European Plastic Convertors association, other recycle, materials associations, CEN & CENELEC, National Standardization Bodies,	The communication and dissemination plan, the open science handbook and the data management plan along with WP1 and WP7 will ensure that related results reach the target group on time and
Technologies, innovations, other resources (education, training materials etc.)	SPIRE stakeholders (through the beta version of search Outputs from SPIRE Projects platform)	Support the effort of the SPIRE and P4P community to document the results of related projects

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[49] European Institute for Gender Equality (EIGE), Gender Statistics Database

Early and open sharing of research results and data: DigInTraCE aims to provide 4 case studies publications in Open Research Europe, one for each of the value chains demonstrated. The first versions will be published upon completion of development phase (M24, WP6) and continuously updated till the end of project.
Measures to ensure reproducibility of research outputs: The strong commitment for providing standardization deliverables of secondary materials, the harmonisation of data as well as the decentralized based digital activities involving and coordinating different actors of the value chains with different information, will ensure that research outputs follow common standards and support mainstream and widespread reporting and documentation methodologies.
Providing open access to research outputs and Participation in open peer-review: DigInTraCE consortium partners are committed to use the Open Research Europe scholarly fully open access publishing service for Horizon Europe to enable rapid publication times and publication outputs that support research integrity, reproducibility, transparency and enable open science practices. The project has dedicated part of its budget to cover the fees associated with open-access publication to facilitate dissemination and reuse of the project's results. Project participants, for various reasons, may need to submit articles to journals (or proceedings) that only offer a lower level of open access, requiring either parallel publication or an embargo period. In this case, Green Open Access and Gold Open Access will be chosen depending on each publication. For green open access, developers will deposit the final peer-reviewed manuscript in a public repository of their choice, ensuring open access to the publication within the embargo period of six-months maximum.

1.2.9. Data management and research outputs

DigInTraCE is expected to collect data in the scope of the project activities and the value chains to assess the technologies and effectiveness of the proposed solutions in real life conditions of the supply chains. All the data made available directly by the partners involved or research results like models, algorithms, scripts, code and analysis will be aggregated and anonymized according to their desired visibility and security level defined. As for experimental measurements and data sets (raw, processed, logs etc.) regarding the materials and the products during the real traceability demonstration of the value chains, will be anonymized, safe and secured in the blockchain DigInTraCE platform. The exact protocols and data types will be defined according to the specific needs. The necessity of publishing all the data generated in the project, if it does not affect exploitation of results or competitiveness, is aligned with ethic issues and protects the privacy of data. This, aligned with Open science, DigInTraCE will make data not only "as open as possible: but also "as closed as necessary" to encourage collaboration across disciplines for solving today's complex societal challenges. So, with respect in confidentiality, most of research data generated by the project will be made open and will be offered to the Open Research Data Pilot (Open AIRE). DigInTraCE will generate research outputs and data which will be managed in line with the **FAIR** principles. Data and research results will be **Findable**: with open science research described, proper identifiers e.g. DOIs and standards used; **Accessible**: integrated to databases -created by initiatives such as these in 1.2.8- and continuously expanding; **Interoperable and Reusable**: standardized, harmonised using common vocabularies, open standards and sharing principles to be easily adjusted and used in further research in several fields. Open science guidelines and exact data management plan will be created by M6 as parts of D1.1, D1.2 respectively. The data within project's work space will be maintained after the conclusion of the project for at least 2 years, so that the knowledge management will abide by the principles of discoverability, accessibility, intelligibility and availability also beyond the end of the project. An indicative list of created data and research outputs and their relation to FAIR management is provided hereafter:

Data/Research Output	Size	Findability	Accessibility	Interoperability	Reusability	Curation
Case Study publications for each Value Chain (one per demo case)	4 Publications	OpenResearch Europe	OpenAccess	n/a	n/a	NTUA
Case studies per Technology	10 Case Studies published	OpenResearch Europe	TDB Per Technology	n/a	n/a	NTUA, UST
Data sets of monitoring of parameters at the pilots	*per Pilot	Zenodo for sample. Full dataset upon request	Not disclosed Anonymized data	JSON-LD files	Under Licence	IRIS

Optimisation algorithms for wood and plastics	1 algorithm	Zenodo	OpenAccess	n/a	OpenData	ICCS, TECNL, UVQ, DGS

1.2.10. The "Do not harm Principle"

DigInTraCE will ensure that it is fully compliant with the Do No Significant Harm (DNSH) Principle since its activities as well as the expected life cycle impact of the innovations at a commercialization stage, substantially contribute in a positive way and/or do not significantly harm any of the six environmental objectives set out in Art. 9 and 17 of the EU Sustainable Finance Taxonomy Regulation (EU) 2020/852. DigInTraCE will apply a life-cycle oriented approach (validated through the utilization of relevant LCA tool when possible) to ensure that proposed solutions are sustainable and consisting of 3 steps: i) Identification of actual and potential adverse impacts during the life cycle of the project activities ii) Prevention or mitigation of adverse impacts iii) Accounting for how adverse impacts are addressed by tracking performance and communicating results. This also applies for the proposed ICT technologies which will be validated with Green ICT principles. Especially, an LCA analysis is to be performed for energy intensive technologies like blockchain. DigInTraCE activities are expected to have a direct positive impact to the environmental objectives defined: Climate Change Mitigation by reducing GHG emissions (use Recommendation 2013/179/EU or *ETSI ES 203 199*[63], *ISO 14067:2018* or *ISO 14064-2:2019* to calculate lifecycle GHG and net emissions), to Pollution prevention and control through waste reduction and identifying harmless substances and contaminants in each pilot, to Transition to Circular Economy through the establishment of symbiotic and circular flows within same or different industries for materials and wastes whereas is not expected any negative impact to the sustainable use and protection of water and marine resources, protection and restoration of biodiversity and ecosystems and the Climate Change adaptation.

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2. Impact

2.1. *Pathways towards Impacts*

In this section, an overview is provided on how the DigInTraCE results generate Outcomes and Wider Impacts. A summary of project results in relation to ***research*** outputs and target groups in presented in Table 2. Following target groups have been identified for the DigInTraCE: I.P4P stakeholders, II. Industrial stakeholders (including other than P4P) III. Partnerships, Associations, Initiatives IV. Policy makers and financing institutions V. Academia and research community and VI. Citizens and general public.

Table 2 Results and their research outputs in relation to relevant objectives and demonstrator, at which they are validated and target groups.

Results, Outcome and Lead Partner	Partner	Research outputs	Obj	Demo	Group
Platform and Services					
Decentralized Traceability Platform	UST	Publ.(methodology, results, All agg. Data)		All	I-II,V
Dynamically updated DPP	ICCS	Publ.(methodology, results, agg. Data)	All	All	I-II,V
Online LCA computation tool based on Ex-Ante LCA	CIRCE	Algorithms, aggregated data	1,5,6	All	I-II,V
Recycled content identification, certification scheme	CTB	Certification scheme	2,5,6	All	I-VI
Algorithms and Prototypes					
AI based wood by products optimization algorithms	CHIMAR/ ICCS	Algorithms, aggregated data	1,5,6	GR	I-II,V
CLSC management toolkit optimization plastic recycling	DGS	Algorithms, aggregated data	1,5,6	IT	I-II,V
Machine vision-based wood re-design toolkit	TECNL	Algorithms, aggregate data	1,5,6	ES	I-II,V
Smart tags for real time sensing	VTT	Prototype	1,4,5,6	GR, BE	I-II,V
Novel spectroscopy techniques and algorithms for plastic waste and wood by-products	IRIS/ ICCS	Prototype, Algorithms	1,3,5,6	IT,GR, ES	I-II,V

Automated Plastic Sorting system	IRIS/ ICCS	Prototype, Algorithms	1,3,5,6	IT	I-II,V
Automated Wood Sorting system	ICCS	Prototype, Algorithms	1,3,5,6	GR, ES	I-II,V
Optimal control and AI package for recycled plastic	UVQ	Algorithms, aggregated data	1,5,6	IT	I-II,V
Protein recovery chemical processes	NTUA	Prototype, publications	Process, 1,5,6	GR	I-II,V
Bio-based adhesives process engineering	CHIMAR	Prototype, publications	Process, 1,5,6	GR	I-II,V

2.1.1. Scientific, Economic/Technological, Societal Outcomes of DigInTraCE

Topic Outcome 1: Demonstrate an increase in the waste reduction by application of digital technologies

DigInTraCE develops a number of digital technologies, and explores and verifies how their application and employment contributes comprehensively to a significant reduction of waste along the value chains in crucial sectors of process industries such as Pulp & Paper (wood value chain) and Chemicals (plastics and textiles). A Set of demo-specific **sensing and sorting technologies** will be applied on the by-products and wastes transported via conveyor belts, thus ensuring maximum uptake of potentially exploitable secondary materials and directly contributing to the minimisation of generated waste and overall waste reduction, an activity with associated environmental, social and economic impact. Moreover, **data-driven solutions**, such as case-specific algorithms, process and recipes optimisation**,** are expected to provide increased productivity and process efficiency, ultimately leading to mitigation of energy demands, decline of the produced residual side streams via effective classification and distribution of secondary plastics and wood residues for the most

63 Final draft ETSI ES 203 199 V1.3.0 (2014-12)

relevant uses based on their characteristics. The horizontal traceability technAssociated with document Ref. Ares(2022)776120
10/11/2022ologies to be applied across all demos, including the **decentralized traceability platform** based **blockchain technology** is expected to assure extended use of the secondary materials across own and other industri thus also contributing to creation of new knowledge across and within disciplines. Additionally, **Smart T technology** for real time sensing will be applied in Greek demonstrator, expanding traceability of products a materials related to storage and transportation conditions, which indicates potential hot spots to be assessed for wa reduction. An online **LCA computation tool** that enables the evaluation of products' environmental footprint will developed and integrated in the overall platform, in order to feed the dynamically updated DPP with environmen information, thus raising consumer awareness related to "green choices". Usage of such emerging technologies business level as well is envisioned to enable the advent of novel products, services and circular business models the market. Moreover, for exhaustive waste reduction, the whole supply chain shall be examined, therefore a **CLS management toolkit** for optimization of plastic recycling will be designed and developed, as a decision making to utilising AI algorithms, supporting the overall decrease of generated waste in the supply chain. Altogether, t **decentralized traceability platform** integrating all the transparent information systems will facilitate closing t loops and therefore significantly strengthen the efforts to increase the waste reduction in line with EU long term goa as described in CEAP, GD and SPI.

Expected Outcomes KPIs	Target
KPI 1.1: Number of active users in the platform	100
KPI 1.2: Possibility to handle different type of users with access rights to the DPP	At least 10 user types
KPI 1.3: Integrated supply chain management between virgin/secondary raw materials and BOMs contributing to the increase of secondary raw materials usage	By 10 %
KPI 1.4: Online LCA integrated in the platform and providing an output within seconds	<20 seconds
KPI 1.5: Percentage increase of secondary raw materials in new BOMs and recipes developed	By 10 %
KPI 1.6: Increase the application of wood by-products in higher-value flows	at least 30%
KPI 1.7 Min. number of batches monitored using SmartTags	1000
KPI 1.8 Number of parameters (temperature, humidity etc.) monitored by SmartTags	At least 2

KPI 1.9 Real time identification of materials parameters not covered with conventional techniques		at least 5 parameters
KPI 1.10 Classification accuracy (wood, plastic)		bigger than 93 %
KPI 1.11 Sorting accuracy (wood, plastic)		min. 90 %
KPI 1.12 Reduce of total waste in the plastic moulding process		20% reduction
Target Groups	**Scale**	**Significance**
ALL	4 Demonstrators	20% Waste reduction

Topic Outcome 2: Demonstrate optimisation of use of secondary raw materials in the value chains

Optimisation of the use of secondary raw materials in DigInTraCE has been approached in an extensive way by deploying digital technologies to enhance their chemical upgrade. At product and process level, optimisation in order to efficiently valorise the materials and consume less energy, by applying novel digital technologies (e.g. data driven optimisation, sensing and sorting mechanisms) and chemical innovations for upcycling (e.g. protein recovery from rapeseed cake for the development of bio-based resins) is to be employed. A set of demo-specific **sensing and sorting technologies** will be applied on the materials transported via conveyor belts to ensure their suitability to particular valorisation routes and contribute directly on the optimisation of secondary materials, while also unlock economic opportunities. Contributions to specific scientific advances are expected to be enhanced by relevant project's results. In particular, the secondary resources optimisation of use approaches will be demonstrated within the Pulp & Paper and Plastics sectors. Recipes will be optimised based on the properties, quality and quantity of the secondary materials, an activity supporting the design and development of added value end products. As a part of this data-driven recipe optimization, investigation for correlating reused secondary materials properties with quality of the final product will be performed, in order to determine the optimal use of by-products. The horizontal use of DPPs offers a possibility to examine how transparency through digital traceability may assist, not only the improvement of use of the specific secondary materials, but also the reinforcement and optimisation of methodologies for tracking materials and products across and within different value chains. The expected outcome is to create new knowledge, new procedures and services, as well as to build on and contribute to standards' setting efforts regarding secondary materials. Finally, the dynamic LCA analysis tool as well as the AI-based CLSC facilitate the identification of possible opportunities for by-products optimisation of use and are designed in such manner in order to boost decision-making procedures for relevant stakeholders.

Expected Outcomes KPIs	**Target**
KPI 2.1: Number of closed loops within the analyzed and participating value chain	10
KPI 2.2: Number of cross-sectoral collaborations achieved between the participating entities	5
KPI 2.3: Number of real time parameters (data types) integrated in the DPP	10
KPI 2.4: Number of ex ANTE LCA Models developed for the online LCA tool and validated	At least 20
KPI 2.5: Number of certification schemes (or methodologies) based on the markers for the recycled content	at least 1
KPI 2.6: Reduce the cost of secondary plastic identification and certification with a new methodology	20% cost reduction

KPI 2.7 Number of optimized recipes for pulp and paper sector Associated with document Ref.		at least 2
KPI 2.8 Increase in recycled plastic use for moulding		20% increase
KPI 2.9 Number of sustainable recovery technologies for proteins validated at TRL 5		3
KPI 2.10 Protein extraction yield		above 40%
KPI 2.11 Increase in the phenol substitution level in adhesives		up to 40 %
Target Groups	**Scale**	**Significance**
ALL	4 Demonstrators	50% Increased use of secondary materials

2.1.2. Wider Impacts

Resilient, sustainable, and secure (critical) raw materials value chains for EU industrial ecosystems, in support of the twin green and digital transformation

The DigInTraCE solutions are demonstrated in 4 different EU industrial ecosystems of Greece, Spain, Belgium and Italy. DigInTraCE is aiming to establish digital traceability on holistically optimized circular and low emission value chains operating at TRL6 at the end of the project and reaching TRL9 by 2030 through continuous research and innovation. DigInTraCE activities will contribute to the twin, green and digital transition by offering novel technologies automated by facilitating access to composition and origin of data (dynamically updated DPP, decentralized traceability platform, smart tags) operating on low waste and emissions value chains optimized by means of technologies (automated sorting systems, AI algorithms, CLSC and LCA tools, etc.) and chemical upgrade (bio-based adhesives, protein recovery, etc.). Innovative products and technologies exploitation strategies, business cases, innovative financial solutions for circular economy, educational services for employees based on co-designed resources and certification and standardization schemes of secondary raw materials, novel products and technologies, that will be integrated into digital tools will be a coherent part of DigInTraCE activities. Proposed solutions are in line with GD, and CEAP contributing to resilient, sustainable and secure value chains of pulp and paper and plastics and chemicals sectors. Both sectors are approached in the context of complete value chains and demonstrated in different parts of Europe to obtain possibly most complete picture per value chain/demonstrator to gain full understanding of all the regional and sectoral specificities. That approach will assure wider impact through smooth replication and transferability to other sectors with priority of those highlighted in P4P roadmap.

New sustainable-by-design materials with enhanced functionalities and applications in a wide range of industrial processes and consumer products.

On the material design level DigInTraCE will develop recipe optimizing algorithms recipe's optimisation based on material properties together with more sustainable use of by-products will lead to products and materials with upgraded functionalities and improved applications. The DigInTraCE materials and products and related processes will be designed with the aim to substitute existing ones and to be used long beyond the project duration at the pilot sites, but also to be replicated and transferred to other sites, value chains and sectors (with P4P sectors prioritized) . The issue of contamination will be also addressed by employing sensing technologies, that will assure that materials are pollutant and contamination free with the composition information available in the dynamically updated DPPs and tracing technologies. Particular wider impact of the project is especially expressed by the contribution to standardization and certification of the materials integrated in the digital tools and through setting sustainability criteria and benchmarks.

Leadership in producing materials that provide solutions for clean, toxic/pollutant free environment, decarbonizing industry, and safeguarding civil infrastructures.

DigInTraCE method to material tracing using digital tools in combination with novel sensing and sorting techniques offers **all-level approach to clean, toxic/pollutant free environment** and at the same time contributing to **reduction of GHG emissions** though **enhanced use of secondary raw materials** (and therefore waste reduction). Starting from substitution of toxic and petroleum-derived components followed by contamination detection (sensing technologies) through enhanced control facilitated by availability of continuously updated information (DPPs) and finally by assuring safety through standardization and certification, DigInTraCE will offer safe solutions available beyond project duration. Leadership will be assured through cross-disciplinary cooperation of experts of different fields to obtain toxic free material cycles supported by transparent information system. Networks of experts created as a part of DigInTraCE activities, supported by business cases, innovative financial solutions and educational services for employees and certification and standardization schemes of secondary raw materials, novel products and technologies are offered to assure long-term leadership.

Leadership in circular economy that strengthens cross-sectorial cooperation along the value chain and enable SMEs to transform their activities and business models.

The fundamental aspect of circularity is efficient cooperation of actors to assure that waste of one process becomes resource of the other. In those settings the availability and access to reliable information is crucial for efficient cross-sectoral cooperation and overall waste reduction. On the other hand, digitalisation and digital technologies, the CE enablers, have the potential to facilitate greatly cross-sectoral collaboration, offering ways to exchange information and data in efficient and secure way. In line with that, DigInTraCE provides advanced transparent information system with central point DPP, that contains information about material's characterization, quality, categorization, composition and history supported by **live update mechanism** and **standardization and certification schemes**, that assure impact maximization. The system is supported by access control to assure that the thin line between

information accessibility and confidentiality is not crossed. In the long run obtaining DPPs for the value chains of crucial process sectors, expanded by replication activities will be the

significant contribution to the leadership in CE. Special focus of DigInTraCE on cross-sectoral collaboration an important aspect of DigInTraCE demos, with the aim to establish a further expand the collaborations between actors from different industries. This will lead to wider impact in a form business models with new economic opportunities addressing the needs of all business partners with special attenti to SMEs. Special facilitating function will be offered by smart contracts, of special interest for SMEs trying to fi solutions for novel circular economy settings.

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Increased adoption of key digital and enabling technologies in industrial value chains and strategic sectors, paying particular attention to SMEs and start-ups.

DigInTraCE focuses on industrial sectors, for which digital treatability importance has been highlighted by P4P roadmap. To assure increased adoption of key digital and enabling technologies and to be in line with EU digital compass to empower citizens and businesses for prosperous digital future DigInTraCE will undertake a number of high-level educational activities and provide financial services especially suitable for circular economy context. In the wider sense those DigInTraCE activities linked to replication and exploitation activities will contribute to increased adoption of novel technologies and enhancement of the digital skills of workforce and citizens. As a result, DigInTraCE will play an active role in the successful digital transformation in Europe by 2030, as proposed in the Communication 'Digital Compass: The European Way for the Digital Decade'. Also, in relation to that document, preliminary business innovation plan, cost-benefit analysis, financial plans and cost revenue models will also contribute to increased adoption of the technologies especially by SMEs and start-ups.

Significance: DigInTraCE will establish digital traceability and upgrades on circular and low emission value chains at TRL6 at the end of the project and TRL9 by 2030

Overall, significance of DigInTraCE to its wider impacts can be related to 3 streams of actions of EU's digital strategy 1) creating technology for people (societal impact), 2) contributing to fair and competitive digital economy (economic impact) and 3) enabling sustainable, open society (environmental, scientific impact). **Societal:** DigInTraCE aims at contributing to successful digital transformation in Europe by 2030. It is expected that DigInTraCE educational services (including co-designed materials available beyond the project) will reach over 1000 users by 2030, contributing to the European Commission targets in the European skills agenda and the digital education action plan, like the one to to ensure 70% of adults having basic digital skills by 2025. It is expected that DigInTraCE standardization deliverables will reach stakeholders in all EU and EFTA countries (measured in number of NSBs that have implemented the standard deliverables nationally), while DigInTraCE decentralized traceability platform is expected to reach 1000 registered users. DigInTraCE services will contribute to long term impacts on the society as described in GD and CEAP. **Environmental**: DigInTraCE will provide crucial digital technologies, having direct impact on the waste and emissions reduction (contribution to overall potential of emissions reduction of 7 times and reduce global emissions by 15 %[50]), and therefore contributing to EU becoming climate neutral by 2050, as set by EGD. At least 5 tracing technologies and 3 sorting technologies, which will reach TRL 9 by 2030 and will expand to at least 5 additional sectors of P4P. DigInTraCE solutions will contribute to increase in the decision-making capacity towards improved use of secondary raw materials. **Economic**: DigInTraCE will provide solutions that facilitate functioning of companies (especially SMEs) throughout the circular transition by providing solutions matching new economic landscape. DigInTraCE through its dissemination and exploitation activities will contribute to the following elements of the level of ambition set by the Digital Compass i) 75 % of European enterprises have engaged in cloud computing services, big data and AI, ii) 90 % of European SMEs reach the basic level of digital intensity. **Scientific:** DigInTraCE findings (including research outputs on 5 services and 9

[50] European Union, 2020, "Supporting the Green Transition. Shaping Europe's Digital Future".

prototypes) will be available to the community for further research and all consortium members are dedicated to follow up actions for circular value chains.

2.1.3. Potential Barriers and Mitigation Measures

Technology and process barriers

Managing the complexity of value chains, products, materials, substances and different technologies and innovation is a great challenge. The traceability of the secondary materials, should be examined sector wise and requires interoperability and related software/hardware infrastructures. Also, individual barriers related to DigInTraCE technologies and processes for recycle, reuse and upgrade of materials are possible to appear during research activities.

Mitigation measures: DigInTraCE intends to involve all the actors of supply chain, from suppliers of primary materials to producers and recyclers, to achieve a coherent and holistic solution, supported by proper managerial and coordination structures and procedures. An effort was made upon proposal to assess current trends using different technical and business criteria to ensure a huge potential for the selected technologies and innovations. All needs, technical capacities and knowhow of companies across supply chains will be combined, as well as the diversity of profile of supply chain actors should be taken into account from the very beginning of the development to assure mitigation of possible technology and process barriers. The high scientific and technical expertise and experience of the consortium is expected to contribute to the mitigation of the possible barriers in line with general sustainability, resilience, circular economy and DNSH principles.

Data and information related barriers

Difficulty on access and acquisition of reliable, relevant and standardized data (even if comprehensive produ information is already available); competition, IPR, security concern lack of data transparency and trust between stakeholders along value chains, finding the balance between sharing a securing information; are important barriers for data and information exchange along the supply chain thus for t update of digitalization and the traceability in the supply chains.

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Mitigation measures: Approaches for data and standards included in the dynamically updated DPP will be fully defined, based on existing product information systems/regulations/initiatives on technological and content level, from multiple sources along the value chain. This multidimensional information flow will ensure data availability. Blockchain Decentralized traceability platform will ensure data reliability, integrity, security, proper management and safety of confidential information, accessible accordingly to different user groups. Also, DigInTraCE will ensure IPR Protection and will focus business cases promoting the cooperation toward sustainability, and degrading competitive culture.

Public Acceptance and Increased prices of recycled materials

Potential barriers related to public acceptance of novel technologies and methodologies might be related to lack of awareness, lack of digital skills, scepticism on using recycled materials and to increased prices of the recycled materials. Industry acceptance might be also affected by the feedstock availability and price stability, as well as occurrence of competing materials. Potential high costs for collection, sensing, sorting, upgrade and recycling processes and infrastructure of multiple technologies compilation might affect the final price of products posing a barrier of wider implementation of project's solutions.
Mitigation measures: Dissemination and communication activities will ensure the promotion of the benefits, results, best practice and performances obtained, providing training, educational activities and awareness and engagement raising though their co-creation. Means such as tutorials, webinars, workshops, use of specialized social media (i.e. Xing, LinkedIn) or tools will be explored. Project will aim to engage industry community by interacting and networking with industry and workers associations. All those activities will contribute to increase in public acceptance of the DigInTraCe solutions. It is expected that high level of transparency provided by the DPPs will help mitigate the remaining barriers related to lack of awareness and scepticism of end users. Mitigation of possible poor acceptance will be further supported by DigInTraCE standardization system as a tool for dissemination of the project results and interaction with the market stakeholders. AI and Blockchain standardization activities are expected to facilitate the launch of the new solutions to the market. Standardization as transparent and formal process has the potential to enhance the trust building and to act as a catalyst. The mitigation measures related to industrial acceptance (e.g. feedstock availability) in DigInTraCE are focused on the establishment of well-functioning cross sectoral collaborations and closing the loops. DigInTraCE implements enhanced financial services (including smart contracts) and will generate strategies against secondary raw material price fluctuation possibility facilitating mitigation of the risk related to increased price of recycled materials. Market needs will be analysed in terms of pricing decision support. Additionally, services will be provided allowing visualization of historic prices and trends on secondary and virgin material prices supported by prediction, fluctuation risk assessment/protection (hedging) services. Financial services in combination with DPPs transparency, are the mitigation measures to barriers arising from possible occurrence of competing materials, through increasing the credibility of DigInTraCE materials.
Legal and institutional barriers
DigInTraCE involves different activities and sectors related to secondary materials, wastes, chemical and plastics, wood and pulp & paper, digitalization, traceability, data etc. As a result, there are multiple and overlapping regulations that have to be taken into consideration. Those regulations might often be conflicting, inconsistent or unclear. Additionally, in some cases, national strategies might not be aligned with the relevant European documents, contributing with different materials classifications or waste management institutional obligations (e.g. local mandatory management). Also, the evolution of EU standards and regulations is expected to be more restrictive and setting highly demanding implementation targets. (e.g. on quality or in the presence of unknown dangerous substances that are no longer allowed). All these may cause obstacles to adopt innovations and best practices or obtain approvals for project activities.
Mitigation measures: DigInTraCE will collect, analyse, follow and plan its actions accordingly with all the related EU strategies already referred and regulations (e.g. waste framework directive 2008/98/EC, Regulation (EC) No 1013/2006 of end-of-waste criteria, Ecodesign Directive 2009/125/EC, GDPR, IRP protection, Cybersecurity directives 2016/1148/EC, 2019/881/EC etc.), national laws and regional/local particular practices and institutions' regulations. DigInTraCE consortium consists of multi-disciplinary group of partners including partners with digital and industrial expertise, standardization and certification bodies, all with wide experience and knowledge on the above practices and regulations that will ensure the required effort. Finally, project will aim to engage member States representatives involved at local level to properly address the challenges by interacting and networking with the relevant ministries and associations, regional and local authorities assuring the mitigation of legal and institutional barriers.

2.2. *Measures to maximize impact*

2.2.1. Dissemination, exploitation, and communication

2.2.1.1. DigInTraCE Dissemination Strategy

While focused on Awareness, Interest, Desire and Action (AIDA) activities, the project's dissemination measures will follow a multi-actor, multi-step and multi-channel approach. Table 3 identifies the target audiences, specific dissemination objectives and target values, tailored to the needs and profile of each target audience.

Scientific Publications	Key Decision Influencers, Decision-making Actors, National Ministries of Energy, Industrial Associations, Environmental Agencies, Academia and Researchers, relevant Consortia.
Scientific Conferences (examples): International Conference on Distributed Ledger Technology and Blockchain, Recycling Congress; 3rd International Conference on Waste Recycling and Management, International Conference on Clothing Sustainability and Textile Processing, International Conference on Textile Exchange Sustainability, International Conference on Wood Biology and Wood Technology, etc. **Peer-Review Open-access Journals (preselected)**: Digital Communications and Networks, Recycling, Resources conservation and recycling, Journal of Materials Processing Technology, Polymer, Composites Science and Technology, Sustainability, Waste management, Wood science and technology	
Target Values: Publishing in Peer-reviewed Journals: > **15**; Presenting in Scientific Conferences > **20**.	
Technical Publications	European Commission Relevant Offices, European Environment Agency, Public and Private Actors.
DigInTraCE will publish and contribute to technical blog posts, articles, position/white papers, catalogues, books or any other reference from technology providers (bottom-up), as well as references related to the application scenarios domains under consideration. Excellent examples of this kind of contribution are the upcoming editions of Institute of Environment and Development Blog and SPIRE database of the outputs from projects of SPIRE portfolio.	
Target Values: Publishing Technical Publications: >**25**; Participation in Industrial Exhibitions with Booths: > **10**.	
Source Code Repository	Ecosystem of Open-Source Developers, Incubators, Associations and Digital Innovation Hubs, Data Spaces for smart agriculture, ICT SMEs, Researchers, relevant Consortia.
Encouraging a transparent and open-source culture, DigInTraCE will make accessible its software repositories through well-known distributed source code platforms. Well-established repositories are GitHub and Bitbucket.	
Target Values: Number of Publicly Available Deliverables: **20;** Source code: publish to >**2** different repositories.	
Q&A Platform	Public Authorities, Associations, Open-source Developers, NGOs, SMEs, relevant Hubs.
Conversations about the project repository would be concentrated in reference sites. The selected Q&A platform developed as a part of project website will be the Stack Overflow, the largest, most trusted online community for developers to learn and share knowledge.	
Target Values: No. of Scientific Conversations to Participate/Contribute: > **25**.	
Workshops and Trainings	Industrial organizations, SPIRE Associations, Authorities, Potential Investors, Funding Organisations, ICT SMEs, Academia.
Special focus will be given to co-design of training activities and lifelong learning programmes to assure that DigInTraCE contributes to establishing digitally skilled population as foreseen by task 7.2.	
Target Values: Technical Workshops: >**10** in **7** different countries; Online Training Tutorials: >**8**; Webinars: >**12**.	
Joint Activities with relevant Initiatives	All Stakeholders with special focus on Advanced Manufacturing, , Policy- and Decisionmaking Actors, Key Environmental Influencing Organisations, relevant Public Bodies.
To further amplify and maximise the impact and outreach of the project's results and to feed them into related international work streams, joint dissemination activities will be contracted. DigInTraCE will explore synergies with other projects and initiatives and organize/participate in events promoting knowledge exchange as described in T7.4	
Target Values: Symposiums/ Fairs to Participate: >9; Banners in each event: **3**; Procures to be Distributed: >**300**.	
Standards and Policy Contribution	Standards Developing Organisations, Certification Committees, Open-Source Communities, Policy Makers, Relevant Public Bodies, Local Agencies.
DigInTraCE will support secondary material, AI and blockchain standardization. Collaboration with relevant standardization Technical Committee (s) will be established to assure that integration into current and future standardization developments. DigInTraCE contribution will be included in the new or revised standards through one of following channels i) providing information to standardization Technical Committees, ii) mutual knowledge sharing iii) technical proposals, iv) promoting the elaboration of new standard documents depending on the maturity	

of specific solution as foreseen by T7.1 and will be available for use by industry. Two policy briefs, each for DigInTraCE sector of analysis (pulp and paper, chemicals and plastics) will be developed as a part of T7.4.
Target Values: Standards Contributions (new or revised): >**5**; Policy Briefs: >**2**

2.2.2. DigInTraCE Communication Strategy

DigInTraCE's communication strategy is built upon the Promotion Mix, the 4th element of the Marketing Mix, focuses on creating awareness and persuading the audience to initiate the engagement, as outlined in Table 4.

Table 4 Communication Plan – Communication Channels/Target Audience/Timeline and Target Values

Project Website	ALL End-users.	From M03 to 3y after
A project website allowing easy retrieval of main action data with few clicks while acting as the core source of information about the project's objectives and activities. The first version w be online by M03, and will be constantly maintained and updated regularly, running also for 3years after project e - ICCS leads the task.		
Target Values: Online by: M03; Unique Visitors from M12: >1,500; Unique Visitors from M36: >5,500.		
Social Media	ALL End-users.	From M03 to M48
DigInTraCE will build and maintain actively its online presence in several social media channels, using Twitter and LinkedIn when interacting with research and innovation communities spreading out new publications, while Facebook, TikTok and Podcasts will be used for wide community engagement, - ICCS leads the task.		
Target Values: Size of online community (by M36): >5,000; No. of impressions (monthly average): >500.		
e-Newsletters and Email Campaigns	ALL End-users.	Quarterly Issues starting from M04
E-newsletters will provide a snapshot of main activities and achievements of DigInTraCE, whereas broadcasting messages to a target pool of contact points via email is a highly effective measure of engagement, especially when promoting events and outcome among different domains. Professional marketing platforms (i.e. Sendinblue and Mailchimp) will be used to automate the distribution among contact points - ICCS leads the task.		
Target Values: e-Newsletters contributed/released: 9; Total Mailing List Contact Points: >2,000.		
Policy Fora and Debate Hubs	Societal Actors, Debaters, Data Management, Materials and Recycle Operations, Environmental and sustainable product Organisations, workers' associations, Climate Alliances,	From M23 to M48
With main objective the Science-Policy dialogue, DigInTraCE will contribute to discussions and make recommendations on how the developed solutions could enhance the sustainable, balanced and inclusive development of secondary material and waste management on different industry sectors and products. DigInTraCE envisions to contribute and support actions of related hubs and forums such as World Economic Forum, IUCN forum, Science Policy Forum, LCM International Conference, EIT RawMaterials Expert Forum, GS1 related to DPP fora etc.		
Target Values: Number of Fora and Hubs to interact: >7; Contributions to Policy Fora/ Debate Hubs: >15.		
Blogs	Digital Innovation Hubs, Relevant Associations, Academia.	From M18 to M48
DigInTraCE will connect and execute periodic broad promotion activities in open communities, DIHs, blogs, nonprofit organisations and channels such as: Gitter Communities, related subjects in Nature.com EIT RawMaterials Innovation Hubs, etc. - ICCS leads the task.		
Target Values: Unique Publications/Contributions to Blogs: >15; Different Blogs to Post: >8.		
Printed Material	ALL End-users attending physical events.	From M03 to M48

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The project will develop a distinct visual identity (logo, key visuals) and reference promotional instrument when participating in events, including brochures, catalogues, posters and any other laid out paper-based resource. Most of the material will be available as e-documents and printing will occur as required. - ICCS leads the task. Target Values: No. of hard copies (i.e., brochures, flyers, posters, etc.) distributed: >1,500; Roll-up Banners: >6.		
Multimedia	ALL End-users focusing on Regional Actors and Local Authorities.	From M09 to M48
The project will produce multimedia material to have a self-explanatory and appealing presentation of the project, leveraging available distribution channels of promotion (e.g., YouTube, Vimeo). Also, a set of video interviews, both from scientific partners, to collect inputs, taking advantage of plenary meetings and events of relevance will be implemented in close collaboration with the DigInTraCE capacity building activities - ICCS leads the task. Target Values: No. of short videos and interviews produced (in total): 7; Number of views: >10,000		

2.2.3. Exploitation Strategy for the Uptake of DigInTraCE Results

Preliminary Market Analysis: Within the framework of circular economy, digital ecosystems consisting of emerging market networks will enable hybrid forms of cooperation and competition. Meanwhile, technology, demographics and government policies are rapidly re-shaping industries. To improve supply chain's performance, agility, cost, and inventory levels in this transformative era, companies have identified the need to restructure their supply chains and operations integrating the latest digital technologies. B2B business entities addressing these needs are expanding their services offering suites of proprietary asset-backed solutions increasingly enabled by such technologies. Solutions extend from end-to-end supply chain strategy, strategic architecture, operational excellence, and supply chain resilience. Subsequently the global digital supply chain market size is rather fragmented with solutions spanning from custom manufacturing traceability software to Manufacturing Execution systems which are increasingly integrated with various other solutions, such as ERP, supply chain management (SCM) or warehouse management system (WMS). Post COVID-19, the **global digital supply chain market size was valued at $3,918.00 million in 2020, and is projected to reach $13,679.00 million by 2030, registering a CAGR of 13.2%**[51]. The current estimation of 2030 is projected to be higher than pre-COVID-19 estimates. COVID-19 pandemic has significantly fostered growth rate of the market owing to the increased shift towards digitization. Major growth drivers of the market include rise in demand for reliable,

fast, and effective order execution; surge in need for cloud-based supply chain management solutions; and increased

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use of industrial-grade digital technology. **In the EU context, the market size for supply chain analytics in Europe reached USD 1000 million in 2020 and is projected to grow at a rate of 10% through 2027 owing to extensive digitization and internet penetration across the world**.[52] Furthermore, new legislation is expected to be a game changer in Europe, gradually enforcing over the next decade a more holistic transformation towards increased industrial traceability and circularity. More specifically, falling under the European Commission's new Circular Economy Action Plan (CEAP) – a key building block of the European Green Deal – the Sustainable Products Initiative (SPI) is a framework regulation comprised of a set of legislative files that aims to boost the circularity of the EU's single market. In contrast to traditional circular economy measures that focus on the end-of-life phase (e.g., returning products to the supply chain instead of the landfill), the SPI aims at building in sustainability throughout the entire product lifecycle, starting at the design phase. Drawing on the success of the 2009 Ecodesign Directive, the SPI will extend minimum sustainability requirements beyond energy-related products to a whole new array of sectors such as electronics & ICT equipment, textiles, furniture, steel, cement & chemicals. These rules will ensure that products on the EU market are designed to not only be energy-efficient, but also repairable, durable, recyclable and free of hazardous chemicals. In addition to product design, the SPI also foresees the introduction of digital product passports, which shall contain information on each product's composition, including material and chemical properties, as well as information on their circularity, such as guidance for reuse and repair operators. In this context, the technologies developed during the project have clear potential to address future market needs that are expected to emerge in EU states as circular economy directives become transposed in national legislation. As the project progresses, relevant market needs and size for the various project technologies shall be constantly evaluated in order to contribute to the maturation of the finally proposed individual and joint exploitation plans.

[51] Allied market research, Global Digital Supply Chain Market 2020-2030
[52] Europe Supply Chain Analytics Market Forecast 2027

A preliminary identification of the foreseen business cases and their adjacent exploitation strategy is provided here. More specifically a categorization of the various individual results to certain preliminary **Key Exploitable Result (KERs) bundles** was made based on the identified markets and customer needs. KERs are single or multiple result solutions with clear target market, addressing cost effectively specific customer needs through identified competitive advantages. Given the low to medium TRL level of the under-development tools, KER characteristics and exploitation routes are to be constantly evaluated and updated as project progresses and engagement with stakeholders provide necessary feedback to the consortium regarding exploitation characteristics and potential.

Traceability Platform and add on services: This is the core offering of the project constituting the main technological solution to be tested and optimized in all 4 demonstrators of the project and **jointly exploited by the involved partner organizations.** It includes the Decentralized Traceability Platform, the Dynamically updated DPPs, the online LCA computational tool and a recycled content identification and certification scheme. These technologies and services are bundled under this KER to deliver its holistic approach of a **cross sectoral digitalization solution** to increase waste reduction and improve use of secondary raw materials. These results will be validated in all four demonstrators. Combining tokenized financial products associated with the blockchain-traced items in the same platform is a novel approach towards a truly circular economy paradigm. In this regard exploitation of the solution shall require subsequent identification of equally innovative business models combining subscription based SaaS together with technical consulting and hardware installation services.
After validation during the project demo phase, work towards commercial exploitation of the DigInTraCE integrated solution will be initiated through a joint venture (JV) involving at least the key consortium partners that are necessary for the successful continuous development, marketization and commercialization of the system (UST,ICCS,CTB,CIRCE).As project results are targeting a TRL6 at the end of the project, further maturisation and development shall be required towards validation of an initial marketable version of the integrated solution and valueadding services. To support this, a preliminary version of a technology maturisation plan shall be developed during the project and reported in D7.7. Scope shall be to identify all the necessary technological development steps as well as resources needed towards increasing TRL to level 9 thus a closer to market state. In the report, a minimum viable product version may also be described for which development will be less resource intensive. This will permit the DigInTraCE JV to project towards a more short term product development and deployment cycle for which specific and more predictable revenues can be anticipated therefore rendering the venture more appealing for investors. Further funding needs will be reported together with an initial mapping of available funding schemes and alternatives that the venture may target after the project ends. The minimum viable product or products of the venture shall be formed based on extensive market and competition analysis implemented during the project as well as on the validation and impact assessment procedures that will have taken place in all 4 project demos. Furthermore, based on the exploitation activities implemented during the project, a comprehensive understanding of the solution/service benefits for all relevant stakeholders will be available . In any case,,the mixture of revenue sources is envisaged to spread the financial

burden to various types of stakeholders, to hedge the risks of certain stakeholders' failure and to share, among the
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involved partners, common development, commercial deployment, marketing and other costs. Commercial operations may even be initiated with pilot partners that shall be charged licensing fees for implementing DigInTraCE results plus ongoing software and hardware tool customization and maintenance/consulting and/or training fees. Based on the technology maturisation plan and the anticipated time-to-market of the developed solutions (2-5 years after the project end) an initial financial projections and estimations shall be provided in the final version of the exploitation and business innovation plan that will be delivered at the also in D7.7 (M48). This plan shall consider all the development and operational costs of the venture, the anticipated revenues based on minimum viable product sales projections taking into consideration various market, policy, environment and other positive and negative externalities. This preliminary business plan consolidated by the end of the project shall support the venture in approaching investors and presenting a unique investment opportunity.

Apart from the above mentioned cross sectoral tool suite validated in all demos, certain **industry and process specific prototypes and algorithms** shall also be developed during the project which constitute separate KERs and their exploitation shall follow the individual exploitation plans of the developing partner organizations. These are:

Wood process and sorting toolkit, Plastic process and sorting toolkit: Although DigInTraCE industry specific datadriven solutions and demo-specific sensing and sorting technologies have the potential to be integrated in the traceability platform, they can still be considered as distinct KERs addressing specific needs of the wood based panel and plastic processing industry. As such, involved partners may independently exploit these solutions addressing clear market needs with the demonstrated and validated competitive advantages of their developed technologies. Exploitation may be implemented either directly or through licensing or selling of the IP rights of the technologies to third parties. Protection of the technologies may be implemented by patenting the solutions. **Regarding the relevant market sizes, the global wood based panel market size is estimated to reach USD 247.23 billion by 2027, exhibiting a revenue-based CAGR of 6.9%[53] during the forecast period**. **. Of that, European market size accounts for approximately 25%.** Rising demand for these panels to manufacture residential and home furniture, Ready-to-Assemble (RTA) products, and small accent pieces is anticipated to positively impact the growth. Similarly, the **global Textile Chemicals Market is projected to reach USD 33.1 billion by 2026, at a CAGR of 4.6%[54] from 2021 to 2026**. Driving factor for the Textile Chemicals market is the growing demand for technical textile as well as low VOC and biodegradable materials for textile manufacturing. Further exploitation aspects to be validated during the project is the marketization of the novel wood-based products as well as extruded plastic filaments. This can be implemented either by patenting and licensing the developed manufacturing processes or direct marketization of the novel products by the relevant pilot partners supported by the technology developing entities.

Chemical processes for wood valorisation: Similarly to the previous two KERs, these processes have clear target groups and their exploitation shall be targeted via direct industrial use to improve efficiency and environmental targets or by the development and marketization of novel products.

Training material for digitalization: Skilled workers with digital abilities are necessary to implement, maintain and replicate usage of the digital tools after the project ends. To this end, development of high quality co-designed learning resources has been foreseen within the project. Training methodology and content is considered another Key Exploitable Result for which a relevant exploitation route shall be determined.

Exploitation plans of individual results: In addition to the identified KERs, project partners shall make sure to concentrate on the commercial exploitation of their individual solutions/results produced during the project in order to contribute equally to the achievement of their individual research or business goals and the mid-/long-term goals of the project. Following the IPR management and protection principles of the project as described in the CA signed by project partners, each partner may plan and pursue individual exploitation routes for the developed technologies. These routes can include commercial or further research purposes.

2.2.4. Management of Intellectual property

IPR strategy and exploitation management will be handled in the CA, as well as in a dedicated task (T7.3) in line with Horizon Europe IPR recommendations. The IPR strategy of the project is considered as an integral part of the project exploitation plan. Abiding to CA rules, the consortium will establish a knowledge portfolio management, able to identify relevant background, to collect information on Results that have a potential commercial exploitation, to generate IPRs related with the new knowledge and to update the knowledge portfolio in accordance with the DigInTraCE dissemination and exploitation plans. Ownership: each participant will own the foreground/ results it generates. Background will be clearly identified within the consortium agreement and when applicable, granting of access rights will be clearly specified. Joint ownership: when results are generated jointly, participants will seek to

reach relevant agreements. The CA will define these rules. Research partners will Associated with document Ref. Ares(2022)7761208 - 10/11/2022be granted a fair compensation in the

form of royalties by partners exploiting the results in which the research partners have contributed; they will also be entitled to create spin-offs for the commercialization of their results. Protection, use and dissemination: Results capable of industrial or commercial application must be protected considering legitimate interests. Prior notice of dissemination must be given to other participants. Access rights: Partners may define the background needed in any manner, and may exclude specific background. At a preliminary stage, partners shall agree on open access publishing. However, in the future, partners may opt for gold or green access to peer- reviewed scientific publications, which might result from the project, depending on the type of information to be published. Each partner shall make available

[53] Wood Based Panel Market Size, Share & Trends Analysis Report by Product (Plywood, MDF, HDF, OSB, Particleboard, Hardboard), by Application (Furniture, Construction), by Region, and Segment Forecasts, 2020 - 2027

[54] Textile Chemicals Market by Fiber (Natural, Synthetic), Product Type (Coating & Sizing, Colorants & Auxiliaries, Finishing Agents, Desizing Agents, Surfactants), Application (Apparel, Home Textile, Technical Textile) & Region - Global Forecast to 2026

its results on a royaltyfree basis, unless otherwise agreed to other contractors to the extent that such information is necessary for them to produce their own results within DigInTraCE.

2.3. *Summary*

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SPECIFIC NEEDS	EXPECTED RESULTS	D & E & C MEASURES
Improved use of secondary raw materials Reduction of waste generation Reduction of GHG emissions Availability and access to material data alon the whole value chain in real time Cutting edge technological innovation fo tracing solutions with integrated DPPs an certification schemes Novel sustainable and toxic free materials Holistic solutions towards circular economy Enhanced cross-sectoral collaborations Enhance adoption of key digital technologies Business models embracing novel circular solutions	Decentralized Traceability Platform Dynamically updated DPP CLSC management toolkit optimization plastic recycling g Online LCA computation tool Recycled content identification, certification scheme AI based wood by products optimization algorithms d Machine vision-based wood re-design toolkit Smart tags for real time sensing Novel spectroscopy techniques and algorithms for plastic waste and wood by-products Automated Sorting systems (for wood and plastic) Optimal control and AI package for recycled plastic Protein recovery chemical processes Bio-based adhesives process engineering	**Dissemination (D):** Peer-review Journals >15; Scientific Conferences >20; Technical Publications >25; Participation in Industrial Exhibitions with booths>10; Technical workshops >10; Webinars >12; Online training tutorials>8, Symposiums/ Fairs to Participate: >9 Policy briefs for sectors of focus = 2 **Communication (C):** logo and visual identity, online channels creating and updating (e.g., social media, newsletters, press releases), short videos produced, press media, project promotional material, cooperation with other projects and networking, final event organization, publications in magazines. **Exploitation (E):** DigInTraCE joint venture (JV) of the key consortium partners, individual exploitation roadmaps, exploitation and business innovation plan
TARGET GROUPS	**OUTCOMES**	**IMPACTS**

P4P stakeholders, **Industrial stakeholders (other than P4P;** (producers, process industries, manufacturers, intermediates, recyclers, etc.) **Partnerships, Associations, Initiatives (e,g,** Circular Plastics Alliance, European Plastic Convertors Association, European (e.g. European Committee for Standardisation) and National standardization bodies, **Policy makers and financing institutions** **Academia and research community** **Citizens and general public**	Secondary raw materials use: ↑15- >80 %, Waste↓: 8-50%, GHG ↓: 20 %, by-product value ↑: 15-200 % Traceability Platform: 100 active users, ≥10 user types with different access rights to DPP CLSC toolkit: ↑ secondary raw material use by 10 % LCA tool: output in < 20 sec Secondary raw materials in new BOMs and recipes: ↑ 10% Wood by-products application for higher-value flows: ↑ ≥ 30% SmartTags monitoring: ≥1000 batches; ≥2 parameters ≥5 material parameters real time identification Sorting of wood and Plastic accuracy: >90 % 10 loops closed and more than 5 cross-sectoral collaborations Min. 10 real time parameters integrated in DPP Certification scheme for recycled content: min. 1 Secondary plastic identification and certification cost: ↓20% min. 2 optimized recipes in P&P sector 20% increase in recycled plastic use for moulding 3 sustainable recovery technologies extraction yield: >40% Adhesives: increase up to 40% phenol substitution	**Environmental:** i) 5 tracing & 3 sorting technologies (TRL 9 by 2030) expansion to ≥5 sectors of P4P, ii) contribution to overall potential of emissions reduction (7 times and reduce global emissions by 15 %[55]) and to EU becoming climate neutral by 2050 iii) ↑ decision-making capacity for improved use of secondary raw materials. **Societal:** i) educational services (incl. co-designed materials): ≥ 1000 users (2030) ii) decentralized traceability platform:1000 registered users iii) standardization deliverables to reach stakeholders in all EU and EFTA countries, iv) contribution to digital transformation in EU by 2030, to EC targets (European skills agenda and the digital education action plan): 70% adults with basic digital skills by 2025 **Scientific:** 5 services & 9 prototypes with research documentation available to scientific community for further enhancements of concepts, scaling and transferability **Economic:** i) facilitate functioning of companies (esp. SMEs) in circular transition ii) contribute to level of ambition (Digital Compass): 75 % Eu enterprises engaged in cloud computing services, big data and AI, 90 % of Eu SMEs with basic level of digital intensity

[55] European Union, 2020, "Supporting the Green Transition. Shaping Europe's Digital Future".

3.1. *Work Plan and Resource*

DigInTraCE's implementation is realised through a set of seven work packages (WPs). Planned duration of the project is 48 Months as presented in the Gantt chart in Figure 7. Research and innovation actions of DigInTraCE are organized around the 4 largescale pilots serving as demonstration of all these activities. This is reflected in the structure of the WPs and their interrelations as presented in Figure 6.

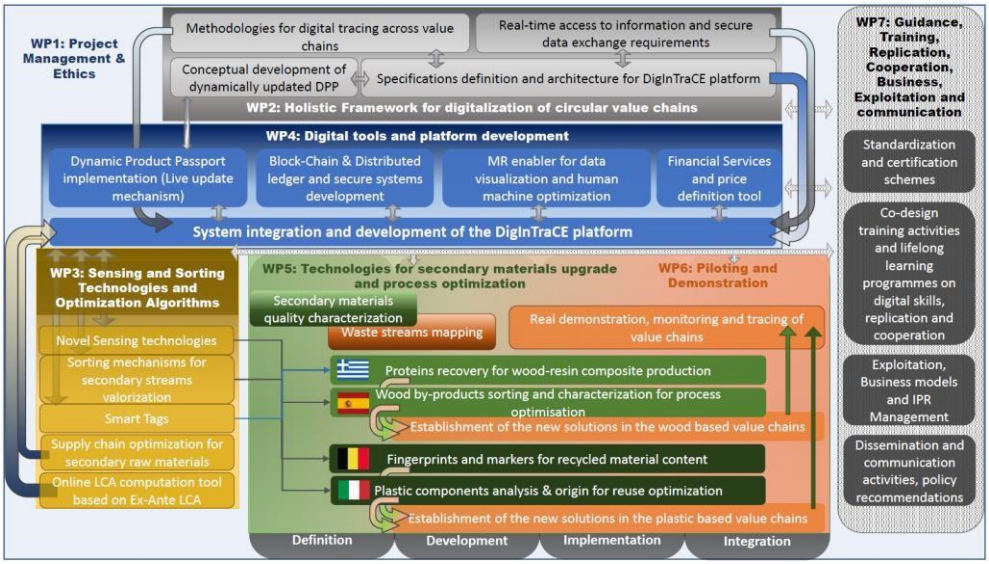

Figure 6 DigInTraCE pert chart

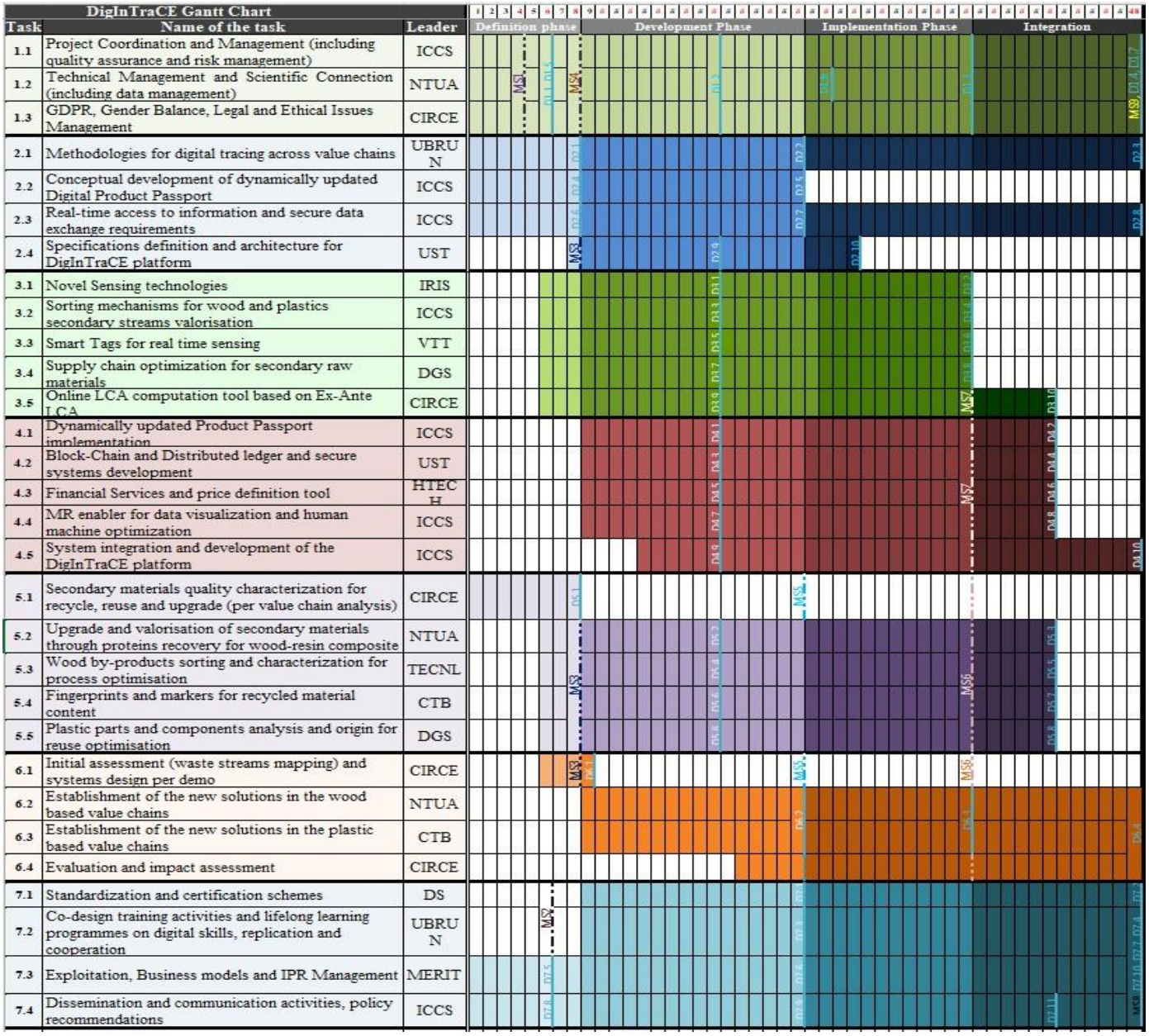

Figure 7 DigInTraCE Gantt chart

3.2. *Capacity of participants and consortium as a whole* Associated with document Ref. Ares(2022)7761208 - 10/11/2022

DigInTraCE consortium includes **22 organisations** (21 partners, 1 affiliated entity) from **6 EU members and 1 Associated country**, that together form a complete group uniting and utilizing the necessary expertise, skills, interdisciplinary knowledge and resources to constitute representative value chains and capable to achieve objectives and project goals. The consortium is multidisciplinary including (based on their main role in the project): **8** Academic and Research Technology Organisations, **6** Industrial partners, **3** novel digital technology providers, **1** Standardization, **2** certification (association & affiliated body) and **2** pioneer SMEs. Partners have been carefully selected to provide complementary skills and competences, which adequately cover and fully deliver all the project objectives and activities, starting from ideas to co-design needs and specifications, to implementation, system integration, validation, piloting and

demonstration, performance evaluation, social acceptance, capacity building and standardisation, impact maximisation through wide dissemination and communication, exploitation and policy outreach. Past collaborations and related experience were also considered upon selection. The consortium will address the objectives as indicated:

Objective 1: requires experts in up-cycling, reuse and upgrade technologies for secondary materials and waste reduction (NTUA, TECNL, CTB) as well as experience in digital traceability, DDPs (UST, ICCS, UBRUN) and data-driven and AI algorithms -supporting sensing and sorting- innovations (IRIS, ICCS, DGS, UVQ). **Obj2:** requires strong experience in the concepts of traceability and DDPs (UBRUN) and in blockchain based decentralized, data exchange and digital platforms (UST, ICCS) as well as experts in certification (CTB, TNLCERT) and standardization (DS). **Obj3:** requires experts in sensing and sorting innovations (IRIS, ICCS). **Obj4:** requires strong experience in SmartTags (VTT), Smart Contracts (UST) and financial services (HTECH) and open software and hardware (ICCS, UST, CIRCE). **Obj5:** requires industry and stakeholders of the 4 value chains (CHIMAR, MXS, AGRIST, ASTI, SIGIT), research activities experts (NTUA, TECNL, CTB, UST, ICCS, UBRUN, IRIS), wide experienced partners to validation, performance and impact evaluation (CIRCE) as well as standardization, certification and exploitation experts (DS, CTB, TNLCERT, MERIT, EUPC). **Obj6:** requires partners experienced in co-design, knowledge transfer, learning, educational and dissemination activities (UBRUN, ICCS, CIRCE, EUPC). All partners have been allocated with adequate resources based on their expertise and capacity to successfully fulfil their obligations.

Consortium composition and the exact role, individual contributions and access to infrastructure are presented in Table 5. The Internal Management Structure to be established will be complemented by external Advisory Board managed in WP1.

Table 5 DigInTraCE Consortium Complentarity.	**HORIZONTAL PARTNERS or Partners with horizontal activities**	
ICCS: Coordinator, in charge of project management, dissemination & communication, platform and ICT development. It has access to state-of-theart ICT equipment within its premises and will exploit the platform for Industry 4.0 and Circular solutions. Responsible partner for all sorting mechanisms and techniques.	**NTUA**: Scientific & technical manager, will be responsible for organizing and planning of technical activities. As experts in physical processes and life cycle assessment, NTUA will exploit research for creation of educational material & potentially a spinoff company regarding novel protein recovery processes.	**IRIS:** Experts in real-time monitoring techniques and industrial AI solutions for improved process quality and control. They will be in charge of all the plastic components identification and characterisation by utilisation of spectroscopy-based analyzers.
HTECH: Research performer, in charge of design, development, and testing of the financial services toolkit to be integrated into the decentralized traceability platform.	**UBRUN:** Research performer & technology developer, will be in charge of defining methodologies to clearly define unit of material for tracing, while also will be in charge of training activities.	**VTT:** Specialists in Smart Tags, will provide related research expertise, technology infrastructure and methodologies for the detection of environment conditions' shifts.
CIRCE: Research performer, responsible for design, testing & validation of LCA tool to be integrated into DPP and decentralized traceability platform. Performance evaluation & assessments.	**MERIT:** Exploitation strategy, plan and technology market transfer experts, responsible for contribute to materials and technologies standardization activities.	**UST:** Blockchain leaders, in standardization co-definitioncharge of blockchain work and market needs and IPR management.
	DS: the official national business organization in Denmark, will DLT technologies.	
	TNLCERT: affiliated entity of TECNL offering certification services, will contribute to certification schemes.	**EUPC:** association of vertical plastics' certification and engagement of EU plastic converting stakeholders.

Wood-based value chains		**Chemicals-based value chains**	
Greek Demonstrator		**Belgian Demonstrator**	
	NTUA is leader & main responsible for R&I activities. **AGRST** and **MXS** are suppliers of by-products (oilseedcake and wood), **NTUA** performs extraction of proteins from the oilseed crops that are supplied to **CHIMAR** (industrial partner with in-house R&D services), to produce bio-based adhesives and wood composites, ultimately provided to **MXS** for commercial applications.		**CTB** is the coordinator and main accountable for R&I activities, while also providing lab facilities for fingerprinting and PET upcycling and textile extrusion, as well as refinement of certification schemes of recycled content. **SIOEN** allows for testing and valiation of upscaling the processes and for textile production in its premises. **CIRCE** supports this pilot in the form of value chain mapping. Supplier of secondary materials will be EREMA, an entity that has ensured its support via a letter of support, although not joining the consortium.

Spanish Demonstrator		Italian Demonstrator	
	ASTI is the supplier of secondary raw materials and provides its processing facilities for demonstration. **TECNL** is the main responsible for R&I activities and valorization pathways.		**SIGIT** provides the secondary raw materials and the required recycling process infrastructure for experimental try-out and lab for chemical analysis, whereas **DGS** is the leader & main responsible for R&I activities and demonstrating prototypes. **UVQ,** as research performing partner, will test and validate the developed algorithms for process optimisation.

4. Ethics Self-Assessment

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The consortium confirms compliance with the ethical principles and the applicable EU, international, and national laws for the ethics issues identified in the Ethics Summary Report and any additional ethics issues that may emerge in the course of the grant, will be ensured. The Consortium also confirms that for any applicable ethics issue, the guidance provided in the European Commission Ethics Self-Assessment Guidelines will be rigorously followed.

4.1. Ethical dimension of the objectives, methodology and likely impact

Each member of the DigInTraCE leading or participating in research task that involves human participants will ensure the respect for people and for human dignity, fair distribution of the benefits and burden of research, and the protection of the values, rights and interests of the research participants. Ethical standards and guidelines will be rigorously followed and applied, irrespectively of the country in which the research is carried out. The necessary ethics approvals must be obtained, if required, as well as free and fully informed consent of the research participants. It is the responsibility of each member of the consortium leading an activity to obtain the relevant approvals and to ensure that their research complies with the laws and regulations in all the EU Member States in which the research takes place. Human participants will be required to provide information e.g. via interviews, online forms and questionnaires that will be used for user engagement, stakeholder's identification, definition of scenarios, and gathering inputs and feedback for the project and the new solutions and technologies to be developed. Participation is voluntary and before any relevant research activity that involves humans, each individual will signify their agreement to participate with a clear affirmative act, DigInTraCE will apply clear informed consent procedures with a dedicated consent form. Although no adverse side effects are expected, a plan for collecting, reporting, and managing solicited and spontaneously reported adverse events and other unintended effects of projects' interventions will be undertaken. All partners are fully aware of this Charter and Directive and have abided by these regulations in past and on-going research projects and any individual partners belonging to professional bodies will abide by the ethics rules and regulations of these bodies. DigInTraCE consortium will keep a record of all potential ethical issues that arise as project is developed and evaluated, and through discussion with the stakeholders. Ethics approval for the project will be sought from all necessary and applicable review panels and relevant committees in all partner countries undertaking any work. Consortium will examine the following aspects in every related activity:

•Composition, roles, and responsibilities of the partners in charge;
•Description of research questions, specific objectives and justification for undertaking the study or analysis;
•Description of study design, including type, settings and sites where data will be collected;
•Inclusion and exclusion criteria for participants;
•Time schedule of enrolment and assessments;
•Estimated number of participants needed to achieve study objectives and how these were determined;
•Plans for assessment and collection of outcome, baseline, and other trial data and study instruments;
•Who will obtain informed consent or assent from potential trial participants or authorized surrogates, and how;
•How personal information about potential and enrolled participants will be collected, shared, erased/anonymized if necessary and maintained;

4.2. Compliance with ethical principles and relevant legislations

All the local legislations for virgin raw or secondary materials, waste treatment and digitization will be analyzed and respected in collaboration with the local authorities and Environmental or other relevant Ministries, in harmony with the respective European directives and regulations. DigInTraCE will ensure that it is fully compliant with the Do No Significant Harm (DNSH) Principle since its activities as well as the expected life cycle impact of the innovations at a commercialization stage, substantially contribute in a positive way and/or do not significantly harm any of the six

environmental objectives set out in Art. 9 and 17 of the EU Sustainable Finance Taxonomy Regulation (EU) 2020/852. DigInTraCE will apply a life-cycle oriented approach (validated through the utilization of relevant LCA tool when possible) to ensure that proposed solutions are sustainable and consisting of 3 steps:

i)Identification of actual and potential adverse impacts during the life cycle of the project activities

ii)Prevention or mitigation of adverse impacts

iii)Accounting for how adverse impacts are addressed by tracking performance and communicating results. DigInTraCE activities are expected to have a direct positive impact to the environmental objectives defined: Climate Change Mitigation by reducing GHG emissions (use Recommendation 2013/179/EU or ETSI ES 203 199,ISO 14067:2018 or ISO 14064-2:2019 to calculate lifecycle GHG and net emissions), to Pollution prevention & control through waste reduction and identifying harmless substances and contaminants in each pilot, to Transition to Circular Economy through the establishment of symbiotic and circular flows within same or different industries for materials and wastes whereas is not expected any negative impact to the sustainable use and protection of water and marine resources, protection and restoration of biodiversity & ecosystems and the Climate Change adaptation.

Specifically, DigInTraCE will make sure that the related DNSH criteria (on Climate Adaption; climate adaptation; water; circular economy; pollution prevention and biodiversity) from the EU Taxonomy Compass will be applied for the below project related activities:

-Data processing, hosting and related activities

-Data-driven ICT solutions for GHG emissions reduction

-Computer programming, consultancy and related activities -Close to market research, development and innovation And if applicable in the below as well:
-Manufacture of aluminium
-Manufacture of iron and steel
-Manufacture of organic basic chemicals
-Manufacture of plastics in primary form
-Engineering activities and related technical consultancy dedicated to adaptation to climate change
-Any other activity that may arise during the project.